PROPERTY DUE DILIGENCE

Comprehensive Lot Due-Diligence Report

Twelve Texas lots · parcel, terrain, flood, soil and site analysis

#	Property	County	Acres	FEMA zone
1	652 Pony Trl, Angleton, TX 77515	Brazoria County	1.25	AE
2	543 Pony Trl, Angleton, TX 77515	Brazoria County	0.84	AE
3	274 Pony Trl, Angleton, TX 77515	Brazoria County	0.86	AE
4	220 Pony Trl, Angleton, TX 77515	Brazoria County	0.76	AE
5	1601 Old Henry Ct, Angleton, TX 77515	Brazoria County	0.80	AE
6	Private Road 52415, Leesburg, TX 75451	Camp County	5.77	X (likely)
7	119 Dolly Rd, Sulphur Springs, TX 75482	—		*not identified*
8	15 Harrell Cemetery Rd, Coldspring, TX 77331	San Jacinto County	1.35	A + X
9	0 Saint Lawrence River Rd, Montgomery, TX 77316 **[identity unresolved]**	Montgomery County	0.00	AE + X
10	141 Ward Rd, Justiceburg, TX 79330	Garza County	1.94	not mapped
11	80 Ward Rd, Snyder, TX 79549	Kent County	1.91	not mapped
12	2883 W Arrowhead Dr, Wichita Falls, TX 76310	Clay County	1.57	X (likely)

LOTS SUBMITTED	PARCELS IDENTIFIED	INSIDE A FEMA SFHA	ANALYSIS DATE
12	11	7	06 October 2026

Read the method section first
Findings are labelled **Official**, **Model-derived** or **Needs confirmation**. Elevations and contours in this report are derived from a 1 m national elevation model and are **not survey grade**. They must not be used to set a finished-floor elevation against a Base Flood Elevation — that requires a surveyor's Elevation Certificate. No HOA, zoning, utility or dock document was obtainable from public data; every such item is flagged rather than estimated.

How to read this report

Every factual line in this report is labelled with where it came from, because the difference between a published flood zone and a computed contour matters when money is at stake. Three labels are used throughout:

OFFICIAL	Published by the agency of record and reproduced unchanged: FEMA flood zones and FIRM panels, county appraisal-roll parcel data, recorded legal descriptions.
MODEL-DERIVED	Computed here from authoritative inputs. Elevations, contours, slopes and cross-sections come from the USGS 3DEP 1 m bare-earth DEM; soil properties come from USDA NRCS map units. These are good engineering estimates and they are **not survey grade**.
NEEDS CONFIRMATION	Cannot be established remotely at all. HOA covenants, zoning, utility availability, dock rights, lake depth, school assignment and short-term-rental rules sit here. The report names the question and the office that holds the answer rather than guessing.

What this report is not

It is not a survey, a geotechnical report, an elevation certificate, a flood determination, a title opinion or an appraisal. DEM-derived contours and elevations carry roughly a foot of vertical uncertainty and must never be used to size a finished floor against a Base Flood Elevation. For that you need a licensed surveyor's Elevation Certificate.

Where the numbers come from

Dataset	Publisher	Used for
StratMap statewide land parcels	Texas Geographic Information Office, rolled up from county appraisal districts	Parcel boundary, legal description, acreage, APN, owner, appraised value
National Flood Hazard Layer	FEMA	Flood zone, SFHA status, FIRM panel and effective date, Base Flood Elevation lines
3DEP bare-earth DEM (1 m)	USGS	Elevations, relief, slope statistics, contours, cross-sections, 3D terrain
Soil Survey Geographic Database	USDA NRCS	Soil series, texture, drainage, shrink-swell, pH, septic and dwelling ratings
National Hydrography Dataset	USGS	Waterbody position and distance from the parcel boundary
ERA5 daily reanalysis	ECMWF, via the Open-Meteo archive	30-year climate normals, growing season, freeze frequency
Geocoder and geographies	US Census Bureau	County and tract identification

Listing imagery, prices and listing history are not reproduced: Zillow blocks automated access, so none was retrieved. Request photographs and price history from the listing agent.

Executive summary

Twelve lots were submitted. 11 were matched to a parcel in the Texas statewide appraisal-roll layer and analysed; 1 could not be identified from the address alone. The findings that most affect value are these:

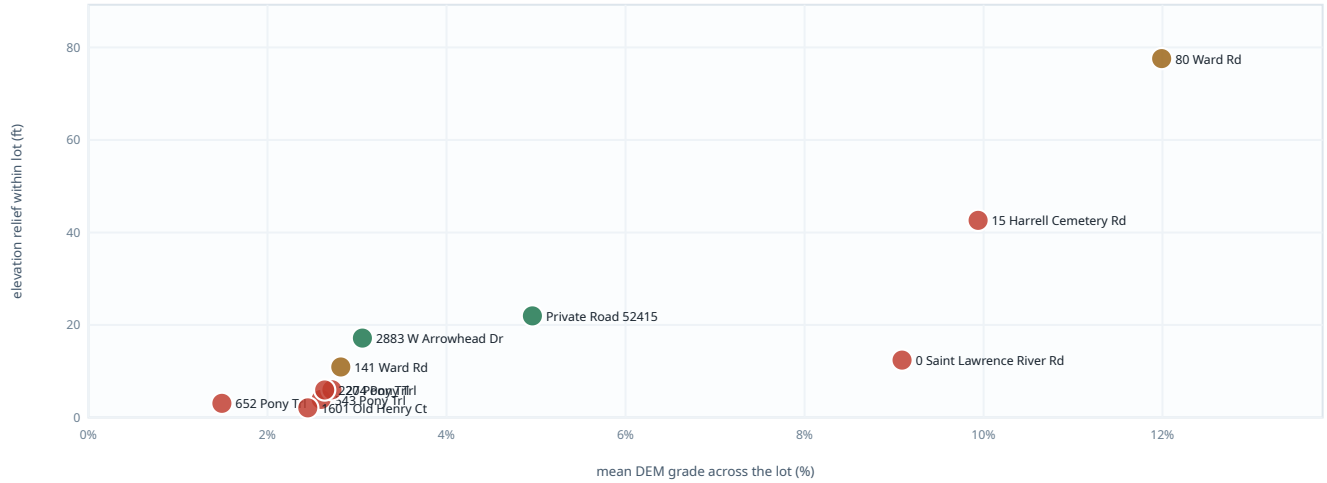
LOTS ANALYSED	IN A FEMA SFHA	HIGH SHRINK-SWELL SOIL	ABUT MAPPED WATER
11 / 12	**7** lots	**5** lots	**5** lots

- **7 lots sit inside a FEMA Special Flood Hazard Area** (652 Pony Trl, 543 Pony Trl, 274 Pony Trl, 220 Pony Trl, 1601 Old Henry Ct, 15 Harrell Cemetery Rd, 0 Saint Lawrence River Rd). For the Bar X Ranch lots in Angleton the measured ground surface lies *below* the indicative Base Flood Elevation, which means an elevated finished floor, floodplain permitting and mandatory flood insurance. This is the single largest cost and feasibility factor in the portfolio.
- **2 lots have no effective FEMA digital flood mapping** (141 Ward Rd, 80 Ward Rd). That is not the same as low risk: the zone is formally undetermined and lenders may treat it conservatively.
- **5 lots carry high or very high soil shrink-swell** (543 Pony Trl, 274 Pony Trl, 1601 Old Henry Ct, 15 Harrell Cemetery Rd, 2883 W Arrowhead Dr). Expansive clay drives foundation cost and is the usual cause of structural distress in these regions. Assume an engineered foundation and a geotechnical report.
- **Terrain varies enormously across the portfolio.** The Angleton lots are nearly flat with 2–6 ft of relief, whereas the Oak Canyon lot at Lake Alan Henry falls about 78 ft across the parcel and reaches roughly 60% grade over a 50 ft run. Those two extremes imply completely different build budgets.
- **Nothing about HOA covenants, zoning, utilities, dock rights or rental rules could be verified.** Those documents are not open data. Each is capable on its own of blocking a second dwelling, a dock, poultry or short-term letting, so they should be obtained before any offer becomes unconditional.

Portfolio comparison

Lot	County	Acres	FEMA zone	Relief ft	Mean grade	Shrink-swell	Septic rating	Water ft
652 Pony Trl	Brazoria	1.25	**AE**	3.0	1.49%	Moderate	Very limited	3,370
543 Pony Trl	Brazoria	0.84	**AE**	3.9	2.60%	**Very high**	Very limited	3,688
274 Pony Trl	Brazoria	0.86	**AE**	6.0	2.72%	**Very high**	Very limited	24
220 Pony Trl	Brazoria	0.76	**AE**	6.0	2.64%	Moderate	Very limited	29
1601 Old Henry Ct	Brazoria	0.80	**AE**	2.1	2.45%	**Very high**	Very limited	1,066
Private Road 52415	Camp	5.77	X (likely)	22.0	4.96%	Moderate	Very limited	652
119 Dolly Rd	not identified — no analysis attributed							
15 Harrell Cemetery Rd	San Jacinto	1.35	**A+X**	42.6	9.94%	**High**	Very limited	25
0 Saint Lawrence River Rd	Montgomery	0.00	**AE+X**	12.4	9.09%	Moderate	Very limited	336
141 Ward Rd	Garza	1.94	**not mapped**	10.9	2.82%	Moderate	Very limited	587
80 Ward Rd	Kent	1.91	**not mapped**	77.6	11.99%	Low	Somewhat limited	56
2883 W Arrowhead Dr	Clay	1.57	X (likely)	17.2	3.06%	**High**	Very limited	26

elevated risk undetermined Water ft = distance from the parcel boundary to the nearest mapped waterbody



Terrain difficulty against flood exposure. Horizontal axis is the mean DEM grade across the lot, vertical axis is the elevation relief within the lot. Red = inside a mapped SFHA, amber = no effective FEMA mapping, green = no SFHA on the parcel. Lots toward the upper right are the most expensive to build on.

652 Pony Trl, Angleton, TX 77515

Lot 1 of 12 | ID confidence: high | Bar X Ranch | Brazoria County

A1 Property identification & record data OFFICIAL

Situs address of record	652 PONY TRAIL , , TX 77515
Legal description	BAR X RANCH SEC 11 (A0053 & A0097) BLK 3 LOT 66 ACRES 1.27
Subdivision	Bar X Ranch
County	Brazoria County (FIPS 48039)
Parcel ID / account	185738
Geographic ID (APN)	1543-0066-000
Owner of record	8 MILE PARK LP
Acreage — appraisal roll (legal)	1.27 ac
Acreage — GIS polygon	1.249 ac
Acreage — recomputed from boundary	1.253 ac
Appraised land value	$34,190
Appraised improvement value	$0
Appraised market value	$34,190
Centroid (WGS84)	29.113202, -95.563507
Longest axis / width	544 ft × 201 ft (axis bearing 232°)

Source: Texas StratMap statewide land parcels (county appraisal roll). Matched by situs address match. Appraisal-roll acreage and a GIS polygon rarely agree exactly; neither replaces a survey.

A2 · Listing and plat material not retrievable here NEEDS CONFIRMATION

Current asking price, days on market and listing history.
Obtain from: Listing agent / MLS

Listing photographs: lot frontage, shoreline, road access and surroundings.
Obtain from: Listing agent (Zillow blocks automated retrieval, so no listing imagery is reproduced here)

Seller disclosure notice and any known defects.
Obtain from: Listing agent

Recorded plat and Esri/county GIS parcel exhibit for comparison with this report.
Obtain from: County Clerk plat records and the county GIS portal

B5 Terrain, contours and true-scale site plan MODEL-DERIVED

LOWEST POINT	HIGHEST POINT	RELIEF	MEAN ELEVATION
23.2 ft	26.2 ft	3.0 ft	24.4 ft



True-scale site plan. Parcel boundary from the county GIS polygon; contours at 1.0 ft intervals derived from the USGS 3DEP 1 m bare-earth DEM. A–B is the cross-section line. Horizontal and vertical map scales are equal, so shapes and distances are undistorted.

MEAN GRADE	MEDIAN GRADE	90TH PCT GRADE	MAX GRADE
1.49 %	1.10 %	3.04 %	12.46 %

Mean slope angle	0.85°	**90th pct angle**	1.74°	**Max angle**	7.10°	**Grid cells**	1,156

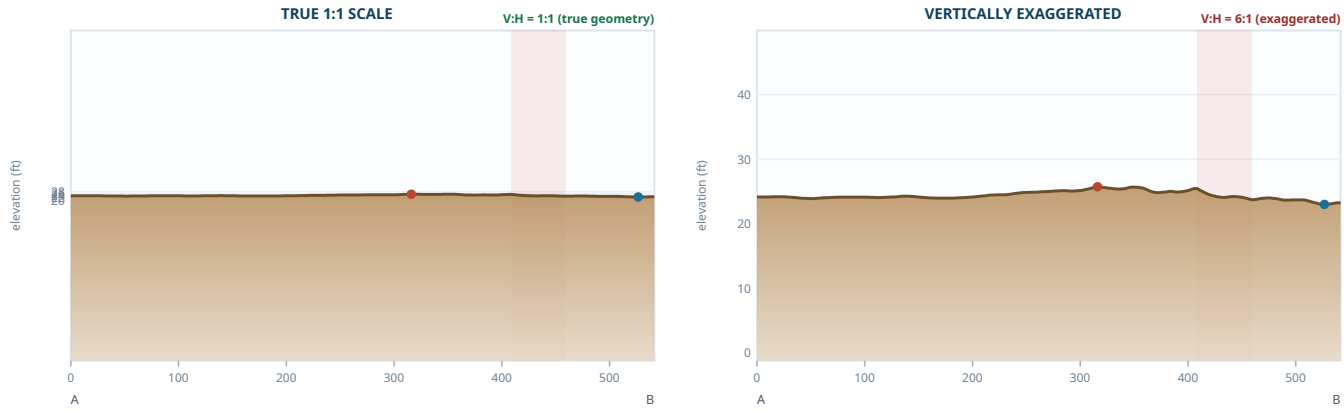
B7 · Terrain classification

Nearly flat. Minimal grading. Establish fall away from the structure.

Basis: mean DEM grade 1.49% (90th pct 3.04%), relief 3.04 ft

Grading expectation: Minimal grading

B6 Longitudinal cross-section MODEL-DERIVED



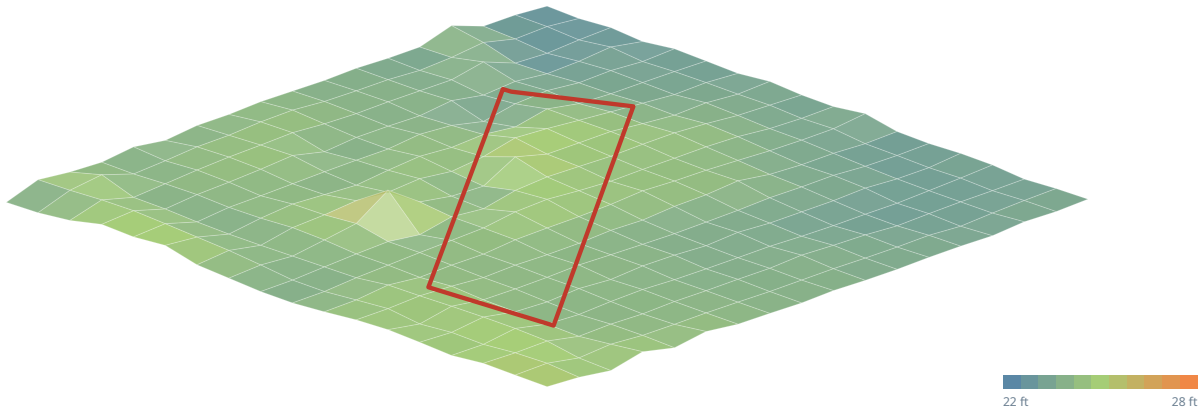
Run A–B	542 ft	**Lowest**	23.0 ft	**Highest**	25.7 ft
Elevation difference	2.7 ft	**Average grade**	0.5%	**Direct rising angle**	0.3°
Steepest 50 ft	**3.4%** (1.9°) over a 50 ft run				

Reading: Over 542 ft the ground rises only 2.7 ft, a direct angle of 0.3°. At true 1:1 scale the profile is almost a straight line, which is the correct reading: this lot is effectively flat. The exaggerated panel is the same data and should not be read as slope.

Left panel is the real geometry at 1:1 vertical-to-horizontal scale. The right panel exaggerates the vertical axis, which is how slope is conventionally drawn on maps but overstates steepness. The shaded band marks the steepest 50 ft run. Profile sampled from the 3DEP DEM along the lot's longest axis.

B8 Three-dimensional terrain MODEL-DERIVED

Sampled window relief 6.4 ft · vertical scale exaggerated for legibility



Isometric view of the sampled DEM window; the red outline is the parcel boundary draped on the surface. Vertical scale is exaggerated to reveal form — use the 1:1 cross-section above for true steepness.

C9 FEMA flood-zone determination OFFICIAL

Zone: AE | Inside Special Flood Hazard Area

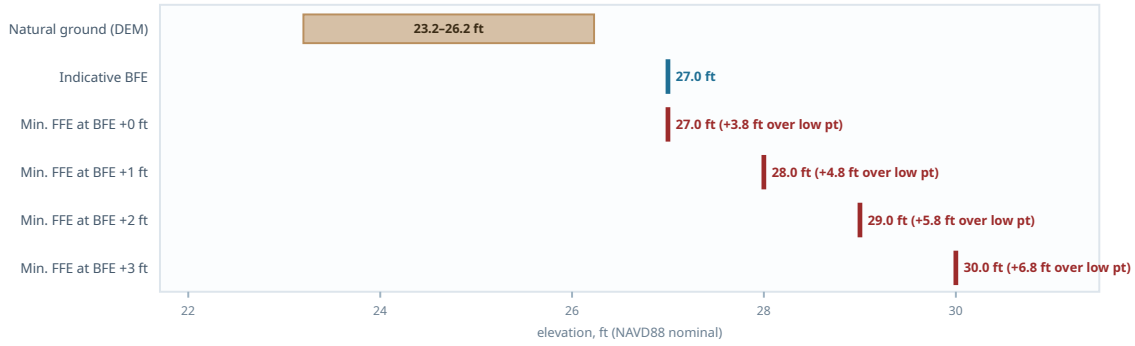
Zones intersecting the parcel	AE
Special Flood Hazard Area	Yes — mandatory flood insurance for a federally backed mortgage
FIRM panel	48039C0580K · effective 2020-12-30
Mapping status	—

Parcel intersects mapped FEMA flood-hazard polygon(s); see zones above.

C11-13 Ground elevation, BFE and minimum finished floor OFFICIAL MODEL-DERIVED

Lowest natural ground (DEM)	23.20 ft
Highest natural ground (DEM)	26.23 ft
Mean ground	24.42 ft
Indicative BFE	27.0 ft
Lowest ground relative to BFE	**-3.80 ft**
Highest ground relative to BFE	-0.77 ft

BFE basis: highest of 1 FEMA BFE line value(s) within 2,000 ft ([27.0]) - indicative only, the governing BFE must be read at the site from the FIRM/FIS profile



Freeboard scenarios. The brown band is the measured spread of natural ground across the lot. Red ticks are the minimum finished-floor elevation at BFE plus 0, 1, 2 and 3 ft of freeboard. **The freeboard the county actually enforces is not established in this report** — the scenarios show the consequence of each, they do not assert a requirement.

Scenario	Min. finished floor (ft)	Rise above lowest ground (ft)	Rise above highest ground (ft)
BFE + 0 ft	27.00	3.80	0.77
BFE + 1 ft	28.00	4.80	1.77
BFE + 2 ft	29.00	5.80	2.77
BFE + 3 ft	30.00	6.80	3.77

DEM ground across the lot runs 23.2-26.23 ft against an indicative BFE of 27.0 ft. The lot therefore sits 3.8 ft below BFE at its lowest point. Any habitable floor must be set above BFE plus the freeboard the county actually enforces - that freeboard figure is NOT established here.

Do not assume fill is permitted
Raising a pad with fill inside a mapped floodplain is frequently restricted or requires compensatory storage and a floodplain development permit. Elevating on piers or a stem wall is often the permitted route instead. This must be confirmed with the county floodplain administrator before any design assumption is made.

C · Flood items still to confirm NEEDS CONFIRMATION

Official flood determination and the effective FIRM panel read at the parcel.
Obtain from: County floodplain administrator or a flood certification service

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Enforced freeboard above BFE, fill rules and compensatory-storage requirements.
Obtain from: County floodplain administrator

First Street / third-party forward-looking flood risk and estimated flood depth.
Obtain from: First Street Foundation (riskfactor.com) for the confirmed address

Flood insurance quote under current NFIP pricing.
Obtain from: Insurance agent, using an Elevation Certificate

Site drainage: swales, culverts, outfall and upstream runoff from neighbouring lots.
Obtain from: Civil engineer site visit; county drainage criteria

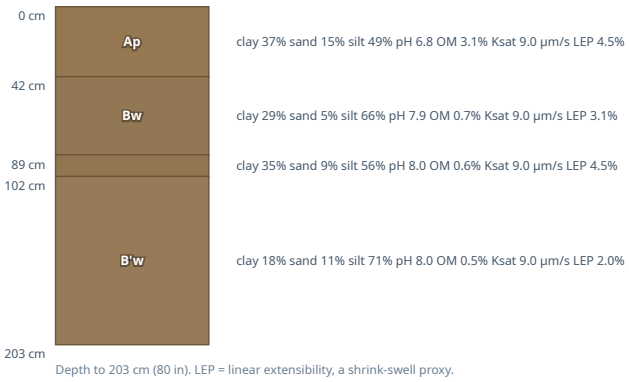
D14 Soil characteristics MODEL-DERIVED

NRCS map unit intersecting the lot	Share of lot (sampled)
Asa silty clay loam, 0 to 1 percent slopes, rarely flooded	54.5%
Pledger clay, 0 to 1 percent slopes, rarely flooded	45.5%

Component	%	Drainage	Hyd.grp	LEP %	Shrink-swell	pH	Clay %	Flooding
Asa	85%	Well drained	B	4.5	Moderate	6.8	37	Rare
Pledger	85%	Moderately well drained	D	19.0	Very high	7.0	70	Rare

LEP = linear extensibility (shrink-swell proxy). Hydrologic group D indicates the slowest infiltration. Values are NRCS representative values for the map unit, not measurements on this lot.

Asa — representative profile



Representative horizon sequence for the dominant component (Asa).

D15 Foundation suitability MODEL-DERIVED

Dominant soil shrink-swell potential is **Moderate** (peak LEP 4.5%). Stiffened/post-tensioned slab or moisture-conditioned subgrade is typical.

Slab-on-grade	Stiffened / post-tensioned
Crawlspace	Suitable
Pier and beam	Suitable
Basement	Not advised
Stepped footing	Only if sloped
NRCS rating, dwellings without basements	—
Geotechnical investigation advised	Yes

Foundation selection is an engineering decision. NRCS map-unit shrink-swell indicates risk; it does not substitute for a geotechnical investigation.

D16 Septic / on-site sewage suitability MODEL-DERIVED

Septic Tank Absorption Fields	Somewhat limited
Septic Tank Leaching Chamber (TX)	Very limited
Septic Tank, Subsurface Drip Irrigation (TX)	Very limited

Constraints identified from the soil and terrain data

- Mapped flooding frequency 'Rare' is a direct constraint on on-site sewage facilities.

Permitted bedroom count: NOT ESTABLISHED
Permitted bedroom count follows from a site-specific soil evaluation (TCEQ Ch. 285 class, loading rate, usable drainfield area and setbacks). It cannot be derived from map-unit data. A licensed site evaluator must run the percolation/soil profile test.

NRCS ratings describe the map unit, not the lot. A TCEQ-compliant site evaluation and percolation test is required for any permit.

D17 Gardening suitability MODEL-DERIVED

SURFACE PH	CLAY	SAND	ORGANIC MATTER
6.8	37 %	15 %	3.1 %

Drainage class	Well drained
pH range through profile	6.8 – 8.0
NRCS lawn / landscape rating	Somewhat limited
Raised beds advised	Yes

Heavy clay / restricted drainage / high shrink-swell in the surface horizon makes raised beds with imported loam the lower-risk option.

Likely amendments

- Coarse organic matter and gypsum-style conditioning to improve tilth in heavy clay
- Mulch, to buffer soil temperature and moisture

Based on NRCS map-unit averages. A lab soil test is required before committing to any amendment programme.

D · Soil and planting items to confirm NEEDS CONFIRMATION

Laboratory soil test: pH, macro and micronutrients, organic matter, texture.
Obtain from: Texas A&M AgriLife Soil Testing Lab, from samples taken on the lot

Plant hardiness zone confirmation and a species list matched to the tested soil.
Obtain from: USDA hardiness map plus the county AgriLife extension office

E19-23 Utilities and infrastructure NEEDS CONFIRMATION

Utility availability is not published as open geospatial data at parcel level. Nothing is inferred here: each item below must come from the serving utility in writing, because a missing water tap or a long electric extension can change the build budget materially.

Water supply: municipal, water-supply-corporation, or private well; is a tap at the road?
Obtain from: Serving water supply corporation / county; well logs via TX Water Development Board

Sewer: is public sewer available, or is an on-site facility required? Distance to any main.
Obtain from: County / city utility department or MUD

Electricity: provider, whether service exists at the lot line, overhead vs underground, connection cost.
Obtain from: Retail electric provider / local co-op (service availability letter)

Natural gas or propane availability and existing infrastructure.
Obtain from: Gas utility or local propane suppliers

Internet: fibre, cable, fixed wireless, DSL or satellite, with advertised speeds.
Obtain from: FCC National Broadband Map plus a direct serviceability check by address

F24-25 Waterfront position MODEL-DERIVED

Interior lot - no adjacent mapped water

Nearest mapped waterbody	unnamed waterbody
Distance from parcel boundary	3,370 ft
Waterbody area	0.002 km²

NHD polygons are hydrographic mapping, not a boundary survey. Whether the lot carries riparian frontage, deeded access or only common-area access is a title/plat question and is NOT established here.

F26-29 · Dock, depth, boating and fishing NEEDS CONFIRMATION

Whether a dock is permittable at this shoreline, and the governing authority.
Obtain from: Controlling lake / river authority, plus the HOA

Dock size limits, shoreline setbacks, shared-dock rules and permit cost.
Obtain from: Lake authority rules and permit schedule

Whether riparian or shoreline rights pass with the lot, or the shoreline is a common reserve.
Obtain from: Title commitment and recorded plat

Lake water-level regime and any shoreline-management plan.
Obtain from: Lake/river authority or reservoir operator

Bathymetry: average and maximum depth, and depth near this shoreline.
Obtain from: Lake authority survey, TPWD reservoir report, or a sonar survey

Seasonal water-level variation and drawdown history.
Obtain from: Reservoir operator gauge records / USGS gauges

Boating rules: permitted craft, motor limits, wake rules, ramps and marinas.
Obtain from: Lake authority and TPWD

Fishery: species present, stocking, licence requirements and catch limits.
Obtain from: Texas Parks & Wildlife reservoir survey reports

G30-33 HOA, covenants, zoning and a second dwelling NEEDS CONFIRMATION

The parcel sits in **Bar X Ranch** according to the recorded legal description, so a declaration of covenants very likely exists. The documents themselves are not public geospatial data and were not retrieved, so no dues figure, restriction or amenity is asserted here.

Whether an HOA or POA exists and is mandatory, with its name and management contact.
Obtain from: Title commitment, recorded Declaration, seller disclosure

Annual dues, monthly equivalent, transfer fee, and any special assessment.
Obtain from: HOA or management company estoppel / resale certificate

Declaration, CC&Rs, architectural guidelines, rules and all amendments.
Obtain from: HOA or title company

Restrictions on docks, fences, outbuildings, RVs, boats, livestock and poultry.
Obtain from: Recorded CC&Rs and architectural guidelines

Minimum dwelling size and any construction time limit after purchase.
Obtain from: Recorded CC&Rs

Amenities actually included: clubhouse, pool, courts, parks, lake and boat facilities.
Obtain from: HOA documents and budget

Zoning designation, or confirmation that the county is unzoned.
Obtain from: County or city development services

Minimum lot size, building coverage, setbacks, height limits, accessory-structure rules.
Obtain from: County/city development standards or subdivision regulations

Whether a second independent dwelling or in-law suite (ADU) is permitted, and on what terms.
Obtain from: County/city development services plus the HOA

Permitting path, driveway/culvert permit and any development fee.
Obtain from: County permitting office

H34 / I35 Hobby farming and short-term rental NEEDS CONFIRMATION

Whether chickens, ducks, rabbits, goats or bees are allowed, and in what numbers.
Obtain from: County rules plus the recorded CC&Rs (either can prohibit)

Any agricultural or wildlife valuation currently on the parcel, and rollback-tax exposure if use changes.
Obtain from: County appraisal district

Whether short-term rental is permitted at this location.
Obtain from: County/city ordinance and the HOA

Any STR permit, occupancy cap, parking or noise condition.
Obtain from: Local ordinance

Hotel occupancy tax registration and remittance obligations.
Obtain from: Texas Comptroller plus local taxing authority

J36-42 Environmental and geological due diligence MODEL-DERIVED

Ground-penetrating radar penetration (NRCS rating)	Very low penetration Indicates how useful GPR would be here; clay-rich soils attenuate the signal.
Sinkhole / karst exposure	Not indicated by the soil map units returned for this parcel. No restrictive bedrock or karst feature was reported. CONFIRM WITH TX BUREAU OF ECONOMIC GEOLOGY
Radon	EPA publishes radon zones by county, not by parcel. LOOK UP THE COUNTY ZONE AND TEST ANY BUILT STRUCTURE

Existing FEMA Elevation Certificate for the lot.
Obtain from: Seller, county floodplain office or a surveyor

Recent boundary, ALTA or topographic survey.
Obtain from: Seller or a licensed Texas surveyor (RPLS)

Geotechnical / soil boring report.
Obtain from: Seller or a geotechnical engineer

On-site sewage facility site evaluation and percolation test.
Obtain from: Licensed site evaluator / county OSSF office

Any ground-penetrating-radar or other subsurface investigation.
Obtain from: Seller; commission if subsurface features are suspected

K43-45 Residential construction feasibility MODEL-DERIVED

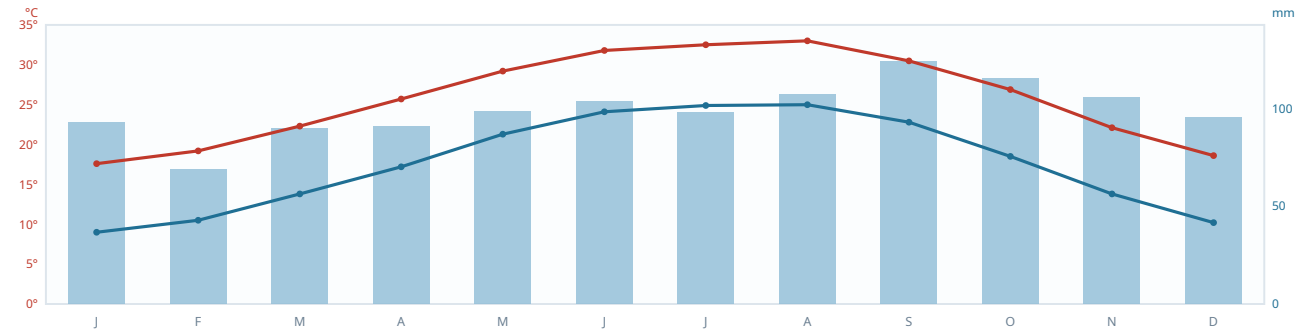
Lot area available	1.249 ac (54,399 sq ft)
Usable building envelope	NOT COMPUTABLE Setbacks, easements and any lake or drainage reserve come from the recorded plat and the HOA architectural guidelines, none of which were available.
Terrain constraint	Nearly flat — Minimal grading
Foundation implication	Moderate shrink-swell. Stiffened/post-tensioned slab or moisture-conditioned subgrade is typical.
Flood implication	Inside a mapped SFHA, so finished-floor elevation, flood venting and insurance all bind the design.

Construction cost
No build cost is estimated in this report. A defensible figure for a four-bedroom house depends on the HOA's minimum size and material rules, the finished-floor elevation forced by the flood zone, the foundation type the soil requires, and whether water and sewer must be created on site. Three of those four inputs are still unconfirmed for this lot, so any number quoted now would be arbitrary. Price it from a builder bid once the plat, CC&Rs, floodplain requirement and soil report are in hand.

L46 Climate MODEL-DERIVED

MEAN ANNUAL TEMP	MEAN SUMMER HIGH	MEAN WINTER LOW	ANNUAL RAINFALL
21.7 °C	32.4 °C	9.9 °C	1,189 mm

Record high	41.8 °C	**Record low**	-9.4 °C	**Rainfall**	46.8 in
Growing season	~342 days	**Days at or below 0 °C**	2.7 /yr	**Period**	1994-2023 (30 years)



Monthly normals, 1994-2023 (30 years). Bars are mean monthly rainfall (mm, right axis); lines are mean daily maximum and minimum temperature (°C, left axis). Source: ERA5 daily reanalysis via Open-Meteo archive API.

L47 Natural hazards

Flood	Mapped Special Flood Hazard Area
Other perils	Hurricane and tropical wind, tornado, hail, severe convective storm, lightning, extreme heat, drought and wildfire all vary by location within Texas. QUANTIFY FROM FEMA NATIONAL RISK INDEX FOR THE CONFIRMED COUNTY AND FROM AN INSURANCE QUOTE

L48-55 Community, schools, healthcare, shopping and access NEEDS CONFIRMATION

Demographics not retrieved
Census API requires an API key (keyless request rejected) Free key from https://api.census.gov/data/key_signup.html, then set CENSUS_API_KEY and re-run 05_context.py. Or read the figures directly from data.census.gov.

Driving distance and time to the nearest hospital, emergency department and urgent care.
Obtain from: Direct routing check from the confirmed parcel location

Distance to grocery, major retail, pharmacy, bank, fuel and hardware.
Obtain from: Direct routing check

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Assigned school district and campuses, with distances and current ratings.
Obtain from: TEA school-district locator and the district attendance-zone map

Nearest Indian / South Asian grocery, restaurants and places of worship.
Obtain from: Direct search from the confirmed parcel location

Nearest international airport, driving time and route.
Obtain from: Direct routing check

Walk / Bike / Transit scores.
Obtain from: Walk Score for the confirmed address

M56 Documented strengths

- Lot of 1.25 acres, which is generous for a single dwelling and leaves room for outbuildings and a garden.
- Gentle mean grade of 1.49%, so routine grading should suffice.
- Long growing season of roughly 342 frost-free days supports year-round production.
- Septic absorption is rated only "somewhat limited" rather than "very limited", which is the better of the two common outcomes here.

M57 Limitations and risks

- **Inside a FEMA Special Flood Hazard Area (AE).** This forces an elevated finished floor, triggers floodplain permitting, and makes flood insurance mandatory with a federally backed mortgage.
- Lowest natural ground sits about 3.8 ft below the indicative BFE, so the pad must be raised or the house elevated.
- Septic absorption is rated "very limited" for the dominant soil, so a conventional drainfield may not be permittable and an engineered or aerobic system may be required at higher cost.
- Mapped flooding frequency 'Rare' is a direct constraint on on-site sewage facilities.
- HOA covenants, zoning, utility availability, dock rights and short-term-rental rules are all unconfirmed. Any of them can independently prevent the intended use.

M59 Construction feasibility conclusion

Main house	Likely The lot is large enough and the terrain is workable. Finished-floor elevation and foundation type are the governing variables, not land area.
Second in-law suite / ADU	Unresolved Depends entirely on county rules and the HOA declaration; neither was available. Do not assume a second dwelling is permitted.
Garage	Likely No physical constraint identified, subject to setbacks and architectural guidelines.
Septic system	Unresolved Soil ratings are unfavourable; permittability and bedroom count require a site evaluation.
Garden	Conditional Workable with raised beds and imported loam; native surface soil is restrictive.
Hobby farming / poultry	Unresolved Governed by county rules and CC&Rs; unconfirmed.
Dock	Unresolved The lot does not adjoin mapped water, so a private dock is unlikely; shared access may exist.
Short-term rental	Unresolved Governed by local ordinance and CC&Rs; unconfirmed.
Pool / outdoor recreation	Conditional Space is available; flood-zone and soil-movement considerations affect pool design where applicable.

M60 Key measurements summary

Parameter	Finding
Lot size	1.249 ac (54,399 sq ft)
Lot dimensions (long axis × width)	544 ft × 201 ft
Lowest elevation	23.20 ft
Highest elevation	26.23 ft
Elevation difference	3.04 ft

... continued

Parameter	Finding
Average slope	1.49% (0.85°)
Maximum slope	12.46% (7.10°)
Steepest 50 ft run	3.4% (1.9°)
FEMA zone	AE
BFE	27.0 ft (indicative)
Minimum finished-floor elevation	28.0 ft at +1 ft / 29.0 ft at +2 ft freeboard COUNTY FIGURE UNCONFIRMED
Fill or elevation required?	Yes — lowest ground is 3.8 ft below BFE
Sewer	NEEDS CONFIRMATION
Water	NEEDS CONFIRMATION
Electricity	NEEDS CONFIRMATION
Gas	NEEDS CONFIRMATION
Internet	NEEDS CONFIRMATION
Septic required?	NEEDS CONFIRMATION — soil absorption rated Very limited
Soil	Asa — Well drained, hydrologic group B, Moderate shrink-swell
HOA	Platted subdivision (Bar X Ranch), so covenants are likely. NEEDS CONFIRMATION
HOA fee	NEEDS CONFIRMATION
Zoning	NEEDS CONFIRMATION
Dock possible?	NEEDS CONFIRMATION
Lake depth	NEEDS CONFIRMATION
Boating	NEEDS CONFIRMATION
Fishing	NEEDS CONFIRMATION
Hobby farming	NEEDS CONFIRMATION
Second in-law suite	NEEDS CONFIRMATION
Short-term rental	NEEDS CONFIRMATION
Nearest hospital	NEEDS CONFIRMATION
Nearest international airport	NEEDS CONFIRMATION
Distance to nearest mapped water	3,370 ft to unnamed waterbody
Mean annual temperature	21.7 °C
Mean annual rainfall	1,189 mm (46.8 in)
Appraised market value	$34,190

543 Pony Trl, Angleton, TX 77515

Lot 2 of 12 | ID confidence: high | Bar X Ranch | Brazoria County

A1 Property identification & record data OFFICIAL

Situs address of record	543 PONY TRAIL , , TX 77515
Legal description	BAR X RANCH SEC 11 (A0053 & A0097) BLK 3 LOT 23 ACRES 0.91
Subdivision	Bar X Ranch
County	Brazoria County (FIPS 48039)
Parcel ID / account	185695
Geographic ID (APN)	1543-0023-000
Owner of record	KOVACS MICHAEL A
Acreage — appraisal roll (legal)	0.91 ac
Acreage — GIS polygon	0.843 ac
Acreage — recomputed from boundary	0.846 ac
Appraised land value	$38,220
Appraised improvement value	$0
Appraised market value	$38,220
Centroid (WGS84)	29.114749, -95.568849
Longest axis / width	463 ft × 146 ft (axis bearing 223°)

Source: Texas StratMap statewide land parcels (county appraisal roll). Matched by situs address match. Appraisal-roll acreage and a GIS polygon rarely agree exactly; neither replaces a survey.

A2 · Listing and plat material not retrievable here NEEDS CONFIRMATION

Current asking price, days on market and listing history.
Obtain from: Listing agent / MLS

Listing photographs: lot frontage, shoreline, road access and surroundings.
Obtain from: Listing agent (Zillow blocks automated retrieval, so no listing imagery is reproduced here)

Seller disclosure notice and any known defects.
Obtain from: Listing agent

Recorded plat and Esri/county GIS parcel exhibit for comparison with this report.
Obtain from: County Clerk plat records and the county GIS portal

B5 Terrain, contours and true-scale site plan MODEL-DERIVED

LOWEST POINT	HIGHEST POINT	RELIEF	MEAN ELEVATION
21.4 ft	25.3 ft	3.9 ft	24.0 ft



True-scale site plan. Parcel boundary from the county GIS polygon; contours at 1.0 ft intervals derived from the USGS 3DEP 1 m bare-earth DEM. A–B is the cross-section line. Horizontal and vertical map scales are equal, so shapes and distances are undistorted.

MEAN GRADE	MEDIAN GRADE	90TH PCT GRADE	MAX GRADE
2.60 %	1.48 %	6.09 %	19.68 %

Mean slope angle	1.49°	**90th pct angle**	3.49°	**Max angle**	11.13°	**Grid cells**	1,156

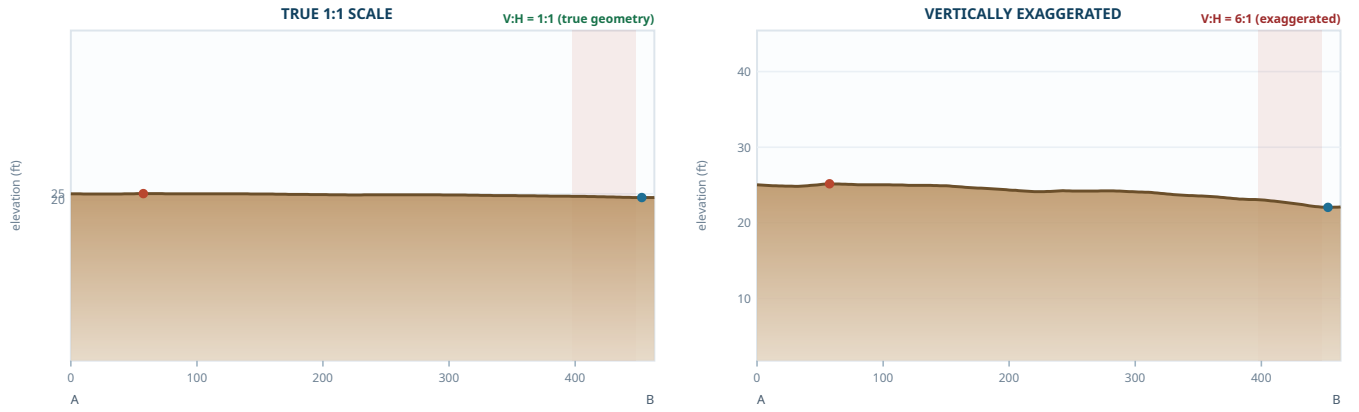
B7 · Terrain classification

Gentle slope. Standard residential grading is normally sufficient.

Basis: mean DEM grade 2.6% (90th pct 6.09%), relief 3.91 ft

Grading expectation: Minimal grading; Routine cut-and-fill

B6 Longitudinal cross-section MODEL-DERIVED

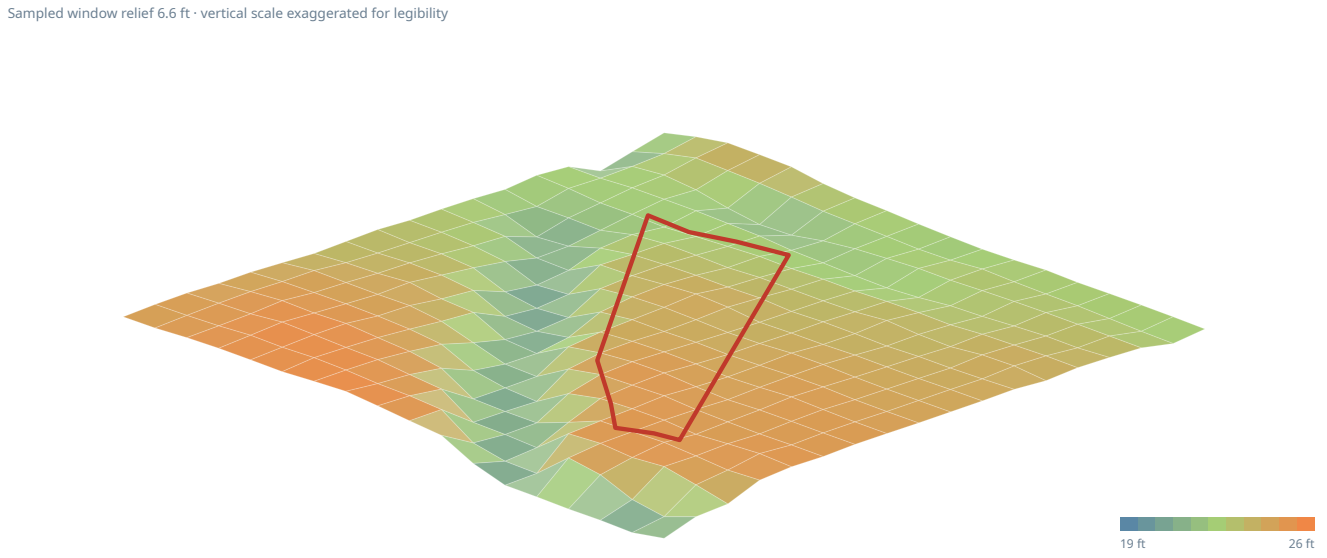


Run A–B	463 ft	**Lowest**	22.1 ft	**Highest**	25.2 ft
Elevation difference	3.1 ft	**Average grade**	0.7%	**Direct rising angle**	0.4°
Steepest 50 ft	**1.9%** (1.1°) over a 50 ft run				

Reading: Over 463 ft the ground rises only 3.1 ft, a direct angle of 0.4°. At true 1:1 scale the profile is almost a straight line, which is the correct reading: this lot is effectively flat. The exaggerated panel is the same data and should not be read as slope.

Left panel is the real geometry at 1:1 vertical-to-horizontal scale. The right panel exaggerates the vertical axis, which is how slope is conventionally drawn on maps but overstates steepness. The shaded band marks the steepest 50 ft run. Profile sampled from the 3DEP DEM along the lot's longest axis.

B8 Three-dimensional terrain MODEL-DERIVED



Isometric view of the sampled DEM window; the red outline is the parcel boundary draped on the surface. Vertical scale is exaggerated to reveal form — use the 1:1 cross-section above for true steepness.

C9 FEMA flood-zone determination OFFICIAL

Zone: AE | Inside Special Flood Hazard Area

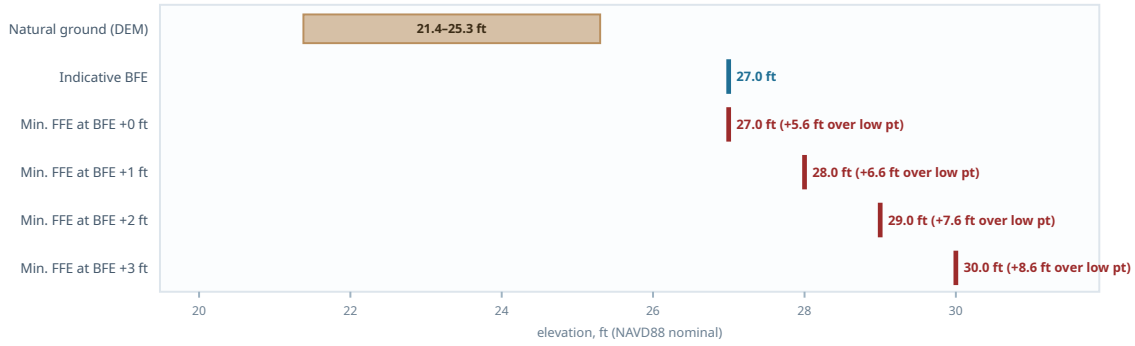
Zones intersecting the parcel	AE
Special Flood Hazard Area	Yes — mandatory flood insurance for a federally backed mortgage
FIRM panel	48039C0580K · effective 2020-12-30
Mapping status	—

Parcel intersects mapped FEMA flood-hazard polygon(s); see zones above.

C11-13 Ground elevation, BFE and minimum finished floor OFFICIAL MODEL-DERIVED

Lowest natural ground (DEM)	21.38 ft
Highest natural ground (DEM)	25.30 ft
Mean ground	24.02 ft
Indicative BFE	27.0 ft
Lowest ground relative to BFE	**-5.62 ft**
Highest ground relative to BFE	-1.70 ft

BFE basis: highest of 1 FEMA BFE line value(s) within 2,000 ft ([27.0]) - indicative only, the governing BFE must be read at the site from the FIRM/FIS profile



Freeboard scenarios. The brown band is the measured spread of natural ground across the lot. Red ticks are the minimum finished-floor elevation at BFE plus 0, 1, 2 and 3 ft of freeboard. **The freeboard the county actually enforces is not established in this report** — the scenarios show the consequence of each, they do not assert a requirement.

Scenario	Min. finished floor (ft)	Rise above lowest ground (ft)	Rise above highest ground (ft)
BFE + 0 ft	27.00	5.62	1.70
BFE + 1 ft	28.00	6.62	2.70
BFE + 2 ft	29.00	7.62	3.70
BFE + 3 ft	30.00	8.62	4.70

DEM ground across the lot runs 21.38-25.3 ft against an indicative BFE of 27.0 ft. The lot therefore sits 5.6 ft below BFE at its lowest point. Any habitable floor must be set above BFE plus the freeboard the county actually enforces - that freeboard figure is NOT established here.

Do not assume fill is permitted
Raising a pad with fill inside a mapped floodplain is frequently restricted or requires compensatory storage and a floodplain development permit. Elevating on piers or a stem wall is often the permitted route instead. This must be confirmed with the county floodplain administrator before any design assumption is made.

C · Flood items still to confirm NEEDS CONFIRMATION

Official flood determination and the effective FIRM panel read at the parcel.
Obtain from: County floodplain administrator or a flood certification service

... continued

Enforced freeboard above BFE, fill rules and compensatory-storage requirements.
Obtain from: County floodplain administrator

First Street / third-party forward-looking flood risk and estimated flood depth.
Obtain from: First Street Foundation (riskfactor.com) for the confirmed address

Flood insurance quote under current NFIP pricing.
Obtain from: Insurance agent, using an Elevation Certificate

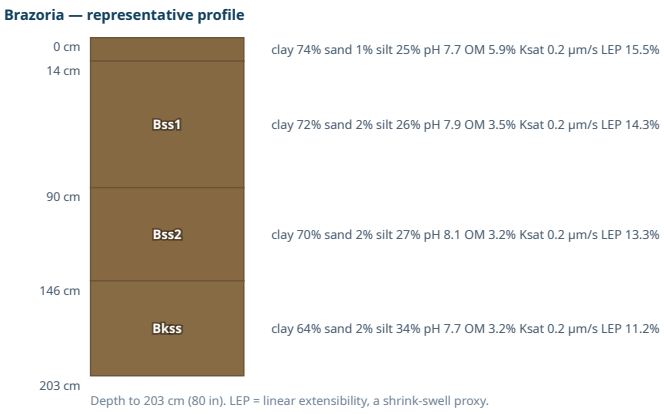
Site drainage: swales, culverts, outfall and upstream runoff from neighbouring lots.
Obtain from: Civil engineer site visit; county drainage criteria

D14 Soil characteristics MODEL-DERIVED

NRCS map unit intersecting the lot	Share of lot (sampled)
Brazoria clay, 1 to 3 percent slopes, rarely flooded	—
Asa silty clay loam, 0 to 1 percent slopes, rarely flooded	75.0%
Pledger clay, 0 to 1 percent slopes, rarely flooded	25.0%

Component	%	Drainage	Hyd.grp	LEP %	Shrink-swell	pH	Clay %	Flooding
Brazoria	80%	Moderately well drained	D	15.5	Very high	7.7	74	Rare
Asa	85%	Well drained	B	4.5	Moderate	6.8	37	Rare
Pledger	85%	Moderately well drained	D	19.0	Very high	7.0	70	Rare

LEP = linear extensibility (shrink-swell proxy). Hydrologic group D indicates the slowest infiltration. Values are NRCS representative values for the map unit, not measurements on this lot.



Representative horizon sequence for the dominant component (Asa).

D15 Foundation suitability MODEL-DERIVED

Dominant soil shrink-swell potential is **Very high** (peak LEP 15.5%). Severe movement potential. Deep piers or heavily engineered slab; geotechnical design required.

Slab-on-grade	Engineered design mandatory
Crawlspace	Deep piers required
Pier and beam	Preferred, deep piers
Basement	Not advised
Stepped footing	Only if sloped
NRCS rating, dwellings without basements	—
Geotechnical investigation advised	Yes

Foundation selection is an engineering decision. NRCS map-unit shrink-swell indicates risk; it does not substitute for a geotechnical investigation.

D16 Septic / on-site sewage suitability MODEL-DERIVED

Septic Tank Absorption Fields	Very limited
Septic Tank Leaching Chamber (TX)	Very limited
Septic Tank, Subsurface Drip Irrigation (TX)	Very limited

Constraints identified from the soil and terrain data

- Hydrologic group D indicates slow infiltration, which constrains conventional drainfields.
- Mapped flooding frequency 'Rare' is a direct constraint on on-site sewage facilities.

Permitted bedroom count: NOT ESTABLISHED
Permitted bedroom count follows from a site-specific soil evaluation (TCEQ Ch. 285 class, loading rate, usable drainfield area and setbacks). It cannot be derived from map-unit data. A licensed site evaluator must run the percolation/soil profile test.

NRCS ratings describe the map unit, not the lot. A TCEQ-compliant site evaluation and percolation test is required for any permit.

D17 Gardening suitability MODEL-DERIVED

SURFACE PH	CLAY	SAND	ORGANIC MATTER
7.7	74 %	1 %	5.9 %

Drainage class	Moderately well drained
pH range through profile	7.7 – 8.1
NRCS lawn / landscape rating	Very limited
Raised beds advised	Yes

Heavy clay / restricted drainage / high shrink-swell in the surface horizon makes raised beds with imported loam the lower-risk option.

Likely amendments

- Elemental sulphur or acidifying organic matter to lower pH
- Coarse organic matter and gypsum-style conditioning to improve tilth in heavy clay
- Mulch, to buffer soil temperature and moisture

Based on NRCS map-unit averages. A lab soil test is required before committing to any amendment programme.

D · Soil and planting items to confirm NEEDS CONFIRMATION

Laboratory soil test: pH, macro and micronutrients, organic matter, texture.
Obtain from: Texas A&M AgriLife Soil Testing Lab, from samples taken on the lot

Plant hardiness zone confirmation and a species list matched to the tested soil.
Obtain from: USDA hardiness map plus the county AgriLife extension office

E19-23 Utilities and infrastructure NEEDS CONFIRMATION

Utility availability is not published as open geospatial data at parcel level. Nothing is inferred here: each item below must come from the serving utility in writing, because a missing water tap or a long electric extension can change the build budget materially.

Water supply: municipal, water-supply-corporation, or private well; is a tap at the road?
Obtain from: Serving water supply corporation / county; well logs via TX Water Development Board

Sewer: is public sewer available, or is an on-site facility required? Distance to any main.
Obtain from: County / city utility department or MUD

Electricity: provider, whether service exists at the lot line, overhead vs underground, connection cost.
Obtain from: Retail electric provider / local co-op (service availability letter)

Natural gas or propane availability and existing infrastructure.
Obtain from: Gas utility or local propane suppliers

... continued

Internet: fibre, cable, fixed wireless, DSL or satellite, with advertised speeds.
Obtain from: FCC National Broadband Map plus a direct serviceability check by address

F24-25 Waterfront position MODEL-DERIVED

Interior lot - no adjacent mapped water

Nearest mapped waterbody	unnamed waterbody
Distance from parcel boundary	3,688 ft
Waterbody area	0.823 km²

NHD polygons are hydrographic mapping, not a boundary survey. Whether the lot carries riparian frontage, deeded access or only common-area access is a title/plat question and is NOT established here.

F26-29 · Dock, depth, boating and fishing NEEDS CONFIRMATION

Whether a dock is permittable at this shoreline, and the governing authority.
Obtain from: Controlling lake / river authority, plus the HOA

Dock size limits, shoreline setbacks, shared-dock rules and permit cost.
Obtain from: Lake authority rules and permit schedule

Whether riparian or shoreline rights pass with the lot, or the shoreline is a common reserve.
Obtain from: Title commitment and recorded plat

Lake water-level regime and any shoreline-management plan.
Obtain from: Lake/river authority or reservoir operator

Bathymetry: average and maximum depth, and depth near this shoreline.
Obtain from: Lake authority survey, TPWD reservoir report, or a sonar survey

Seasonal water-level variation and drawdown history.
Obtain from: Reservoir operator gauge records / USGS gauges

Boating rules: permitted craft, motor limits, wake rules, ramps and marinas.
Obtain from: Lake authority and TPWD

Fishery: species present, stocking, licence requirements and catch limits.
Obtain from: Texas Parks & Wildlife reservoir survey reports

G30-33 HOA, covenants, zoning and a second dwelling NEEDS CONFIRMATION

The parcel sits in **Bar X Ranch** according to the recorded legal description, so a declaration of covenants very likely exists. The documents themselves are not public geospatial data and were not retrieved, so no dues figure, restriction or amenity is asserted here.

Whether an HOA or POA exists and is mandatory, with its name and management contact.
Obtain from: Title commitment, recorded Declaration, seller disclosure

Zoning designation, or confirmation that the county is unzoned.
Obtain from: County or city development services

Minimum lot size, building coverage, setbacks, height limits, accessory-structure rules.
Obtain from: County/city development standards or subdivision regulations

Whether a second independent dwelling or in-law suite (ADU) is permitted, and on what terms.
Obtain from: County/city development services plus the HOA

Permitting path, driveway/culvert permit and any development fee.
Obtain from: County permitting office

... continued

Annual dues, monthly equivalent, transfer fee, and any special assessment.
Obtain from: HOA or management company estoppel / resale certificate

Declaration, CC&Rs, architectural guidelines, rules and all amendments.
Obtain from: HOA or title company

Restrictions on docks, fences, outbuildings, RVs, boats, livestock and poultry.
Obtain from: Recorded CC&Rs and architectural guidelines

Minimum dwelling size and any construction time limit after purchase.
Obtain from: Recorded CC&Rs

Amenities actually included: clubhouse, pool, courts, parks, lake and boat facilities.
Obtain from: HOA documents and budget

H34 / I35 Hobby farming and short-term rental NEEDS CONFIRMATION

Whether chickens, ducks, rabbits, goats or bees are allowed, and in what numbers.
Obtain from: County rules plus the recorded CC&Rs (either can prohibit)

Any agricultural or wildlife valuation currently on the parcel, and rollback-tax exposure if use changes.
Obtain from: County appraisal district

Whether short-term rental is permitted at this location.
Obtain from: County/city ordinance and the HOA

Any STR permit, occupancy cap, parking or noise condition.
Obtain from: Local ordinance

Hotel occupancy tax registration and remittance obligations.
Obtain from: Texas Comptroller plus local taxing authority

J36-42 Environmental and geological due diligence MODEL-DERIVED

Ground-penetrating radar penetration (NRCS rating)	Very low penetration Indicates how useful GPR would be here; clay-rich soils attenuate the signal.
Sinkhole / karst exposure	Not indicated by the soil map units returned for this parcel. No restrictive bedrock or karst feature was reported. CONFIRM WITH TX BUREAU OF ECONOMIC GEOLOGY
Radon	EPA publishes radon zones by county, not by parcel. LOOK UP THE COUNTY ZONE AND TEST ANY BUILT STRUCTURE

Existing FEMA Elevation Certificate for the lot.
Obtain from: Seller, county floodplain office or a surveyor

Recent boundary, ALTA or topographic survey.
Obtain from: Seller or a licensed Texas surveyor (RPLS)

Geotechnical / soil boring report.
Obtain from: Seller or a geotechnical engineer

On-site sewage facility site evaluation and percolation test.
Obtain from: Licensed site evaluator / county OSSF office

Any ground-penetrating-radar or other subsurface investigation.
Obtain from: Seller; commission if subsurface features are suspected

K43-45 Residential construction feasibility MODEL-DERIVED

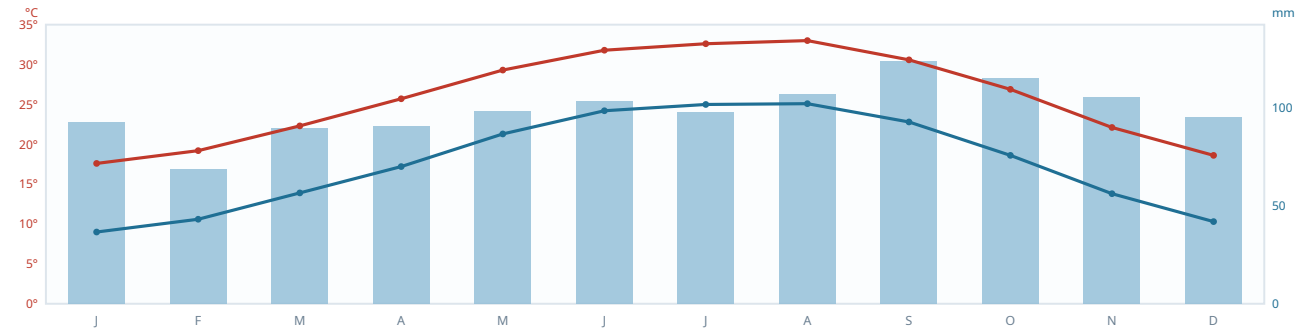
Lot area available	0.843 ac (36,704 sq ft)
Usable building envelope	NOT COMPUTABLE Setbacks, easements and any lake or drainage reserve come from the recorded plat and the HOA architectural guidelines, none of which were available.
Terrain constraint	Gentle slope — Minimal grading; Routine cut-and-fill
Foundation implication	Very high shrink-swell. Severe movement potential. Deep piers or heavily engineered slab; geotechnical design required.
Flood implication	Inside a mapped SFHA, so finished-floor elevation, flood venting and insurance all bind the design.

Construction cost
No build cost is estimated in this report. A defensible figure for a four-bedroom house depends on the HOA's minimum size and material rules, the finished-floor elevation forced by the flood zone, the foundation type the soil requires, and whether water and sewer must be created on site. Three of those four inputs are still unconfirmed for this lot, so any number quoted now would be arbitrary. Price it from a builder bid once the plat, CC&Rs, floodplain requirement and soil report are in hand.

L46 Climate MODEL-DERIVED

MEAN ANNUAL TEMP	MEAN SUMMER HIGH	MEAN WINTER LOW	ANNUAL RAINFALL
21.8 °C	32.5 °C	9.9 °C	1,189 mm

Record high	41.8 °C	**Record low**	-9.4 °C	**Rainfall**	46.8 in
Growing season	~342 days	**Days at or below 0 °C**	2.6 /yr	**Period**	1994-2023 (30 years)



Monthly normals, 1994-2023 (30 years). Bars are mean monthly rainfall (mm, right axis); lines are mean daily maximum and minimum temperature (°C, left axis). Source: ERA5 daily reanalysis via Open-Meteo archive API.

L47 Natural hazards

Flood	Mapped Special Flood Hazard Area
Other perils	Hurricane and tropical wind, tornado, hail, severe convective storm, lightning, extreme heat, drought and wildfire all vary by location within Texas. QUANTIFY FROM FEMA NATIONAL RISK INDEX FOR THE CONFIRMED COUNTY AND FROM AN INSURANCE QUOTE

L48-55 Community, schools, healthcare, shopping and access NEEDS CONFIRMATION

Demographics not retrieved
Census API requires an API key (keyless request rejected) Free key from https://api.census.gov/data/key_signup.html, then set CENSUS_API_KEY and re-run 05_context.py. Or read the figures directly from data.census.gov.

Driving distance and time to the nearest hospital, emergency department and urgent care.
Obtain from: Direct routing check from the confirmed parcel location

Distance to grocery, major retail, pharmacy, bank, fuel and hardware.
Obtain from: Direct routing check

Assigned school district and campuses, with distances and current ratings.
Obtain from: TEA school-district locator and the district attendance-zone map

Nearest Indian / South Asian grocery, restaurants and places of worship.
Obtain from: Direct search from the confirmed parcel location

Nearest international airport, driving time and route.
Obtain from: Direct routing check

Walk / Bike / Transit scores.
Obtain from: Walk Score for the confirmed address

M56 Documented strengths

- Gentle mean grade of 2.60%, so routine grading should suffice.
- Long growing season of roughly 342 frost-free days supports year-round production.

M57 Limitations and risks

- **Inside a FEMA Special Flood Hazard Area (AE).** This forces an elevated finished floor, triggers floodplain permitting, and makes flood insurance mandatory with a federally backed mortgage.
- Lowest natural ground sits about 5.6 ft below the indicative BFE, so the pad must be raised or the house elevated.
- **Very high soil shrink-swell.** Expansive clay is the leading cause of foundation distress in this region; an engineered foundation and a geotechnical report should be assumed.
- Septic absorption is rated "very limited" for the dominant soil, so a conventional drainfield may not be permittable and an engineered or aerobic system may be required at higher cost.
- Mapped flooding frequency 'Rare' is a direct constraint on on-site sewage facilities.
- HOA covenants, zoning, utility availability, dock rights and short-term-rental rules are all unconfirmed. Any of them can independently prevent the intended use.

M59 Construction feasibility conclusion

Main house	Likely The lot is large enough and the terrain is workable. Finished-floor elevation and foundation type are the governing variables, not land area.
Second in-law suite / ADU	Unresolved Depends entirely on county rules and the HOA declaration; neither was available. Do not assume a second dwelling is permitted.
Garage	Likely No physical constraint identified, subject to setbacks and architectural guidelines.
Septic system	Unresolved Soil ratings are unfavourable; permittability and bedroom count require a site evaluation.
Garden	Conditional Workable with raised beds and imported loam; native surface soil is restrictive.
Hobby farming / poultry	Unresolved Governed by county rules and CC&Rs; unconfirmed.
Dock	Unresolved The lot does not adjoin mapped water, so a private dock is unlikely; shared access may exist.
Short-term rental	Unresolved Governed by local ordinance and CC&Rs; unconfirmed.
Pool / outdoor recreation	Conditional Space is available; flood-zone and soil-movement considerations affect pool design where applicable.

M60 Key measurements summary

Parameter	Finding
Lot size	0.843 ac (36,704 sq ft)
Lot dimensions (long axis × width)	463 ft × 146 ft
Lowest elevation	21.38 ft
Highest elevation	25.30 ft
Elevation difference	3.91 ft
Average slope	2.60% (1.49°)
Maximum slope	19.68% (11.13°)
Steepest 50 ft run	1.9% (1.1°)
FEMA zone	AE
BFE	27.0 ft (indicative)
Minimum finished-floor elevation	28.0 ft at +1 ft / 29.0 ft at +2 ft freeboard COUNTY FIGURE UNCONFIRMED
Fill or elevation required?	Yes — lowest ground is 5.6 ft below BFE
Sewer	NEEDS CONFIRMATION
Water	NEEDS CONFIRMATION
Electricity	NEEDS CONFIRMATION
Gas	NEEDS CONFIRMATION

... continued

Parameter	Finding
Internet	NEEDS CONFIRMATION
Septic required?	NEEDS CONFIRMATION — soil absorption rated Very limited
Soil	Asa — Moderately well drained, hydrologic group D, Very high shrink-swell
HOA	Platted subdivision (Bar X Ranch), so covenants are likely. NEEDS CONFIRMATION
HOA fee	NEEDS CONFIRMATION
Zoning	NEEDS CONFIRMATION
Dock possible?	NEEDS CONFIRMATION
Lake depth	NEEDS CONFIRMATION
Boating	NEEDS CONFIRMATION
Fishing	NEEDS CONFIRMATION
Hobby farming	NEEDS CONFIRMATION
Second in-law suite	NEEDS CONFIRMATION
Short-term rental	NEEDS CONFIRMATION
Nearest hospital	NEEDS CONFIRMATION
Nearest international airport	NEEDS CONFIRMATION
Distance to nearest mapped water	3,688 ft to unnamed waterbody
Mean annual temperature	21.8 °C
Mean annual rainfall	1,189 mm (46.8 in)
Appraised market value	$38,220

274 Pony Trl, Angleton, TX 77515

Lot 3 of 12 | ID confidence: high | Bar X Ranch | Brazoria County

A1 Property identification & record data OFFICIAL

Situs address of record	274 PONY TRAIL , , TX 77515
Legal description	BAR X RANCH SEC 10 (A0053 S CARTER) BLK 1 LOT 19 ACRES 0.84
Subdivision	Bar X Ranch
County	Brazoria County (FIPS 48039)
Parcel ID / account	185556
Geographic ID (APN)	1542-0019-001
Owner of record	CASSKEN LLC
Acreage — appraisal roll (legal)	0.84 ac
Acreage — GIS polygon	0.863 ac
Acreage — recomputed from boundary	0.867 ac
Appraised land value	$35,280
Appraised improvement value	$0
Appraised market value	$35,280
Centroid (WGS84)	29.123251, -95.575519
Longest axis / width	377 ft × 201 ft (axis bearing 69°)

Source: Texas StratMap statewide land parcels (county appraisal roll). Matched by situs address match. Appraisal-roll acreage and a GIS polygon rarely agree exactly; neither replaces a survey.

A2 · Listing and plat material not retrievable here NEEDS CONFIRMATION

Current asking price, days on market and listing history.
Obtain from: Listing agent / MLS

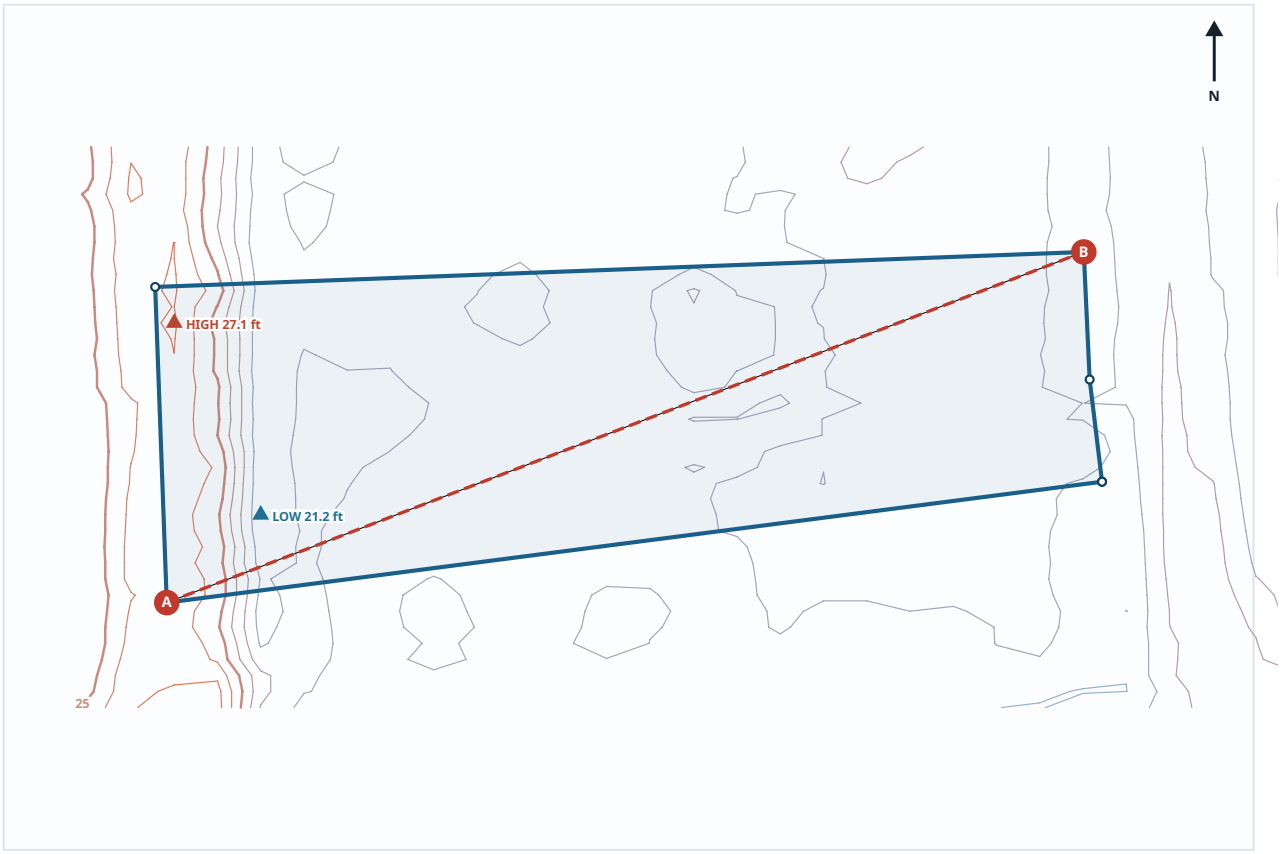
Listing photographs: lot frontage, shoreline, road access and surroundings.
Obtain from: Listing agent (Zillow blocks automated retrieval, so no listing imagery is reproduced here)

Seller disclosure notice and any known defects.
Obtain from: Listing agent

Recorded plat and Esri/county GIS parcel exhibit for comparison with this report.
Obtain from: County Clerk plat records and the county GIS portal

B5 Terrain, contours and true-scale site plan MODEL-DERIVED

LOWEST POINT	HIGHEST POINT	RELIEF	MEAN ELEVATION
21.2 ft	27.1 ft	6.0 ft	22.4 ft



True-scale site plan. Parcel boundary from the county GIS polygon; contours at 1.0 ft intervals derived from the USGS 3DEP 1 m bare-earth DEM. A–B is the cross-section line. Horizontal and vertical map scales are equal, so shapes and distances are undistorted.

MEAN GRADE	MEDIAN GRADE	90TH PCT GRADE	MAX GRADE
2.72 %	1.12 %	7.69 %	20.96 %

Mean slope angle	1.56°	**90th pct angle**	4.40°	**Max angle**	11.84°	**Grid cells**	1,156

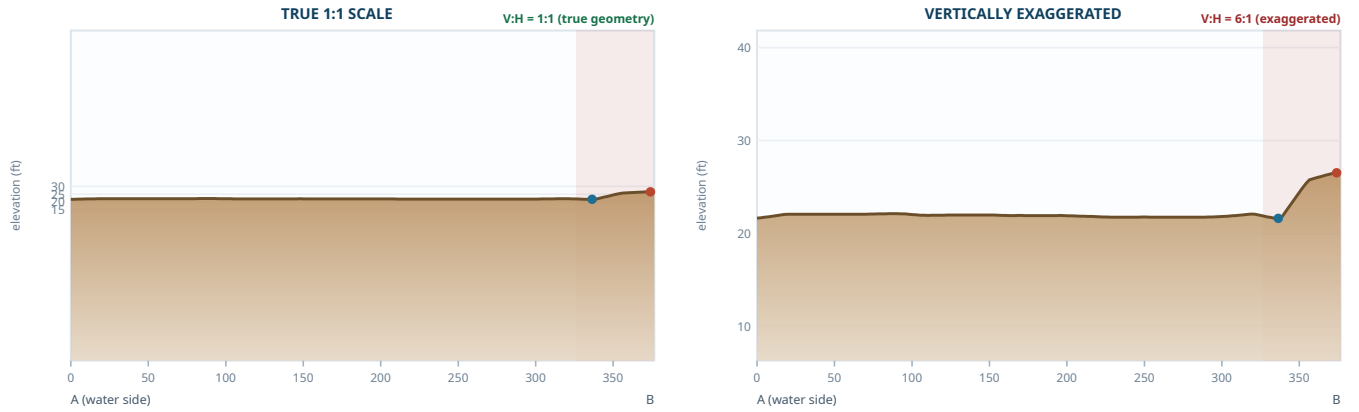
B7 · Terrain classification

Gentle slope. Standard residential grading is normally sufficient.

Basis: mean DEM grade 2.72% (90th pct 7.69%), relief 5.99 ft

Grading expectation: Minimal grading; Routine cut-and-fill

B6 Longitudinal cross-section MODEL-DERIVED

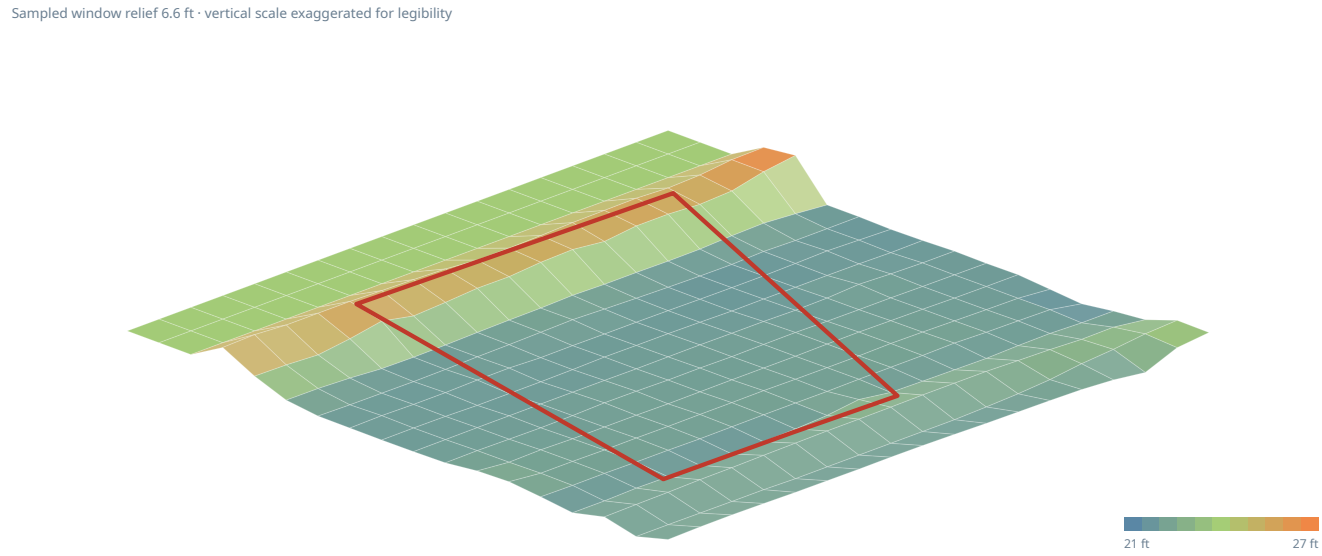


Run A–B	377 ft	**Lowest**	21.6 ft	**Highest**	26.5 ft
Elevation difference	4.9 ft	**Average grade**	1.3%	**Direct rising angle**	0.7°
Steepest 50 ft	**9.1%** (5.2°) over a 50 ft run				

Reading: Over 377 ft the ground rises only 4.9 ft, a direct angle of 0.7°. At true 1:1 scale the profile is almost a straight line, which is the correct reading: this lot is effectively flat. The exaggerated panel is the same data and should not be read as slope.

Left panel is the real geometry at 1:1 vertical-to-horizontal scale. The right panel exaggerates the vertical axis, which is how slope is conventionally drawn on maps but overstates steepness. The shaded band marks the steepest 50 ft run. Profile sampled from the 3DEP DEM along the lot's longest axis.

B8 Three-dimensional terrain MODEL-DERIVED



Isometric view of the sampled DEM window; the red outline is the parcel boundary draped on the surface. Vertical scale is exaggerated to reveal form — use the 1:1 cross-section above for true steepness.

C9 FEMA flood-zone determination OFFICIAL

Zone: AE | Inside Special Flood Hazard Area

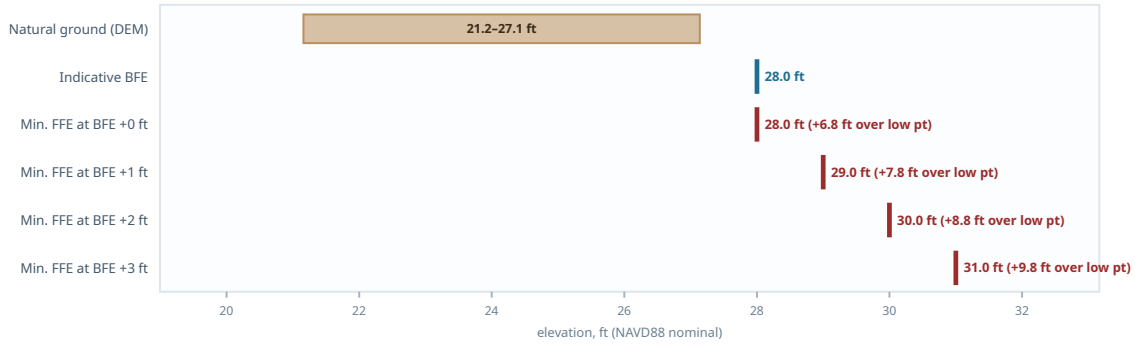
Zones intersecting the parcel	AE
Special Flood Hazard Area	Yes — mandatory flood insurance for a federally backed mortgage
FIRM panel	48039C0580K · effective 2020-12-30
Mapping status	—

Parcel intersects mapped FEMA flood-hazard polygon(s); see zones above.

C11-13 Ground elevation, BFE and minimum finished floor OFFICIAL MODEL-DERIVED

Lowest natural ground (DEM)	21.16 ft
Highest natural ground (DEM)	27.14 ft
Mean ground	22.38 ft
Indicative BFE	28.0 ft
Lowest ground relative to BFE	**-6.84 ft**
Highest ground relative to BFE	-0.86 ft

BFE basis: highest of 1 FEMA BFE line value(s) within 2,000 ft ([28.0]) - indicative only, the governing BFE must be read at the site from the FIRM/FIS profile



Freeboard scenarios. The brown band is the measured spread of natural ground across the lot. Red ticks are the minimum finished-floor elevation at BFE plus 0, 1, 2 and 3 ft of freeboard. **The freeboard the county actually enforces is not established in this report** — the scenarios show the consequence of each, they do not assert a requirement.

Scenario	Min. finished floor (ft)	Rise above lowest ground (ft)	Rise above highest ground (ft)
BFE + 0 ft	28.00	6.84	0.86
BFE + 1 ft	29.00	7.84	1.86
BFE + 2 ft	30.00	8.84	2.86
BFE + 3 ft	31.00	9.84	3.86

DEM ground across the lot runs 21.16-27.14 ft against an indicative BFE of 28.0 ft. The lot therefore sits 6.8 ft below BFE at its lowest point. Any habitable floor must be set above BFE plus the freeboard the county actually enforces - that freeboard figure is NOT established here.

Do not assume fill is permitted
Raising a pad with fill inside a mapped floodplain is frequently restricted or requires compensatory storage and a floodplain development permit. Elevating on piers or a stem wall is often the permitted route instead. This must be confirmed with the county floodplain administrator before any design assumption is made.

C · Flood items still to confirm NEEDS CONFIRMATION

Official flood determination and the effective FIRM panel read at the parcel.
Obtain from: County floodplain administrator or a flood certification service

... continued

Enforced freeboard above BFE, fill rules and compensatory-storage requirements.
Obtain from: County floodplain administrator

First Street / third-party forward-looking flood risk and estimated flood depth.
Obtain from: First Street Foundation (riskfactor.com) for the confirmed address

Flood insurance quote under current NFIP pricing.
Obtain from: Insurance agent, using an Elevation Certificate

Site drainage: swales, culverts, outfall and upstream runoff from neighbouring lots.
Obtain from: Civil engineer site visit; county drainage criteria

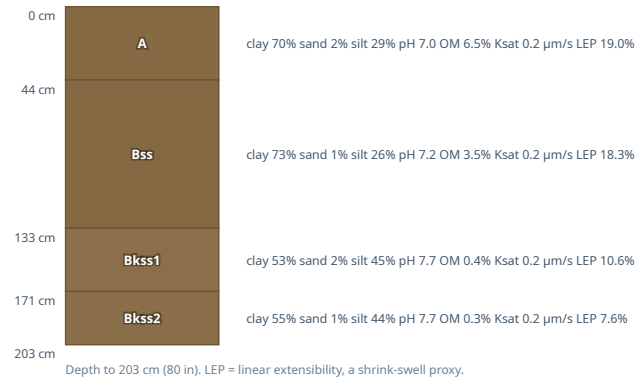
D14 Soil characteristics MODEL-DERIVED

NRCS map unit intersecting the lot	Share of lot (sampled)
Pledger clay, 0 to 1 percent slopes, rarely flooded	100.0%
Water, fresh	—

Component	%	Drainage	Hyd.grp	LEP %	Shrink-swell	pH	Clay %	Flooding
Pledger	85%	Moderately well drained	D	19.0	Very high	7.0	70	Rare
Water	100%	—	D	—	—	—	—	—

LEP = linear extensibility (shrink-swell proxy). Hydrologic group D indicates the slowest infiltration. Values are NRCS representative values for the map unit, not measurements on this lot.

Pledger — representative profile



Representative horizon sequence for the dominant component (Pledger).

D15 Foundation suitability MODEL-DERIVED

Dominant soil shrink-swell potential is **Very high** (peak LEP 19.0%). Severe movement potential. Deep piers or heavily engineered slab; geotechnical design required.

Slab-on-grade	Engineered design mandatory
Crawlspace	Deep piers required
Pier and beam	Preferred, deep piers
Basement	Not advised
Stepped footing	Only if sloped
NRCS rating, dwellings without basements	—
Geotechnical investigation advised	Yes

Foundation selection is an engineering decision. NRCS map-unit shrink-swell indicates risk; it does not substitute for a geotechnical investigation.

D16 Septic / on-site sewage suitability MODEL-DERIVED

Septic Tank Absorption Fields	Very limited
Septic Tank Leaching Chamber (TX)	Very limited
Septic Tank, Subsurface Drip Irrigation (TX)	Very limited

Constraints identified from the soil and terrain data

- Hydrologic group D indicates slow infiltration, which constrains conventional drainfields.
- Mapped flooding frequency 'Rare' is a direct constraint on on-site sewage facilities.
- Mapped water about 23.6 ft away; TCEQ/county setbacks from water will restrict drainfield placement.

Permitted bedroom count: NOT ESTABLISHED
Permitted bedroom count follows from a site-specific soil evaluation (TCEQ Ch. 285 class, loading rate, usable drainfield area and setbacks). It cannot be derived from map-unit data. A licensed site evaluator must run the percolation/soil profile test.

NRCS ratings describe the map unit, not the lot. A TCEQ-compliant site evaluation and percolation test is required for any permit.

D17 Gardening suitability MODEL-DERIVED

SURFACE PH	CLAY	SAND	ORGANIC MATTER
7.0	70 %	2 %	6.5 %

Drainage class	Moderately well drained
pH range through profile	7.0 – 7.7
NRCS lawn / landscape rating	Very limited
Raised beds advised	Yes

Heavy clay / restricted drainage / high shrink-swell in the surface horizon makes raised beds with imported loam the lower-risk option.

Likely amendments

- Coarse organic matter and gypsum-style conditioning to improve tilth in heavy clay
- Mulch, to buffer soil temperature and moisture

Based on NRCS map-unit averages. A lab soil test is required before committing to any amendment programme.

D · Soil and planting items to confirm NEEDS CONFIRMATION

Laboratory soil test: pH, macro and micronutrients, organic matter, texture.
Obtain from: Texas A&M AgriLife Soil Testing Lab, from samples taken on the lot

Plant hardiness zone confirmation and a species list matched to the tested soil.
Obtain from: USDA hardiness map plus the county AgriLife extension office

E19-23 Utilities and infrastructure NEEDS CONFIRMATION

Utility availability is not published as open geospatial data at parcel level. Nothing is inferred here: each item below must come from the serving utility in writing, because a missing water tap or a long electric extension can change the build budget materially.

Water supply: municipal, water-supply-corporation, or private well; is a tap at the road?
Obtain from: Serving water supply corporation / county; well logs via TX Water Development Board

Sewer: is public sewer available, or is an on-site facility required? Distance to any main.
Obtain from: County / city utility department or MUD

Electricity: provider, whether service exists at the lot line, overhead vs underground, connection cost.
Obtain from: Retail electric provider / local co-op (service availability letter)

Natural gas or propane availability and existing infrastructure.
Obtain from: Gas utility or local propane suppliers

... continued

Internet: fibre, cable, fixed wireless, DSL or satellite, with advertised speeds.
Obtain from: FCC National Broadband Map plus a direct serviceability check by address

F24-25 Waterfront position MODEL-DERIVED

Abuts mapped water - consistent with direct water frontage

Nearest mapped waterbody	unnamed waterbody
Distance from parcel boundary	24 ft
Waterbody area	0.823 km²

NHD polygons are hydrographic mapping, not a boundary survey. Whether the lot carries riparian frontage, deeded access or only common-area access is a title/plat question and is NOT established here.

F26-29 · Dock, depth, boating and fishing NEEDS CONFIRMATION

Whether a dock is permittable at this shoreline, and the governing authority.
Obtain from: Controlling lake / river authority, plus the HOA

Dock size limits, shoreline setbacks, shared-dock rules and permit cost.
Obtain from: Lake authority rules and permit schedule

Whether riparian or shoreline rights pass with the lot, or the shoreline is a common reserve.
Obtain from: Title commitment and recorded plat

Lake water-level regime and any shoreline-management plan.
Obtain from: Lake/river authority or reservoir operator

Bathymetry: average and maximum depth, and depth near this shoreline.
Obtain from: Lake authority survey, TPWD reservoir report, or a sonar survey

Seasonal water-level variation and drawdown history.
Obtain from: Reservoir operator gauge records / USGS gauges

Boating rules: permitted craft, motor limits, wake rules, ramps and marinas.
Obtain from: Lake authority and TPWD

Fishery: species present, stocking, licence requirements and catch limits.
Obtain from: Texas Parks & Wildlife reservoir survey reports

G30-33 HOA, covenants, zoning and a second dwelling NEEDS CONFIRMATION

The parcel sits in **Bar X Ranch** according to the recorded legal description, so a declaration of covenants very likely exists. The documents themselves are not public geospatial data and were not retrieved, so no dues figure, restriction or amenity is asserted here.

Whether an HOA or POA exists and is mandatory, with its name and management contact.
Obtain from: Title commitment, recorded Declaration, seller disclosure

Zoning designation, or confirmation that the county is unzoned.
Obtain from: County or city development services

Minimum lot size, building coverage, setbacks, height limits, accessory-structure rules.
Obtain from: County/city development standards or subdivision regulations

Whether a second independent dwelling or in-law suite (ADU) is permitted, and on what terms.
Obtain from: County/city development services plus the HOA

Permitting path, driveway/culvert permit and any development fee.
Obtain from: County permitting office

... continued

Annual dues, monthly equivalent, transfer fee, and any special assessment.
Obtain from: HOA or management company estoppel / resale certificate

Declaration, CC&Rs, architectural guidelines, rules and all amendments.
Obtain from: HOA or title company

Restrictions on docks, fences, outbuildings, RVs, boats, livestock and poultry.
Obtain from: Recorded CC&Rs and architectural guidelines

Minimum dwelling size and any construction time limit after purchase.
Obtain from: Recorded CC&Rs

Amenities actually included: clubhouse, pool, courts, parks, lake and boat facilities.
Obtain from: HOA documents and budget

H34 / I35 Hobby farming and short-term rental NEEDS CONFIRMATION

Whether chickens, ducks, rabbits, goats or bees are allowed, and in what numbers.
Obtain from: County rules plus the recorded CC&Rs (either can prohibit)

Any agricultural or wildlife valuation currently on the parcel, and rollback-tax exposure if use changes.
Obtain from: County appraisal district

Whether short-term rental is permitted at this location.
Obtain from: County/city ordinance and the HOA

Any STR permit, occupancy cap, parking or noise condition.
Obtain from: Local ordinance

Hotel occupancy tax registration and remittance obligations.
Obtain from: Texas Comptroller plus local taxing authority

J36-42 Environmental and geological due diligence MODEL-DERIVED

Ground-penetrating radar penetration (NRCS rating)	Very low penetration Indicates how useful GPR would be here; clay-rich soils attenuate the signal.
Sinkhole / karst exposure	Not indicated by the soil map units returned for this parcel. No restrictive bedrock or karst feature was reported. CONFIRM WITH TX BUREAU OF ECONOMIC GEOLOGY
Radon	EPA publishes radon zones by county, not by parcel. LOOK UP THE COUNTY ZONE AND TEST ANY BUILT STRUCTURE

Existing FEMA Elevation Certificate for the lot.
Obtain from: Seller, county floodplain office or a surveyor

Recent boundary, ALTA or topographic survey.
Obtain from: Seller or a licensed Texas surveyor (RPLS)

Geotechnical / soil boring report.
Obtain from: Seller or a geotechnical engineer

On-site sewage facility site evaluation and percolation test.
Obtain from: Licensed site evaluator / county OSSF office

Any ground-penetrating-radar or other subsurface investigation.
Obtain from: Seller; commission if subsurface features are suspected

K43-45 Residential construction feasibility MODEL-DERIVED

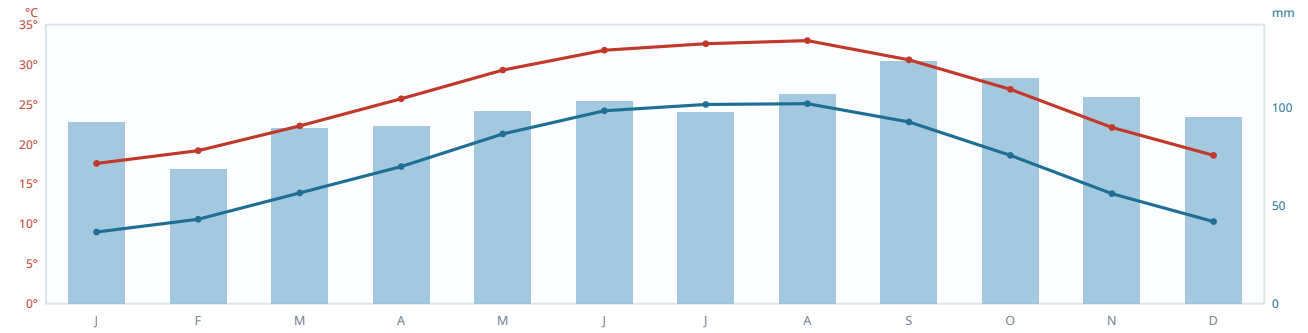
Lot area available	0.863 ac (37,596 sq ft)
Usable building envelope	NOT COMPUTABLE Setbacks, easements and any lake or drainage reserve come from the recorded plat and the HOA architectural guidelines, none of which were available.
Terrain constraint	Gentle slope — Minimal grading; Routine cut-and-fill
Foundation implication	Very high shrink-swell. Severe movement potential. Deep piers or heavily engineered slab; geotechnical design required.
Flood implication	Inside a mapped SFHA, so finished-floor elevation, flood venting and insurance all bind the design.

Construction cost
No build cost is estimated in this report. A defensible figure for a four-bedroom house depends on the HOA's minimum size and material rules, the finished-floor elevation forced by the flood zone, the foundation type the soil requires, and whether water and sewer must be created on site. Three of those four inputs are still unconfirmed for this lot, so any number quoted now would be arbitrary. Price it from a builder bid once the plat, CC&Rs, floodplain requirement and soil report are in hand.

L46 Climate MODEL-DERIVED

MEAN ANNUAL TEMP	MEAN SUMMER HIGH	MEAN WINTER LOW	ANNUAL RAINFALL
21.8 °C	32.5 °C	9.9 °C	1,189 mm

Record high	41.8 °C	**Record low**	-9.4 °C	**Rainfall**	46.8 in
Growing season	~342 days	**Days at or below 0 °C**	2.6 /yr	**Period**	1994-2023 (30 years)



Monthly normals, 1994-2023 (30 years). Bars are mean monthly rainfall (mm, right axis); lines are mean daily maximum and minimum temperature (°C, left axis). Source: ERA5 daily reanalysis via Open-Meteo archive API.

L47 Natural hazards

Flood	Mapped Special Flood Hazard Area
Other perils	Hurricane and tropical wind, tornado, hail, severe convective storm, lightning, extreme heat, drought and wildfire all vary by location within Texas. QUANTIFY FROM FEMA NATIONAL RISK INDEX FOR THE CONFIRMED COUNTY AND FROM AN INSURANCE QUOTE

L48-55 Community, schools, healthcare, shopping and access NEEDS CONFIRMATION

Demographics not retrieved
Census API requires an API key (keyless request rejected) Free key from https://api.census.gov/data/key_signup.html, then set CENSUS_API_KEY and re-run 05_context.py. Or read the figures directly from data.census.gov.

Driving distance and time to the nearest hospital, emergency department and urgent care.
Obtain from: Direct routing check from the confirmed parcel location

Distance to grocery, major retail, pharmacy, bank, fuel and hardware.
Obtain from: Direct routing check

Assigned school district and campuses, with distances and current ratings.
Obtain from: TEA school-district locator and the district attendance-zone map

Nearest Indian / South Asian grocery, restaurants and places of worship.
Obtain from: Direct search from the confirmed parcel location

Nearest international airport, driving time and route.
Obtain from: Direct routing check

Walk / Bike / Transit scores.
Obtain from: Walk Score for the confirmed address

M56 Documented strengths

- Abuts mapped water at roughly 24 ft, consistent with genuine water frontage subject to title confirmation.
- Gentle mean grade of 2.72%, so routine grading should suffice.
- Long growing season of roughly 342 frost-free days supports year-round production.

M57 Limitations and risks

- **Inside a FEMA Special Flood Hazard Area (AE).** This forces an elevated finished floor, triggers floodplain permitting, and makes flood insurance mandatory with a federally backed mortgage.
- Lowest natural ground sits about 6.8 ft below the indicative BFE, so the pad must be raised or the house elevated.
- **Very high soil shrink-swell.** Expansive clay is the leading cause of foundation distress in this region; an engineered foundation and a geotechnical report should be assumed.
- Septic absorption is rated "very limited" for the dominant soil, so a conventional drainfield may not be permittable and an engineered or aerobic system may be required at higher cost.
- Mapped flooding frequency 'Rare' is a direct constraint on on-site sewage facilities.
- HOA covenants, zoning, utility availability, dock rights and short-term-rental rules are all unconfirmed. Any of them can independently prevent the intended use.

M59 Construction feasibility conclusion

Main house	Likely The lot is large enough and the terrain is workable. Finished-floor elevation and foundation type are the governing variables, not land area.
Second in-law suite / ADU	Unresolved Depends entirely on county rules and the HOA declaration; neither was available. Do not assume a second dwelling is permitted.
Garage	Likely No physical constraint identified, subject to setbacks and architectural guidelines.
Septic system	Unresolved Soil ratings are unfavourable; permittability and bedroom count require a site evaluation.
Garden	Conditional Workable with raised beds and imported loam; native surface soil is restrictive.
Hobby farming / poultry	Unresolved Governed by county rules and CC&Rs; unconfirmed.
Dock	Unresolved The lot adjoins water, but dock rights are a title and permitting question that was not resolved.
Short-term rental	Unresolved Governed by local ordinance and CC&Rs; unconfirmed.
Pool / outdoor recreation	Conditional Space is available; flood-zone and soil-movement considerations affect pool design where applicable.

M60 Key measurements summary

Parameter	Finding
Lot size	0.863 ac (37,596 sq ft)
Lot dimensions (long axis × width)	377 ft × 201 ft
Lowest elevation	21.16 ft
Highest elevation	27.14 ft
Elevation difference	5.99 ft
Average slope	2.72% (1.56°)
Maximum slope	20.96% (11.84°)
Steepest 50 ft run	9.1% (5.2°)
FEMA zone	AE
BFE	28.0 ft (indicative)
Minimum finished-floor elevation	29.0 ft at +1 ft / 30.0 ft at +2 ft freeboard COUNTY FIGURE UNCONFIRMED
Fill or elevation required?	Yes — lowest ground is 6.8 ft below BFE
Sewer	NEEDS CONFIRMATION
Water	NEEDS CONFIRMATION
Electricity	NEEDS CONFIRMATION

... continued

Parameter	Finding
Gas	NEEDS CONFIRMATION
Internet	NEEDS CONFIRMATION
Septic required?	NEEDS CONFIRMATION — soil absorption rated Very limited
Soil	Pledger — Moderately well drained, hydrologic group D, Very high shrink-swell
HOA	Platted subdivision (Bar X Ranch), so covenants are likely. NEEDS CONFIRMATION
HOA fee	NEEDS CONFIRMATION
Zoning	NEEDS CONFIRMATION
Dock possible?	NEEDS CONFIRMATION
Lake depth	NEEDS CONFIRMATION
Boating	NEEDS CONFIRMATION
Fishing	NEEDS CONFIRMATION
Hobby farming	NEEDS CONFIRMATION
Second in-law suite	NEEDS CONFIRMATION
Short-term rental	NEEDS CONFIRMATION
Nearest hospital	NEEDS CONFIRMATION
Nearest international airport	NEEDS CONFIRMATION
Distance to nearest mapped water	24 ft to unnamed waterbody
Mean annual temperature	21.8 °C
Mean annual rainfall	1,189 mm (46.8 in)
Appraised market value	$35,280

220 Pony Trl, Angleton, TX 77515

Lot 4 of 12 | ID confidence: high | Bar X Ranch | Brazoria County

A1 Property identification & record data OFFICIAL

Situs address of record	220 PONY TRAIL , , TX 77515
Legal description	BAR X RANCH SEC 10 (A0053 S CARTER) BLK 1 LOT 28 ACRES 0.76
Subdivision	Bar X Ranch
County	Brazoria County (FIPS 48039)
Parcel ID / account	185565
Geographic ID (APN)	1542-0028-001
Owner of record	LOPEZ VICTOR
Acreage — appraisal roll (legal)	0.76 ac
Acreage — GIS polygon	0.763 ac
Acreage — recomputed from boundary	0.766 ac
Appraised land value	$31,920
Appraised improvement value	$0
Appraised market value	$31,920
Centroid (WGS84)	29.125582, -95.575536
Longest axis / width	368 ft × 181 ft (axis bearing 283°)

Source: Texas StratMap statewide land parcels (county appraisal roll). Matched by situs address match. Appraisal-roll acreage and a GIS polygon rarely agree exactly; neither replaces a survey.

A2 · Listing and plat material not retrievable here NEEDS CONFIRMATION

Current asking price, days on market and listing history.
Obtain from: Listing agent / MLS

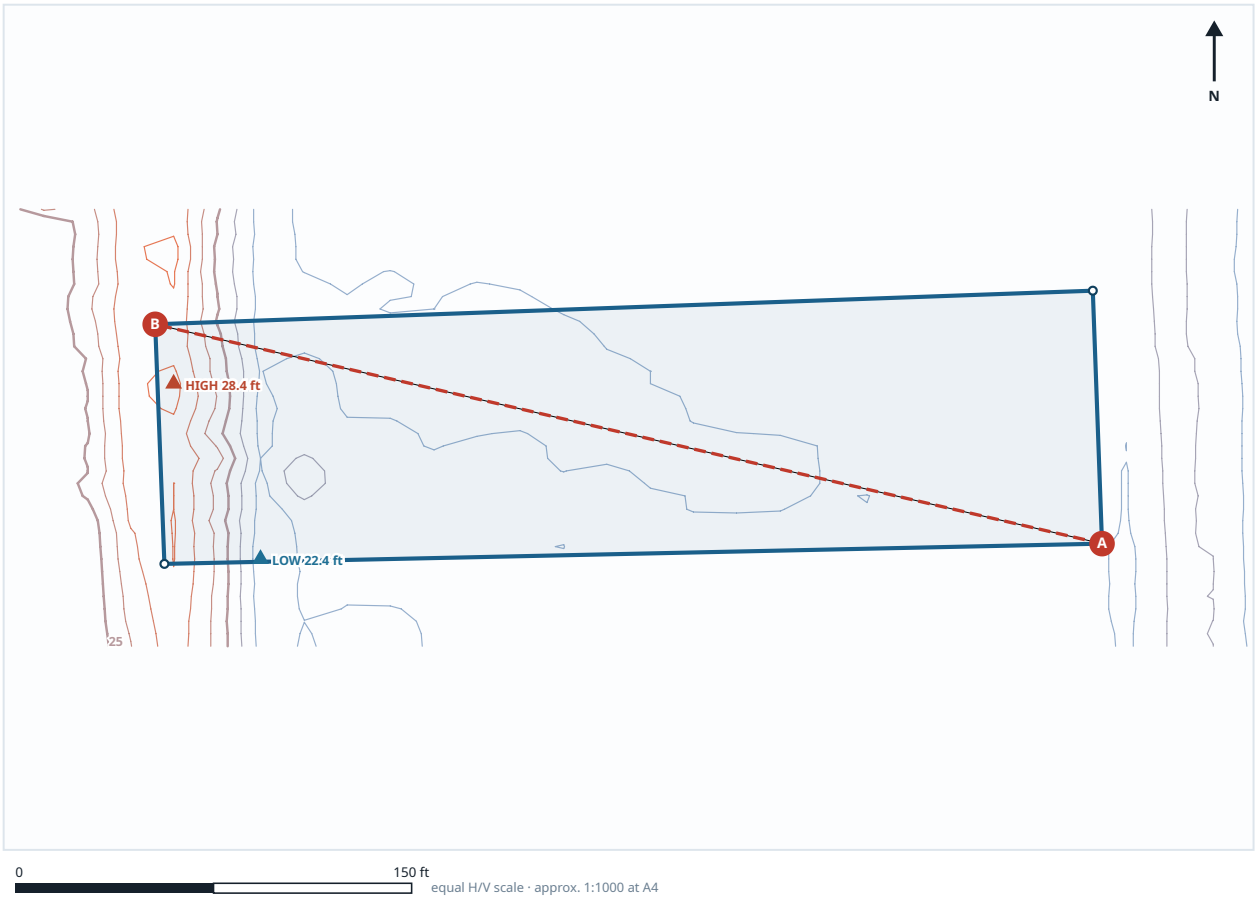
Listing photographs: lot frontage, shoreline, road access and surroundings.
Obtain from: Listing agent (Zillow blocks automated retrieval, so no listing imagery is reproduced here)

Seller disclosure notice and any known defects.
Obtain from: Listing agent

Recorded plat and Esri/county GIS parcel exhibit for comparison with this report.
Obtain from: County Clerk plat records and the county GIS portal

B5 Terrain, contours and true-scale site plan MODEL-DERIVED

LOWEST POINT	HIGHEST POINT	RELIEF	MEAN ELEVATION
22.4 ft	28.4 ft	6.0 ft	23.4 ft



True-scale site plan. Parcel boundary from the county GIS polygon; contours at 1.0 ft intervals derived from the USGS 3DEP 1 m bare-earth DEM. A–B is the cross-section line. Horizontal and vertical map scales are equal, so shapes and distances are undistorted.

MEAN GRADE	MEDIAN GRADE	90TH PCT GRADE	MAX GRADE
2.64 %	1.20 %	7.81 %	21.98 %

Mean slope angle	1.51°	**90th pct angle**	4.47°	**Max angle**	12.40°	**Grid cells**	1,156

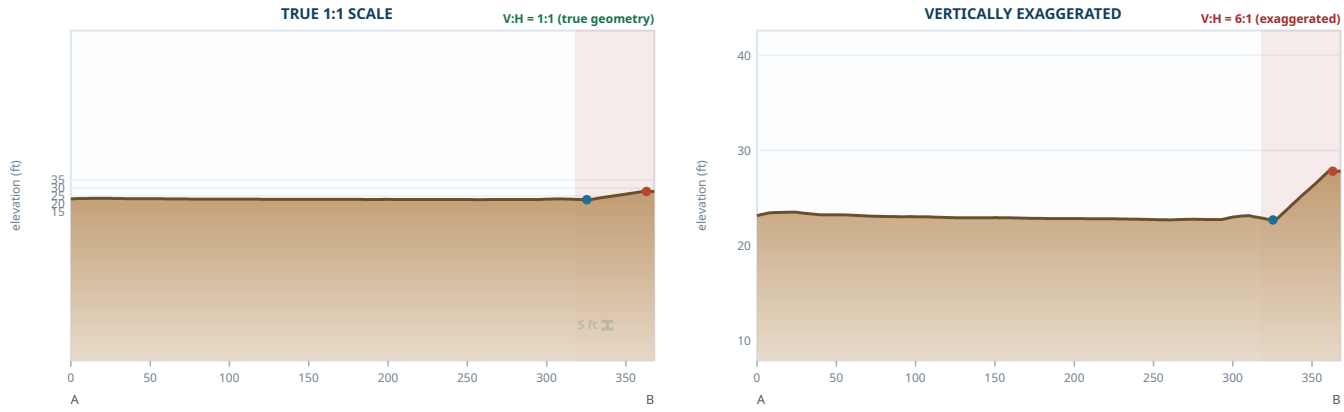
B7 · Terrain classification

Gentle slope. Standard residential grading is normally sufficient.

Basis: mean DEM grade 2.64% (90th pct 7.81%), relief 5.96 ft

Grading expectation: Minimal grading; Routine cut-and-fill

B6 Longitudinal cross-section MODEL-DERIVED



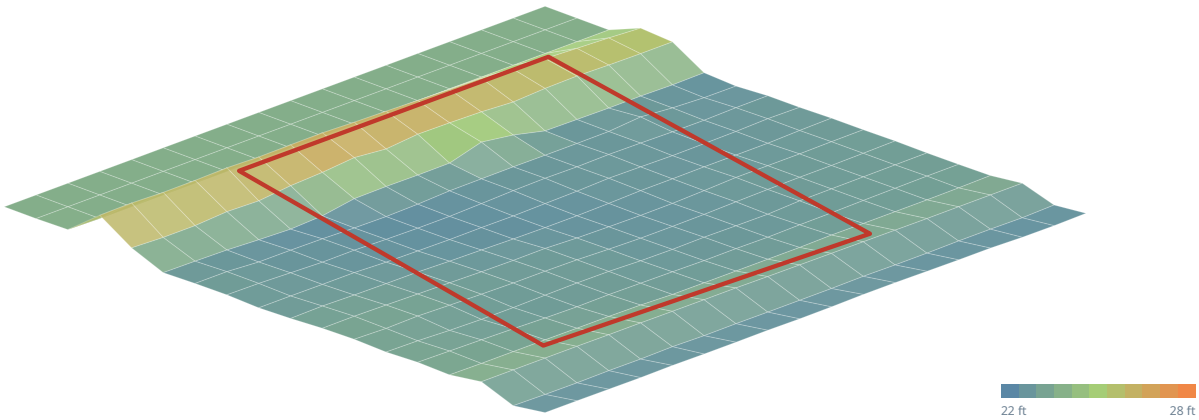
Run A–B	368 ft	**Lowest**	22.7 ft	**Highest**	27.8 ft
Elevation difference	5.1 ft	**Average grade**	1.4%	**Direct rising angle**	0.8°
Steepest 50 ft	**9.8%** (5.6°) over a 50 ft run				

Reading: Over 368 ft the ground rises only 5.1 ft, a direct angle of 0.8°. At true 1:1 scale the profile is almost a straight line, which is the correct reading: this lot is effectively flat. The exaggerated panel is the same data and should not be read as slope.

Left panel is the real geometry at 1:1 vertical-to-horizontal scale. The right panel exaggerates the vertical axis, which is how slope is conventionally drawn on maps but overstates steepness. The shaded band marks the steepest 50 ft run. Profile sampled from the 3DEP DEM along the lot's longest axis.

B8 Three-dimensional terrain MODEL-DERIVED

Sampled window relief 6.0 ft · vertical scale exaggerated for legibility



Isometric view of the sampled DEM window; the red outline is the parcel boundary draped on the surface. Vertical scale is exaggerated to reveal form — use the 1:1 cross-section above for true steepness.

C9 FEMA flood-zone determination OFFICIAL

Zone: AE | Inside Special Flood Hazard Area

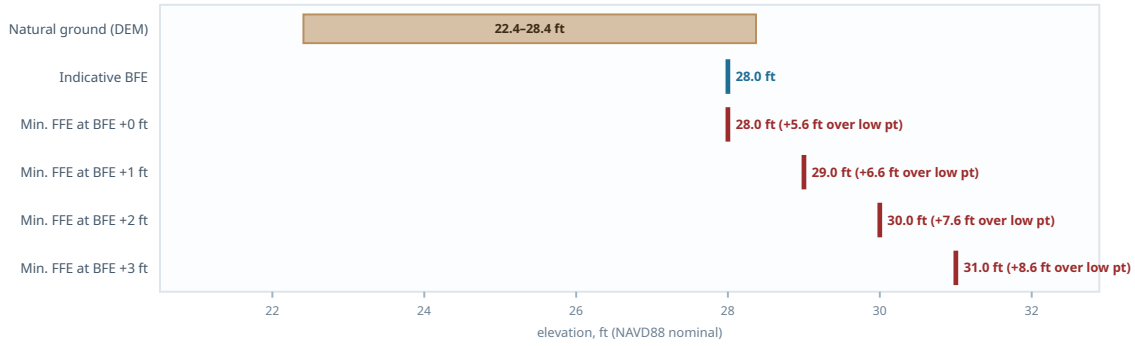
Zones intersecting the parcel	AE
Special Flood Hazard Area	Yes — mandatory flood insurance for a federally backed mortgage
FIRM panel	48039C0415K · effective 2020-12-30
Mapping status	—

Parcel intersects mapped FEMA flood-hazard polygon(s); see zones above.

C11-13 Ground elevation, BFE and minimum finished floor OFFICIAL MODEL-DERIVED

Lowest natural ground (DEM)	22.41 ft
Highest natural ground (DEM)	28.37 ft
Mean ground	23.44 ft
Indicative BFE	28.0 ft
Lowest ground relative to BFE	**-5.59 ft**
Highest ground relative to BFE	+0.37 ft

BFE basis: highest of 1 FEMA BFE line value(s) within 2,000 ft ([28.0]) - indicative only, the governing BFE must be read at the site from the FIRM/FIS profile



Freeboard scenarios. The brown band is the measured spread of natural ground across the lot. Red ticks are the minimum finished-floor elevation at BFE plus 0, 1, 2 and 3 ft of freeboard. **The freeboard the county actually enforces is not established in this report** — the scenarios show the consequence of each, they do not assert a requirement.

Scenario	Min. finished floor (ft)	Rise above lowest ground (ft)	Rise above highest ground (ft)
BFE + 0 ft	28.00	5.59	-0.37
BFE + 1 ft	29.00	6.59	0.63
BFE + 2 ft	30.00	7.59	1.63
BFE + 3 ft	31.00	8.59	2.63

DEM ground across the lot runs 22.41-28.37 ft against an indicative BFE of 28.0 ft. The lot therefore sits 5.6 ft below BFE at its lowest point. Any habitable floor must be set above BFE plus the freeboard the county actually enforces - that freeboard figure is NOT established here.

Do not assume fill is permitted
Raising a pad with fill inside a mapped floodplain is frequently restricted or requires compensatory storage and a floodplain development permit. Elevating on piers or a stem wall is often the permitted route instead. This must be confirmed with the county floodplain administrator before any design assumption is made.

C · Flood items still to confirm NEEDS CONFIRMATION

Official flood determination and the effective FIRM panel read at the parcel.
Obtain from: County floodplain administrator or a flood certification service

... continued

Enforced freeboard above BFE, fill rules and compensatory-storage requirements.
Obtain from: County floodplain administrator

First Street / third-party forward-looking flood risk and estimated flood depth.
Obtain from: First Street Foundation (riskfactor.com) for the confirmed address

Flood insurance quote under current NFIP pricing.
Obtain from: Insurance agent, using an Elevation Certificate

Site drainage: swales, culverts, outfall and upstream runoff from neighbouring lots.
Obtain from: Civil engineer site visit; county drainage criteria

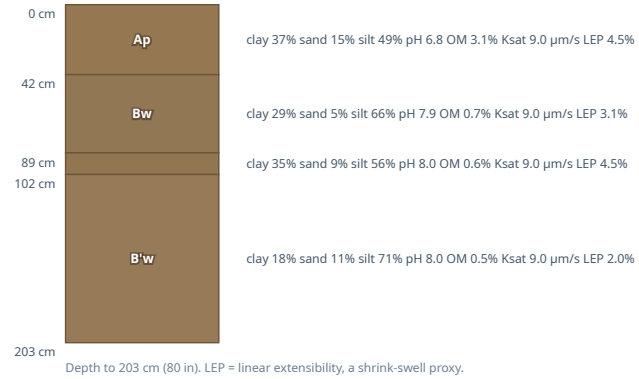
D14 Soil characteristics MODEL-DERIVED

NRCS map unit intersecting the lot	Share of lot (sampled)
Asa silty clay loam, 0 to 1 percent slopes, rarely flooded	100.0%

Component	%	Drainage	Hyd.grp	LEP %	Shrink-swell	pH	Clay %	Flooding
Asa	85%	Well drained	B	4.5	Moderate	6.8	37	Rare

LEP = linear extensibility (shrink-swell proxy). Hydrologic group D indicates the slowest infiltration. Values are NRCS representative values for the map unit, not measurements on this lot.

Asa — representative profile



Representative horizon sequence for the dominant component (Asa).

D15 Foundation suitability MODEL-DERIVED

Dominant soil shrink-swell potential is **Moderate** (peak LEP 4.5%). Stiffened/post-tensioned slab or moisture-conditioned subgrade is typical.

Slab-on-grade	Stiffened / post-tensioned
Crawlspace	Suitable
Pier and beam	Suitable
Basement	Not advised
Stepped footing	Only if sloped
NRCS rating, dwellings without basements	—
Geotechnical investigation advised	Yes

Foundation selection is an engineering decision. NRCS map-unit shrink-swell indicates risk; it does not substitute for a geotechnical investigation.

D16 Septic / on-site sewage suitability MODEL-DERIVED

Septic Tank Absorption Fields	Somewhat limited
Septic Tank Leaching Chamber (TX)	Very limited
Septic Tank, Subsurface Drip Irrigation (TX)	Very limited

Constraints identified from the soil and terrain data

- Mapped flooding frequency 'Rare' is a direct constraint on on-site sewage facilities.
- Mapped water about 29.0 ft away; TCEQ/county setbacks from water will restrict drainfield placement.

Permitted bedroom count: NOT ESTABLISHED
Permitted bedroom count follows from a site-specific soil evaluation (TCEQ Ch. 285 class, loading rate, usable drainfield area and setbacks). It cannot be derived from map-unit data. A licensed site evaluator must run the percolation/soil profile test.

NRCS ratings describe the map unit, not the lot. A TCEQ-compliant site evaluation and percolation test is required for any permit.

D17 Gardening suitability MODEL-DERIVED

SURFACE PH	CLAY	SAND	ORGANIC MATTER
6.8	37 %	15 %	3.1 %

Drainage class	Well drained
pH range through profile	6.8 – 8.0
NRCS lawn / landscape rating	Somewhat limited
Raised beds advised	Yes

Heavy clay / restricted drainage / high shrink-swell in the surface horizon makes raised beds with imported loam the lower-risk option.

Likely amendments

- Coarse organic matter and gypsum-style conditioning to improve tilth in heavy clay
- Mulch, to buffer soil temperature and moisture

Based on NRCS map-unit averages. A lab soil test is required before committing to any amendment programme.

D · Soil and planting items to confirm NEEDS CONFIRMATION

Laboratory soil test: pH, macro and micronutrients, organic matter, texture.
Obtain from: Texas A&M AgriLife Soil Testing Lab, from samples taken on the lot

Plant hardiness zone confirmation and a species list matched to the tested soil.
Obtain from: USDA hardiness map plus the county AgriLife extension office

E19-23 Utilities and infrastructure NEEDS CONFIRMATION

Utility availability is not published as open geospatial data at parcel level. Nothing is inferred here: each item below must come from the serving utility in writing, because a missing water tap or a long electric extension can change the build budget materially.

Water supply: municipal, water-supply-corporation, or private well; is a tap at the road?
Obtain from: Serving water supply corporation / county; well logs via TX Water Development Board

Sewer: is public sewer available, or is an on-site facility required? Distance to any main.
Obtain from: County / city utility department or MUD

Electricity: provider, whether service exists at the lot line, overhead vs underground, connection cost.
Obtain from: Retail electric provider / local co-op (service availability letter)

Natural gas or propane availability and existing infrastructure.
Obtain from: Gas utility or local propane suppliers

... continued

Internet: fibre, cable, fixed wireless, DSL or satellite, with advertised speeds.
Obtain from: FCC National Broadband Map plus a direct serviceability check by address

F24-25 Waterfront position MODEL-DERIVED

Abuts mapped water - consistent with direct water frontage

Nearest mapped waterbody	unnamed waterbody
Distance from parcel boundary	29 ft
Waterbody area	0.823 km²

NHD polygons are hydrographic mapping, not a boundary survey. Whether the lot carries riparian frontage, deeded access or only common-area access is a title/plat question and is NOT established here.

F26-29 · Dock, depth, boating and fishing NEEDS CONFIRMATION

Whether a dock is permittable at this shoreline, and the governing authority.
Obtain from: Controlling lake / river authority, plus the HOA

Dock size limits, shoreline setbacks, shared-dock rules and permit cost.
Obtain from: Lake authority rules and permit schedule

Whether riparian or shoreline rights pass with the lot, or the shoreline is a common reserve.
Obtain from: Title commitment and recorded plat

Lake water-level regime and any shoreline-management plan.
Obtain from: Lake/river authority or reservoir operator

Bathymetry: average and maximum depth, and depth near this shoreline.
Obtain from: Lake authority survey, TPWD reservoir report, or a sonar survey

Seasonal water-level variation and drawdown history.
Obtain from: Reservoir operator gauge records / USGS gauges

Boating rules: permitted craft, motor limits, wake rules, ramps and marinas.
Obtain from: Lake authority and TPWD

Fishery: species present, stocking, licence requirements and catch limits.
Obtain from: Texas Parks & Wildlife reservoir survey reports

G30-33 HOA, covenants, zoning and a second dwelling NEEDS CONFIRMATION

The parcel sits in **Bar X Ranch** according to the recorded legal description, so a declaration of covenants very likely exists. The documents themselves are not public geospatial data and were not retrieved, so no dues figure, restriction or amenity is asserted here.

Whether an HOA or POA exists and is mandatory, with its name and management contact.
Obtain from: Title commitment, recorded Declaration, seller disclosure

Zoning designation, or confirmation that the county is unzoned.
Obtain from: County or city development services

Minimum lot size, building coverage, setbacks, height limits, accessory-structure rules.
Obtain from: County/city development standards or subdivision regulations

Whether a second independent dwelling or in-law suite (ADU) is permitted, and on what terms.
Obtain from: County/city development services plus the HOA

Permitting path, driveway/culvert permit and any development fee.
Obtain from: County permitting office

... continued

Annual dues, monthly equivalent, transfer fee, and any special assessment.
Obtain from: HOA or management company estoppel / resale certificate

Declaration, CC&Rs, architectural guidelines, rules and all amendments.
Obtain from: HOA or title company

Restrictions on docks, fences, outbuildings, RVs, boats, livestock and poultry.
Obtain from: Recorded CC&Rs and architectural guidelines

Minimum dwelling size and any construction time limit after purchase.
Obtain from: Recorded CC&Rs

Amenities actually included: clubhouse, pool, courts, parks, lake and boat facilities.
Obtain from: HOA documents and budget

H34 / I35 Hobby farming and short-term rental NEEDS CONFIRMATION

Whether chickens, ducks, rabbits, goats or bees are allowed, and in what numbers.
Obtain from: County rules plus the recorded CC&Rs (either can prohibit)

Any agricultural or wildlife valuation currently on the parcel, and rollback-tax exposure if use changes.
Obtain from: County appraisal district

Whether short-term rental is permitted at this location.
Obtain from: County/city ordinance and the HOA

Any STR permit, occupancy cap, parking or noise condition.
Obtain from: Local ordinance

Hotel occupancy tax registration and remittance obligations.
Obtain from: Texas Comptroller plus local taxing authority

J36-42 Environmental and geological due diligence MODEL-DERIVED

Ground-penetrating radar penetration (NRCS rating)	Very low penetration Indicates how useful GPR would be here; clay-rich soils attenuate the signal.
Sinkhole / karst exposure	Not indicated by the soil map units returned for this parcel. No restrictive bedrock or karst feature was reported. CONFIRM WITH TX BUREAU OF ECONOMIC GEOLOGY
Radon	EPA publishes radon zones by county, not by parcel. LOOK UP THE COUNTY ZONE AND TEST ANY BUILT STRUCTURE

Existing FEMA Elevation Certificate for the lot.
Obtain from: Seller, county floodplain office or a surveyor

Recent boundary, ALTA or topographic survey.
Obtain from: Seller or a licensed Texas surveyor (RPLS)

Geotechnical / soil boring report.
Obtain from: Seller or a geotechnical engineer

On-site sewage facility site evaluation and percolation test.
Obtain from: Licensed site evaluator / county OSSF office

Any ground-penetrating-radar or other subsurface investigation.
Obtain from: Seller; commission if subsurface features are suspected

K43-45 Residential construction feasibility MODEL-DERIVED

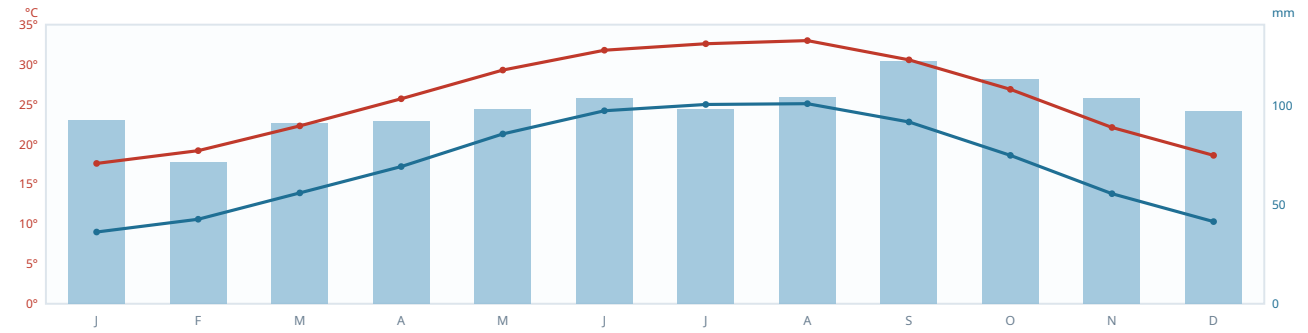
Lot area available	0.763 ac (33,236 sq ft)
Usable building envelope	NOT COMPUTABLE Setbacks, easements and any lake or drainage reserve come from the recorded plat and the HOA architectural guidelines, none of which were available.
Terrain constraint	Gentle slope — Minimal grading; Routine cut-and-fill
Foundation implication	Moderate shrink-swell. Stiffened/post-tensioned slab or moisture-conditioned subgrade is typical.
Flood implication	Inside a mapped SFHA, so finished-floor elevation, flood venting and insurance all bind the design.

Construction cost
No build cost is estimated in this report. A defensible figure for a four-bedroom house depends on the HOA's minimum size and material rules, the finished-floor elevation forced by the flood zone, the foundation type the soil requires, and whether water and sewer must be created on site. Three of those four inputs are still unconfirmed for this lot, so any number quoted now would be arbitrary. Price it from a builder bid once the plat, CC&Rs, floodplain requirement and soil report are in hand.

L46 Climate MODEL-DERIVED

MEAN ANNUAL TEMP	MEAN SUMMER HIGH	MEAN WINTER LOW	ANNUAL RAINFALL
21.8 °C	32.5 °C	9.9 °C	1,191 mm

Record high	41.8 °C	**Record low**	-9.4 °C	**Rainfall**	46.9 in
Growing season	~342 days	**Days at or below 0 °C**	2.6 /yr	**Period**	1994-2023 (30 years)



Monthly normals, 1994-2023 (30 years). Bars are mean monthly rainfall (mm, right axis); lines are mean daily maximum and minimum temperature (°C, left axis). Source: ERA5 daily reanalysis via Open-Meteo archive API.

L47 Natural hazards

Flood	Mapped Special Flood Hazard Area
Other perils	Hurricane and tropical wind, tornado, hail, severe convective storm, lightning, extreme heat, drought and wildfire all vary by location within Texas. QUANTIFY FROM FEMA NATIONAL RISK INDEX FOR THE CONFIRMED COUNTY AND FROM AN INSURANCE QUOTE

L48-55 Community, schools, healthcare, shopping and access NEEDS CONFIRMATION

Demographics not retrieved
Census API requires an API key (keyless request rejected) Free key from https://api.census.gov/data/key_signup.html, then set CENSUS_API_KEY and re-run 05_context.py. Or read the figures directly from data.census.gov.

Driving distance and time to the nearest hospital, emergency department and urgent care.
Obtain from: Direct routing check from the confirmed parcel location

Distance to grocery, major retail, pharmacy, bank, fuel and hardware.
Obtain from: Direct routing check

Assigned school district and campuses, with distances and current ratings.
Obtain from: TEA school-district locator and the district attendance-zone map

Nearest Indian / South Asian grocery, restaurants and places of worship.
Obtain from: Direct search from the confirmed parcel location

Nearest international airport, driving time and route.
Obtain from: Direct routing check

Walk / Bike / Transit scores.
Obtain from: Walk Score for the confirmed address

M56 Documented strengths

- Abuts mapped water at roughly 29 ft, consistent with genuine water frontage subject to title confirmation.
- Gentle mean grade of 2.64%, so routine grading should suffice.
- Long growing season of roughly 342 frost-free days supports year-round production.
- Septic absorption is rated only "somewhat limited" rather than "very limited", which is the better of the two common outcomes here.

M57 Limitations and risks

- **Inside a FEMA Special Flood Hazard Area (AE).** This forces an elevated finished floor, triggers floodplain permitting, and makes flood insurance mandatory with a federally backed mortgage.
- Lowest natural ground sits about 5.6 ft below the indicative BFE, so the pad must be raised or the house elevated.
- Septic absorption is rated "very limited" for the dominant soil, so a conventional drainfield may not be permittable and an engineered or aerobic system may be required at higher cost.
- Mapped flooding frequency 'Rare' is a direct constraint on on-site sewage facilities.
- HOA covenants, zoning, utility availability, dock rights and short-term-rental rules are all unconfirmed. Any of them can independently prevent the intended use.

M59 Construction feasibility conclusion

Main house	Likely The lot is large enough and the terrain is workable. Finished-floor elevation and foundation type are the governing variables, not land area.
Second in-law suite / ADU	Unresolved Depends entirely on county rules and the HOA declaration; neither was available. Do not assume a second dwelling is permitted.
Garage	Likely No physical constraint identified, subject to setbacks and architectural guidelines.
Septic system	Unresolved Soil ratings are unfavourable; permittability and bedroom count require a site evaluation.
Garden	Conditional Workable with raised beds and imported loam; native surface soil is restrictive.
Hobby farming / poultry	Unresolved Governed by county rules and CC&Rs; unconfirmed.
Dock	Unresolved The lot adjoins water, but dock rights are a title and permitting question that was not resolved.
Short-term rental	Unresolved Governed by local ordinance and CC&Rs; unconfirmed.
Pool / outdoor recreation	Conditional Space is available; flood-zone and soil-movement considerations affect pool design where applicable.

M60 Key measurements summary

Parameter	Finding
Lot size	0.763 ac (33,236 sq ft)
Lot dimensions (long axis × width)	368 ft × 181 ft
Lowest elevation	22.41 ft
Highest elevation	28.37 ft
Elevation difference	5.96 ft
Average slope	2.64% (1.51°)
Maximum slope	21.98% (12.40°)
Steepest 50 ft run	9.8% (5.6°)
FEMA zone	AE
BFE	28.0 ft (indicative)
Minimum finished-floor elevation	29.0 ft at +1 ft / 30.0 ft at +2 ft freeboard COUNTY FIGURE UNCONFIRMED
Fill or elevation required?	Yes — lowest ground is 5.6 ft below BFE
Sewer	NEEDS CONFIRMATION
Water	NEEDS CONFIRMATION
Electricity	NEEDS CONFIRMATION

... continued

Parameter	Finding
Gas	NEEDS CONFIRMATION
Internet	NEEDS CONFIRMATION
Septic required?	NEEDS CONFIRMATION — soil absorption rated Very limited
Soil	Asa — Well drained, hydrologic group B, Moderate shrink-swell
HOA	Platted subdivision (Bar X Ranch), so covenants are likely. NEEDS CONFIRMATION
HOA fee	NEEDS CONFIRMATION
Zoning	NEEDS CONFIRMATION
Dock possible?	NEEDS CONFIRMATION
Lake depth	NEEDS CONFIRMATION
Boating	NEEDS CONFIRMATION
Fishing	NEEDS CONFIRMATION
Hobby farming	NEEDS CONFIRMATION
Second in-law suite	NEEDS CONFIRMATION
Short-term rental	NEEDS CONFIRMATION
Nearest hospital	NEEDS CONFIRMATION
Nearest international airport	NEEDS CONFIRMATION
Distance to nearest mapped water	29 ft to unnamed waterbody
Mean annual temperature	21.8 °C
Mean annual rainfall	1,191 mm (46.9 in)
Appraised market value	$31,920

1601 Old Henry Ct, Angleton, TX 77515

Lot 5 of 12 | ID confidence: high | Bar X Ranch | Brazoria County

A1 Property identification & record data OFFICIAL

Situs address of record	1601 OLD HENRY CT, , TX 77515
Legal description	BAR X RANCH SEC 10 (A0053 S CARTER) BLK 3 LOT 66 ACRES 0.76
Subdivision	Bar X Ranch
County	Brazoria County (FIPS 48039)
Parcel ID / account	185603
Geographic ID (APN)	1542-0066-003
Owner of record	ALVAREZ BENITA & CIRO
Acreage — appraisal roll (legal)	0.76 ac
Acreage — GIS polygon	0.797 ac
Acreage — recomputed from boundary	0.800 ac
Appraised land value	$21,280
Appraised improvement value	$0
Appraised market value	$21,280
Centroid (WGS84)	29.125863, -95.572386
Longest axis / width	379 ft × 187 ft (axis bearing 18°)

Source: Texas StratMap statewide land parcels (county appraisal roll). Matched by situs address match. Appraisal-roll acreage and a GIS polygon rarely agree exactly; neither replaces a survey.

A2 · Listing and plat material not retrievable here NEEDS CONFIRMATION

Current asking price, days on market and listing history.
Obtain from: Listing agent / MLS

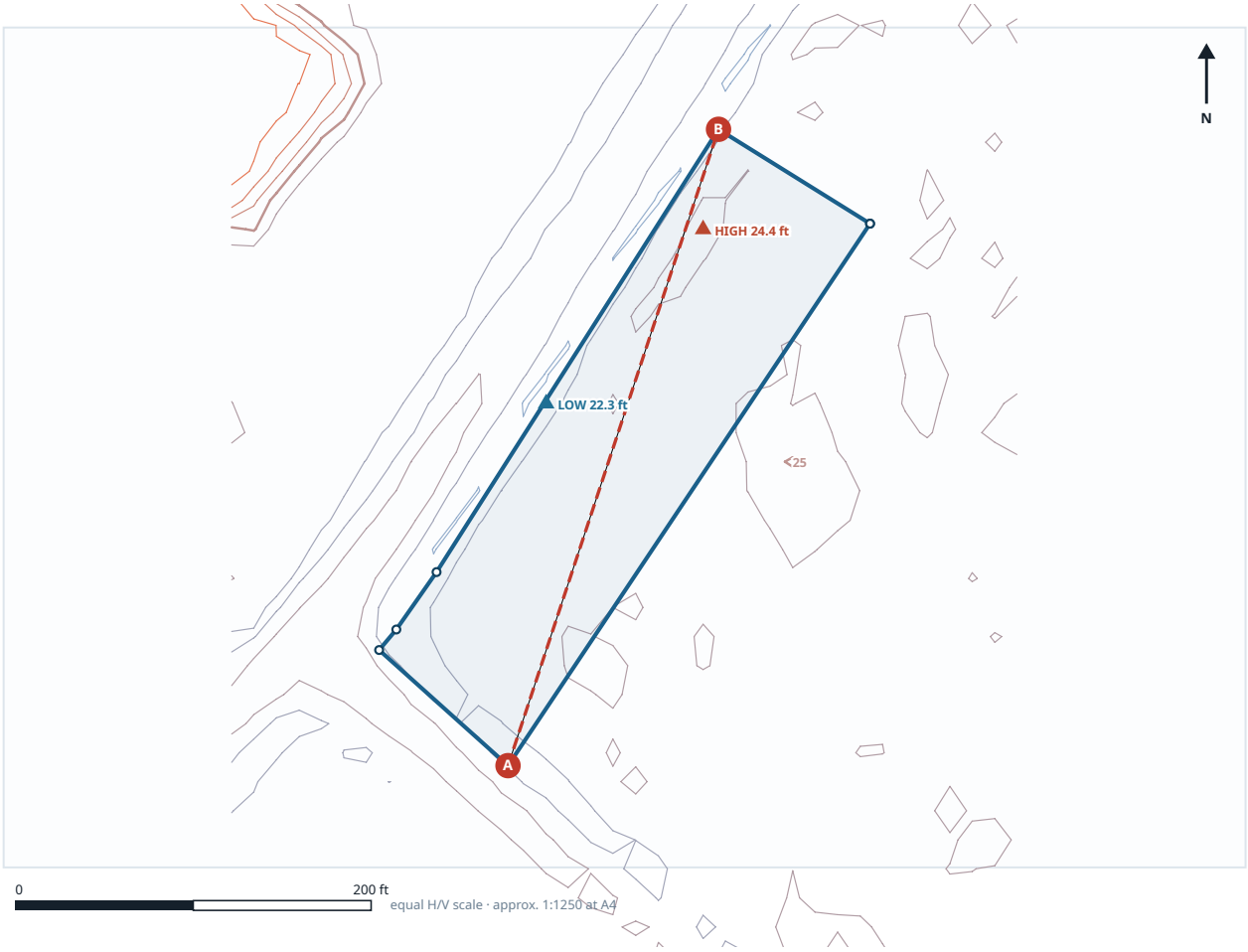
Listing photographs: lot frontage, shoreline, road access and surroundings.
Obtain from: Listing agent (Zillow blocks automated retrieval, so no listing imagery is reproduced here)

Seller disclosure notice and any known defects.
Obtain from: Listing agent

Recorded plat and Esri/county GIS parcel exhibit for comparison with this report.
Obtain from: County Clerk plat records and the county GIS portal

B5 Terrain, contours and true-scale site plan MODEL-DERIVED

LOWEST POINT	HIGHEST POINT	RELIEF	MEAN ELEVATION
22.3 ft	24.4 ft	2.1 ft	23.6 ft



True-scale site plan. Parcel boundary from the county GIS polygon; contours at 1.0 ft intervals derived from the USGS 3DEP 1 m bare-earth DEM. A–B is the cross-section line. Horizontal and vertical map scales are equal, so shapes and distances are undistorted.

MEAN GRADE	MEDIAN GRADE	90TH PCT GRADE	MAX GRADE
2.45 %	1.28 %	6.58 %	17.17 %

Mean slope angle	1.40°	**90th pct angle**	3.77°	**Max angle**	9.74°	**Grid cells**	1,156

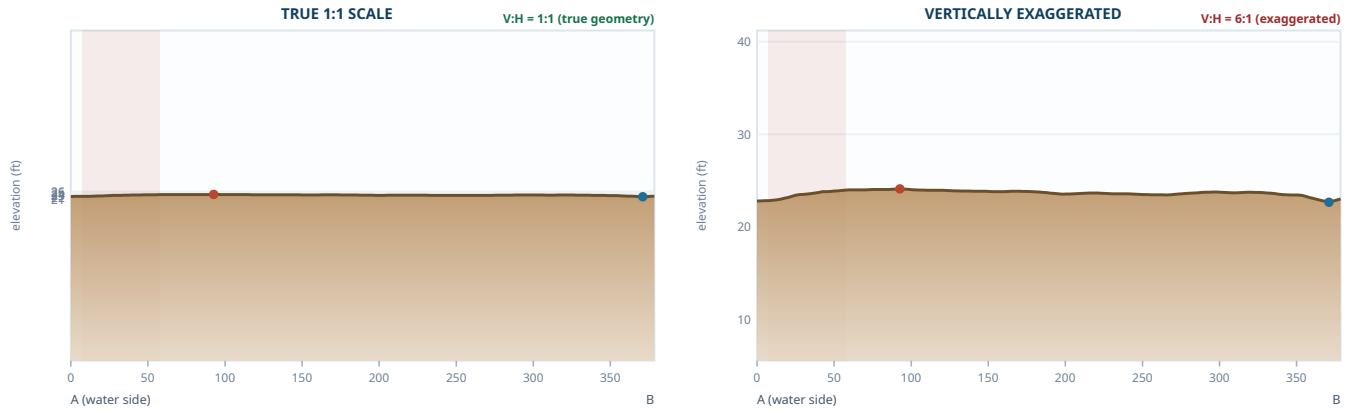
B7 · Terrain classification

Gentle slope. Standard residential grading is normally sufficient.

Basis: mean DEM grade 2.45% (90th pct 6.58%), relief 2.1 ft

Grading expectation: Minimal grading; Routine cut-and-fill

B6 Longitudinal cross-section MODEL-DERIVED

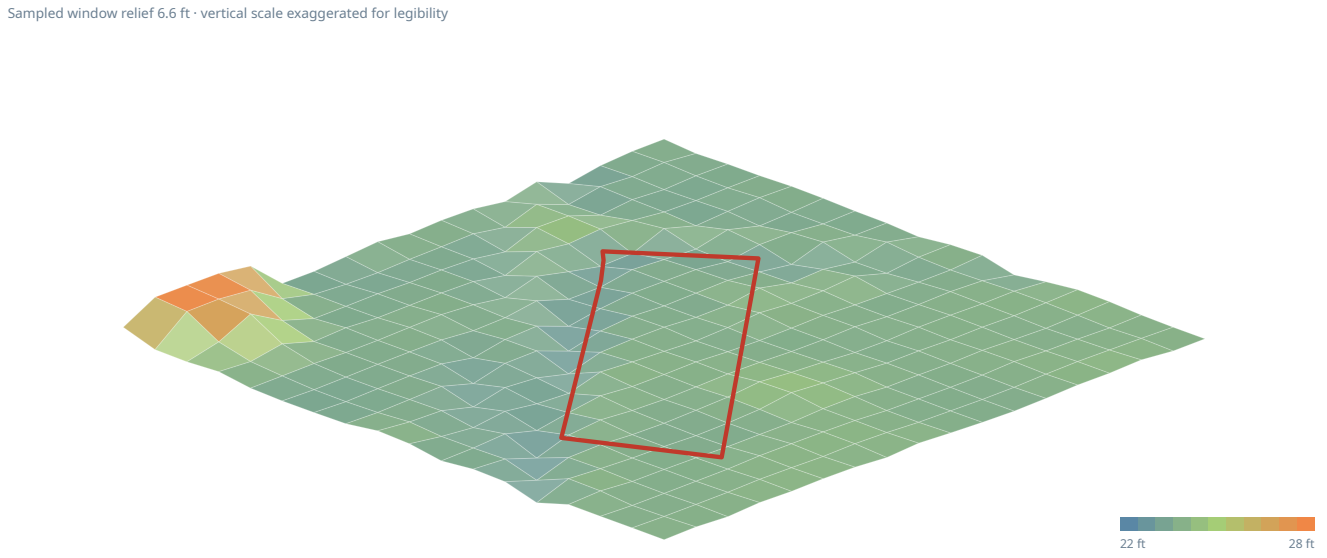


Run A–B	379 ft	**Lowest**	22.7 ft	**Highest**	24.1 ft
Elevation difference	1.4 ft	**Average grade**	0.4%	**Direct rising angle**	0.2°
Steepest 50 ft	**2.3%** (1.3°) over a 50 ft run				

Reading: Over 379 ft the ground rises only 1.4 ft, a direct angle of 0.2°. At true 1:1 scale the profile is almost a straight line, which is the correct reading: this lot is effectively flat. The exaggerated panel is the same data and should not be read as slope.

Left panel is the real geometry at 1:1 vertical-to-horizontal scale. The right panel exaggerates the vertical axis, which is how slope is conventionally drawn on maps but overstates steepness. The shaded band marks the steepest 50 ft run. Profile sampled from the 3DEP DEM along the lot's longest axis.

B8 Three-dimensional terrain MODEL-DERIVED



Isometric view of the sampled DEM window; the red outline is the parcel boundary draped on the surface. Vertical scale is exaggerated to reveal form — use the 1:1 cross-section above for true steepness.

C9 FEMA flood-zone determination OFFICIAL

Zone: AE | Inside Special Flood Hazard Area

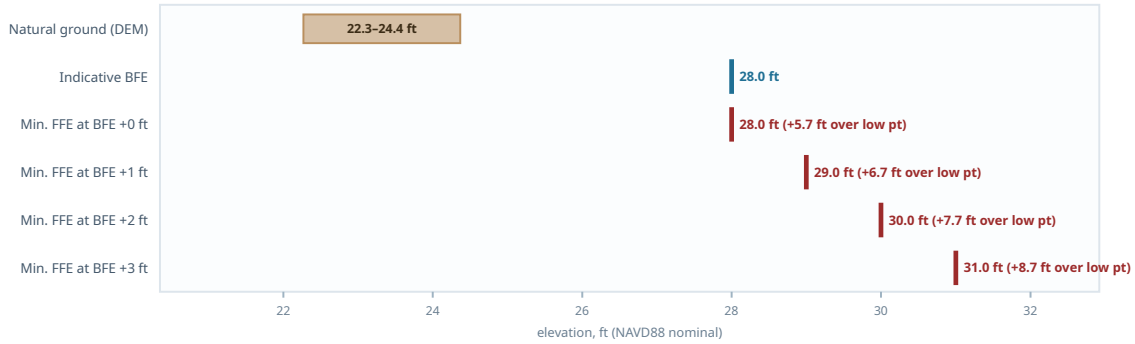
Zones intersecting the parcel	AE
Special Flood Hazard Area	Yes — mandatory flood insurance for a federally backed mortgage
FIRM panel	48039C0415K · effective 2020-12-30
Mapping status	—

Parcel intersects mapped FEMA flood-hazard polygon(s); see zones above.

C11-13 Ground elevation, BFE and minimum finished floor OFFICIAL MODEL-DERIVED

Lowest natural ground (DEM)	22.27 ft
Highest natural ground (DEM)	24.37 ft
Mean ground	23.56 ft
Indicative BFE	28.0 ft
Lowest ground relative to BFE	**-5.73 ft**
Highest ground relative to BFE	-3.63 ft

BFE basis: highest of 1 FEMA BFE line value(s) within 2,000 ft ([28.0]) - indicative only, the governing BFE must be read at the site from the FIRM/FIS profile



Freeboard scenarios. The brown band is the measured spread of natural ground across the lot. Red ticks are the minimum finished-floor elevation at BFE plus 0, 1, 2 and 3 ft of freeboard. **The freeboard the county actually enforces is not established in this report** — the scenarios show the consequence of each, they do not assert a requirement.

Scenario	Min. finished floor (ft)	Rise above lowest ground (ft)	Rise above highest ground (ft)
BFE + 0 ft	28.00	5.73	3.63
BFE + 1 ft	29.00	6.73	4.63
BFE + 2 ft	30.00	7.73	5.63
BFE + 3 ft	31.00	8.73	6.63

DEM ground across the lot runs 22.27-24.37 ft against an indicative BFE of 28.0 ft. The lot therefore sits 5.7 ft below BFE at its lowest point. Any habitable floor must be set above BFE plus the freeboard the county actually enforces - that freeboard figure is NOT established here.

Do not assume fill is permitted
Raising a pad with fill inside a mapped floodplain is frequently restricted or requires compensatory storage and a floodplain development permit. Elevating on piers or a stem wall is often the permitted route instead. This must be confirmed with the county floodplain administrator before any design assumption is made.

C · Flood items still to confirm NEEDS CONFIRMATION

Official flood determination and the effective FIRM panel read at the parcel.
Obtain from: County floodplain administrator or a flood certification service

... continued

Enforced freeboard above BFE, fill rules and compensatory-storage requirements.
Obtain from: County floodplain administrator

First Street / third-party forward-looking flood risk and estimated flood depth.
Obtain from: First Street Foundation (riskfactor.com) for the confirmed address

Flood insurance quote under current NFIP pricing.
Obtain from: Insurance agent, using an Elevation Certificate

Site drainage: swales, culverts, outfall and upstream runoff from neighbouring lots.
Obtain from: Civil engineer site visit; county drainage criteria

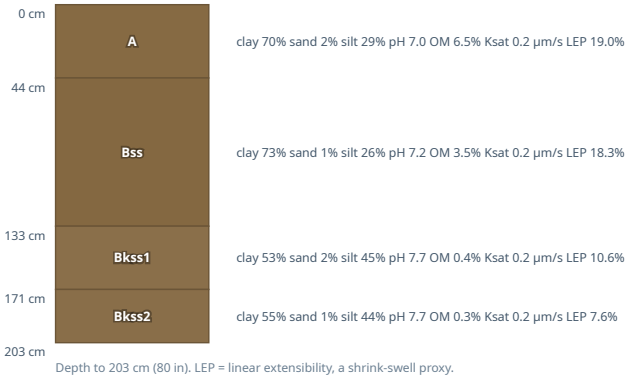
D14 Soil characteristics MODEL-DERIVED

NRCS map unit intersecting the lot	Share of lot (sampled)
Pledger clay, 0 to 1 percent slopes, rarely flooded	100.0%

Component	%	Drainage	Hyd.grp	LEP %	Shrink-swell	pH	Clay %	Flooding
Pledger	85%	Moderately well drained	D	19.0	Very high	7.0	70	Rare

LEP = linear extensibility (shrink-swell proxy). Hydrologic group D indicates the slowest infiltration. Values are NRCS representative values for the map unit, not measurements on this lot.

Pledger — representative profile



Depth to 203 cm (80 in). LEP = linear extensibility, a shrink-swell proxy.

Representative horizon sequence for the dominant component (Pledger).

D15 Foundation suitability MODEL-DERIVED

Dominant soil shrink-swell potential is **Very high** (peak LEP 19.0%). Severe movement potential. Deep piers or heavily engineered slab; geotechnical design required.

Slab-on-grade	Engineered design mandatory
Crawlspace	Deep piers required
Pier and beam	Preferred, deep piers
Basement	Not advised
Stepped footing	Only if sloped
NRCS rating, dwellings without basements	—
Geotechnical investigation advised	Yes

Foundation selection is an engineering decision. NRCS map-unit shrink-swell indicates risk; it does not substitute for a geotechnical investigation.

D16 Septic / on-site sewage suitability MODEL-DERIVED

Septic Tank Absorption Fields	Very limited
Septic Tank Leaching Chamber (TX)	Very limited
Septic Tank, Subsurface Drip Irrigation (TX)	Very limited

Constraints identified from the soil and terrain data

- Hydrologic group D indicates slow infiltration, which constrains conventional drainfields.
- Mapped flooding frequency 'Rare' is a direct constraint on on-site sewage facilities.

Permitted bedroom count: NOT ESTABLISHED
Permitted bedroom count follows from a site-specific soil evaluation (TCEQ Ch. 285 class, loading rate, usable drainfield area and setbacks). It cannot be derived from map-unit data. A licensed site evaluator must run the percolation/soil profile test.

NRCS ratings describe the map unit, not the lot. A TCEQ-compliant site evaluation and percolation test is required for any permit.

D17 Gardening suitability MODEL-DERIVED

SURFACE PH	CLAY	SAND	ORGANIC MATTER
7.0	70 %	2 %	6.5 %

Drainage class	Moderately well drained
pH range through profile	7.0 – 7.7
NRCS lawn / landscape rating	Very limited
Raised beds advised	Yes

Heavy clay / restricted drainage / high shrink-swell in the surface horizon makes raised beds with imported loam the lower-risk option.

Likely amendments

- Coarse organic matter and gypsum-style conditioning to improve tilth in heavy clay
- Mulch, to buffer soil temperature and moisture

Based on NRCS map-unit averages. A lab soil test is required before committing to any amendment programme.

D · Soil and planting items to confirm NEEDS CONFIRMATION

Laboratory soil test: pH, macro and micronutrients, organic matter, texture.
Obtain from: Texas A&M AgriLife Soil Testing Lab, from samples taken on the lot

Plant hardiness zone confirmation and a species list matched to the tested soil.
Obtain from: USDA hardiness map plus the county AgriLife extension office

E19-23 Utilities and infrastructure NEEDS CONFIRMATION

Utility availability is not published as open geospatial data at parcel level. Nothing is inferred here: each item below must come from the serving utility in writing, because a missing water tap or a long electric extension can change the build budget materially.

Water supply: municipal, water-supply-corporation, or private well; is a tap at the road?
Obtain from: Serving water supply corporation / county; well logs via TX Water Development Board

Sewer: is public sewer available, or is an on-site facility required? Distance to any main.
Obtain from: County / city utility department or MUD

Electricity: provider, whether service exists at the lot line, overhead vs underground, connection cost.
Obtain from: Retail electric provider / local co-op (service availability letter)

Natural gas or propane availability and existing infrastructure.
Obtain from: Gas utility or local propane suppliers

... continued

Internet: fibre, cable, fixed wireless, DSL or satellite, with advertised speeds.
Obtain from: FCC National Broadband Map plus a direct serviceability check by address

F24-25 Waterfront position MODEL-DERIVED

Near water - frontage unlikely; access likely via easement or common area

Nearest mapped waterbody	unnamed waterbody
Distance from parcel boundary	1,066 ft
Waterbody area	0.823 km²

NHD polygons are hydrographic mapping, not a boundary survey. Whether the lot carries riparian frontage, deeded access or only common-area access is a title/plat question and is NOT established here.

F26-29 · Dock, depth, boating and fishing NEEDS CONFIRMATION

Whether a dock is permittable at this shoreline, and the governing authority.
Obtain from: Controlling lake / river authority, plus the HOA

Dock size limits, shoreline setbacks, shared-dock rules and permit cost.
Obtain from: Lake authority rules and permit schedule

Whether riparian or shoreline rights pass with the lot, or the shoreline is a common reserve.
Obtain from: Title commitment and recorded plat

Lake water-level regime and any shoreline-management plan.
Obtain from: Lake/river authority or reservoir operator

Bathymetry: average and maximum depth, and depth near this shoreline.
Obtain from: Lake authority survey, TPWD reservoir report, or a sonar survey

Seasonal water-level variation and drawdown history.
Obtain from: Reservoir operator gauge records / USGS gauges

Boating rules: permitted craft, motor limits, wake rules, ramps and marinas.
Obtain from: Lake authority and TPWD

Fishery: species present, stocking, licence requirements and catch limits.
Obtain from: Texas Parks & Wildlife reservoir survey reports

G30-33 HOA, covenants, zoning and a second dwelling NEEDS CONFIRMATION

The parcel sits in **Bar X Ranch** according to the recorded legal description, so a declaration of covenants very likely exists. The documents themselves are not public geospatial data and were not retrieved, so no dues figure, restriction or amenity is asserted here.

Whether an HOA or POA exists and is mandatory, with its name and management contact.
Obtain from: Title commitment, recorded Declaration, seller disclosure

Zoning designation, or confirmation that the county is unzoned.
Obtain from: County or city development services

Minimum lot size, building coverage, setbacks, height limits, accessory-structure rules.
Obtain from: County/city development standards or subdivision regulations

Whether a second independent dwelling or in-law suite (ADU) is permitted, and on what terms.
Obtain from: County/city development services plus the HOA

Permitting path, driveway/culvert permit and any development fee.
Obtain from: County permitting office

... continued

Annual dues, monthly equivalent, transfer fee, and any special assessment.
Obtain from: HOA or management company estoppel / resale certificate

Declaration, CC&Rs, architectural guidelines, rules and all amendments.
Obtain from: HOA or title company

Restrictions on docks, fences, outbuildings, RVs, boats, livestock and poultry.
Obtain from: Recorded CC&Rs and architectural guidelines

Minimum dwelling size and any construction time limit after purchase.
Obtain from: Recorded CC&Rs

Amenities actually included: clubhouse, pool, courts, parks, lake and boat facilities.
Obtain from: HOA documents and budget

H34 / I35 Hobby farming and short-term rental NEEDS CONFIRMATION

Whether chickens, ducks, rabbits, goats or bees are allowed, and in what numbers.
Obtain from: County rules plus the recorded CC&Rs (either can prohibit)

Any agricultural or wildlife valuation currently on the parcel, and rollback-tax exposure if use changes.
Obtain from: County appraisal district

Whether short-term rental is permitted at this location.
Obtain from: County/city ordinance and the HOA

Any STR permit, occupancy cap, parking or noise condition.
Obtain from: Local ordinance

Hotel occupancy tax registration and remittance obligations.
Obtain from: Texas Comptroller plus local taxing authority

J36-42 Environmental and geological due diligence MODEL-DERIVED

Ground-penetrating radar penetration (NRCS rating)	Very low penetration Indicates how useful GPR would be here; clay-rich soils attenuate the signal.
Sinkhole / karst exposure	Not indicated by the soil map units returned for this parcel. No restrictive bedrock or karst feature was reported. CONFIRM WITH TX BUREAU OF ECONOMIC GEOLOGY
Radon	EPA publishes radon zones by county, not by parcel. LOOK UP THE COUNTY ZONE AND TEST ANY BUILT STRUCTURE

Existing FEMA Elevation Certificate for the lot.
Obtain from: Seller, county floodplain office or a surveyor

Recent boundary, ALTA or topographic survey.
Obtain from: Seller or a licensed Texas surveyor (RPLS)

Geotechnical / soil boring report.
Obtain from: Seller or a geotechnical engineer

On-site sewage facility site evaluation and percolation test.
Obtain from: Licensed site evaluator / county OSSF office

Any ground-penetrating-radar or other subsurface investigation.
Obtain from: Seller; commission if subsurface features are suspected

K43-45 Residential construction feasibility MODEL-DERIVED

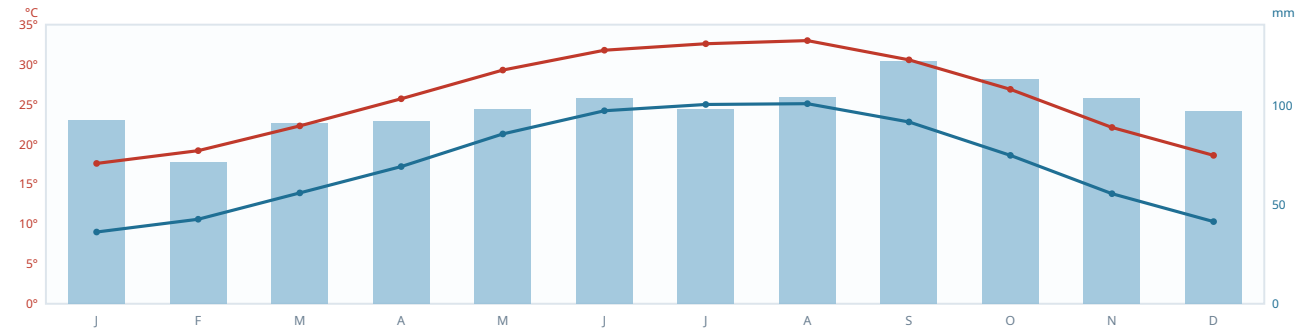
Lot area available	0.797 ac (34,717 sq ft)
Usable building envelope	NOT COMPUTABLE Setbacks, easements and any lake or drainage reserve come from the recorded plat and the HOA architectural guidelines, none of which were available.
Terrain constraint	Gentle slope — Minimal grading; Routine cut-and-fill
Foundation implication	Very high shrink-swell. Severe movement potential. Deep piers or heavily engineered slab; geotechnical design required.
Flood implication	Inside a mapped SFHA, so finished-floor elevation, flood venting and insurance all bind the design.

Construction cost
No build cost is estimated in this report. A defensible figure for a four-bedroom house depends on the HOA's minimum size and material rules, the finished-floor elevation forced by the flood zone, the foundation type the soil requires, and whether water and sewer must be created on site. Three of those four inputs are still unconfirmed for this lot, so any number quoted now would be arbitrary. Price it from a builder bid once the plat, CC&Rs, floodplain requirement and soil report are in hand.

L46 Climate MODEL-DERIVED

MEAN ANNUAL TEMP	MEAN SUMMER HIGH	MEAN WINTER LOW	ANNUAL RAINFALL
21.8 °C	32.5 °C	9.9 °C	1,191 mm

Record high	41.8 °C	**Record low**	-9.4 °C	**Rainfall**	46.9 in
Growing season	~342 days	**Days at or below 0 °C**	2.6 /yr	**Period**	1994-2023 (30 years)



Monthly normals, 1994-2023 (30 years). Bars are mean monthly rainfall (mm, right axis); lines are mean daily maximum and minimum temperature (°C, left axis). Source: ERA5 daily reanalysis via Open-Meteo archive API.

L47 Natural hazards

Flood	Mapped Special Flood Hazard Area
Other perils	Hurricane and tropical wind, tornado, hail, severe convective storm, lightning, extreme heat, drought and wildfire all vary by location within Texas. QUANTIFY FROM FEMA NATIONAL RISK INDEX FOR THE CONFIRMED COUNTY AND FROM AN INSURANCE QUOTE

L48-55 Community, schools, healthcare, shopping and access NEEDS CONFIRMATION

Demographics not retrieved
Census API requires an API key (keyless request rejected) Free key from https://api.census.gov/data/key_signup.html, then set CENSUS_API_KEY and re-run 05_context.py. Or read the figures directly from data.census.gov.

Driving distance and time to the nearest hospital, emergency department and urgent care.
Obtain from: Direct routing check from the confirmed parcel location

Distance to grocery, major retail, pharmacy, bank, fuel and hardware.
Obtain from: Direct routing check

Assigned school district and campuses, with distances and current ratings.
Obtain from: TEA school-district locator and the district attendance-zone map

Nearest Indian / South Asian grocery, restaurants and places of worship.
Obtain from: Direct search from the confirmed parcel location

Nearest international airport, driving time and route.
Obtain from: Direct routing check

Walk / Bike / Transit scores.
Obtain from: Walk Score for the confirmed address

M56 Documented strengths

- Gentle mean grade of 2.45%, so routine grading should suffice.
- Long growing season of roughly 342 frost-free days supports year-round production.

M57 Limitations and risks

- **Inside a FEMA Special Flood Hazard Area (AE).** This forces an elevated finished floor, triggers floodplain permitting, and makes flood insurance mandatory with a federally backed mortgage.
- Lowest natural ground sits about 5.7 ft below the indicative BFE, so the pad must be raised or the house elevated.
- **Very high soil shrink-swell.** Expansive clay is the leading cause of foundation distress in this region; an engineered foundation and a geotechnical report should be assumed.
- Septic absorption is rated "very limited" for the dominant soil, so a conventional drainfield may not be permittable and an engineered or aerobic system may be required at higher cost.
- Mapped flooding frequency 'Rare' is a direct constraint on on-site sewage facilities.
- HOA covenants, zoning, utility availability, dock rights and short-term-rental rules are all unconfirmed. Any of them can independently prevent the intended use.

M59 Construction feasibility conclusion

Main house	Likely The lot is large enough and the terrain is workable. Finished-floor elevation and foundation type are the governing variables, not land area.
Second in-law suite / ADU	Unresolved Depends entirely on county rules and the HOA declaration; neither was available. Do not assume a second dwelling is permitted.
Garage	Likely No physical constraint identified, subject to setbacks and architectural guidelines.
Septic system	Unresolved Soil ratings are unfavourable; permittability and bedroom count require a site evaluation.
Garden	Conditional Workable with raised beds and imported loam; native surface soil is restrictive.
Hobby farming / poultry	Unresolved Governed by county rules and CC&Rs; unconfirmed.
Dock	Unresolved The lot does not adjoin mapped water, so a private dock is unlikely; shared access may exist.
Short-term rental	Unresolved Governed by local ordinance and CC&Rs; unconfirmed.
Pool / outdoor recreation	Conditional Space is available; flood-zone and soil-movement considerations affect pool design where applicable.

M60 Key measurements summary

Parameter	Finding
Lot size	0.797 ac (34,717 sq ft)
Lot dimensions (long axis × width)	379 ft × 187 ft
Lowest elevation	22.27 ft
Highest elevation	24.37 ft
Elevation difference	2.10 ft
Average slope	2.45% (1.40°)
Maximum slope	17.17% (9.74°)
Steepest 50 ft run	2.3% (1.3°)
FEMA zone	AE
BFE	28.0 ft (indicative)
Minimum finished-floor elevation	29.0 ft at +1 ft / 30.0 ft at +2 ft freeboard COUNTY FIGURE UNCONFIRMED
Fill or elevation required?	Yes — lowest ground is 5.7 ft below BFE
Sewer	NEEDS CONFIRMATION
Water	NEEDS CONFIRMATION
Electricity	NEEDS CONFIRMATION
Gas	NEEDS CONFIRMATION

... continued

Parameter	Finding
Internet	NEEDS CONFIRMATION
Septic required?	NEEDS CONFIRMATION — soil absorption rated Very limited
Soil	Pledger — Moderately well drained, hydrologic group D, Very high shrink-swell
HOA	Platted subdivision (Bar X Ranch), so covenants are likely. NEEDS CONFIRMATION
HOA fee	NEEDS CONFIRMATION
Zoning	NEEDS CONFIRMATION
Dock possible?	NEEDS CONFIRMATION
Lake depth	NEEDS CONFIRMATION
Boating	NEEDS CONFIRMATION
Fishing	NEEDS CONFIRMATION
Hobby farming	NEEDS CONFIRMATION
Second in-law suite	NEEDS CONFIRMATION
Short-term rental	NEEDS CONFIRMATION
Nearest hospital	NEEDS CONFIRMATION
Nearest international airport	NEEDS CONFIRMATION
Distance to nearest mapped water	1,066 ft to unnamed waterbody
Mean annual temperature	21.8 °C
Mean annual rainfall	1,191 mm (46.9 in)
Appraised market value	$21,280

Private Road 52415, Leesburg, TX 75451

Lot 6 of 12 | ID confidence: medium-high | Paradise Shores | Camp County

Identification caveat
Exactly one parcel in the county roll carries situs 'PR 52415': Lot 05, Paradise Shores, 5.77 acres. The listing gives no lot number, so confirm the acreage against the listing.

A1 Property identification & record data OFFICIAL

Situs address of record	PR 52415 ,
Legal description	LOT 05 PARADISE SHORES LLC
Subdivision	Paradise Shores
County	Camp County (FIPS 48063)
Parcel ID / account	9932303
Geographic ID (APN)	21000-07000-00004-000005
Owner of record	YU YANG PING
Acreage — appraisal roll (legal)	5.75 ac
Acreage — GIS polygon	5.771 ac
Acreage — recomputed from boundary	5.788 ac
Appraised land value	$80,444
Appraised improvement value	$0
Appraised market value	$80,444
Centroid (WGS84)	33.023559, -95.114320
Longest axis / width	959 ft × 526 ft (axis bearing 300°)

Source: Texas StratMap statewide land parcels (county appraisal roll). Matched by manual pin / legal-desc match. Appraisal-roll acreage and a GIS polygon rarely agree exactly; neither replaces a survey.

A2 · Listing and plat material not retrievable here NEEDS CONFIRMATION

Current asking price, days on market and listing history.
Obtain from: Listing agent / MLS

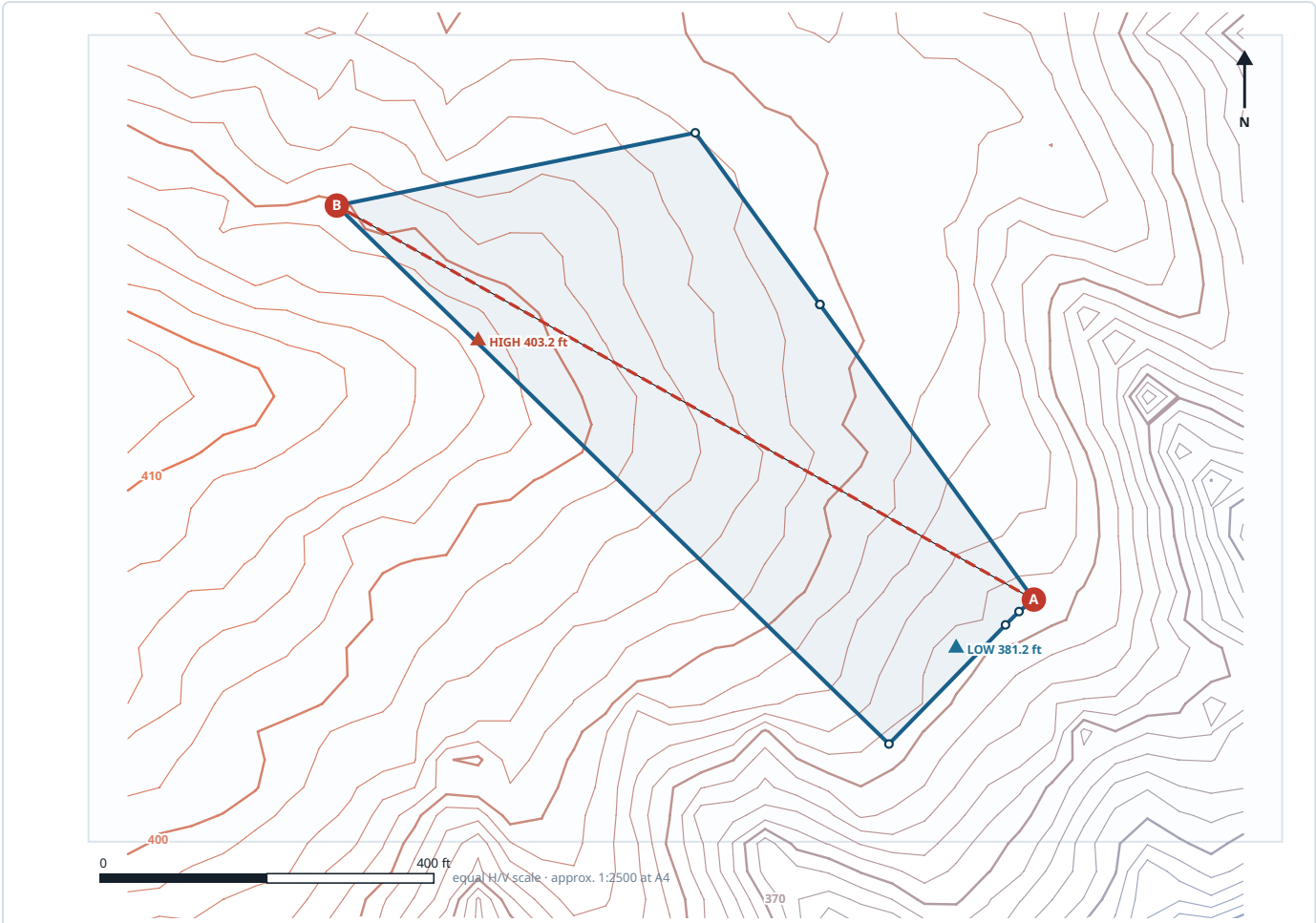
Listing photographs: lot frontage, shoreline, road access and surroundings.
Obtain from: Listing agent (Zillow blocks automated retrieval, so no listing imagery is reproduced here)

Seller disclosure notice and any known defects.
Obtain from: Listing agent

Recorded plat and Esri/county GIS parcel exhibit for comparison with this report.
Obtain from: County Clerk plat records and the county GIS portal

B5 Terrain, contours and true-scale site plan MODEL-DERIVED

LOWEST POINT	HIGHEST POINT	RELIEF	MEAN ELEVATION
381.2 ft	403.2 ft	22.0 ft	392.9 ft



True-scale site plan. Parcel boundary from the county GIS polygon; contours at 2.0 ft intervals derived from the USGS 3DEP 1 m bare-earth DEM. A–B is the cross-section line. Horizontal and vertical map scales are equal, so shapes and distances are undistorted.

MEAN GRADE	MEDIAN GRADE	90TH PCT GRADE	MAX GRADE
4.96 %	4.15 %	9.25 %	17.90 %

Mean slope angle	2.84°	**90th pct angle**	5.29°	**Max angle**	10.15°	**Grid cells**	1,156

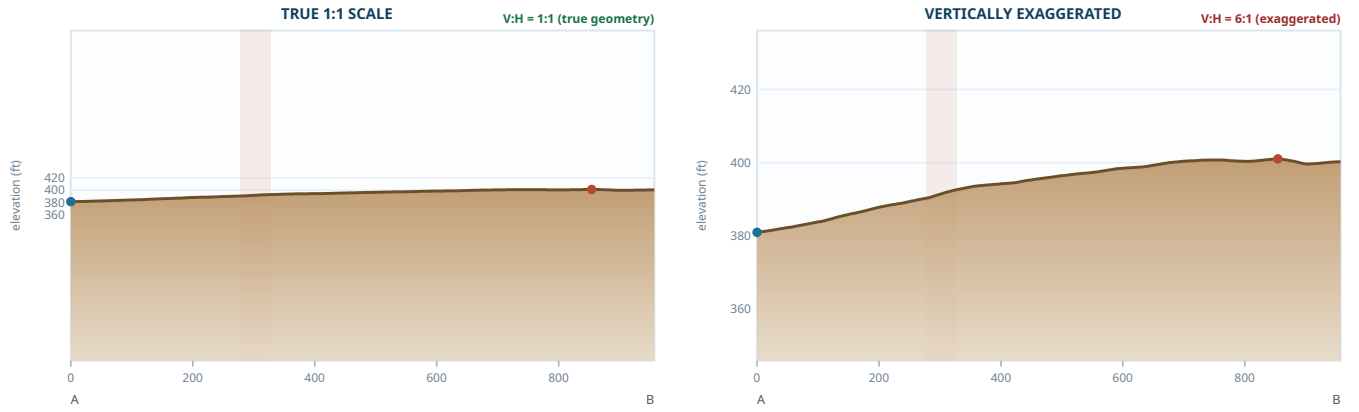
B7 · Terrain classification

Gentle slope. Standard residential grading is normally sufficient.

Basis: mean DEM grade 4.96% (90th pct 9.25%), relief 21.98 ft

Grading expectation: Minimal grading; Routine cut-and-fill

B6 Longitudinal cross-section MODEL-DERIVED



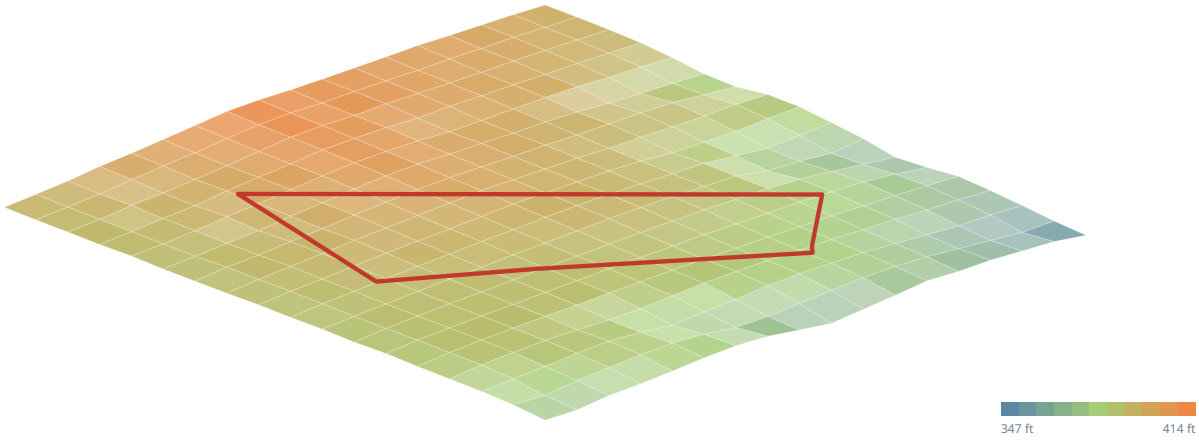
Run A–B	957 ft	**Lowest**	381.0 ft	**Highest**	401.0 ft
Elevation difference	20.0 ft	**Average grade**	2.1%	**Direct rising angle**	1.2°
Steepest 50 ft					**4.8%** (2.7°) over a 50 ft run

Reading: Over 957 ft the ground rises only 20.0 ft, a direct angle of 1.2°. At true 1:1 scale the profile is almost a straight line, which is the correct reading: this lot is effectively flat. The exaggerated panel is the same data and should not be read as slope.

Left panel is the real geometry at 1:1 vertical-to-horizontal scale. The right panel exaggerates the vertical axis, which is how slope is conventionally drawn on maps but overstates steepness. The shaded band marks the steepest 50 ft run. Profile sampled from the 3DEP DEM along the lot's longest axis.

B8 Three-dimensional terrain MODEL-DERIVED

Sampled window relief 66.4 ft · vertical scale exaggerated for legibility



Isometric view of the sampled DEM window; the red outline is the parcel boundary draped on the surface. Vertical scale is exaggerated to reveal form — use the 1:1 cross-section above for true steepness.

C9 FEMA flood-zone determination OFFICIAL

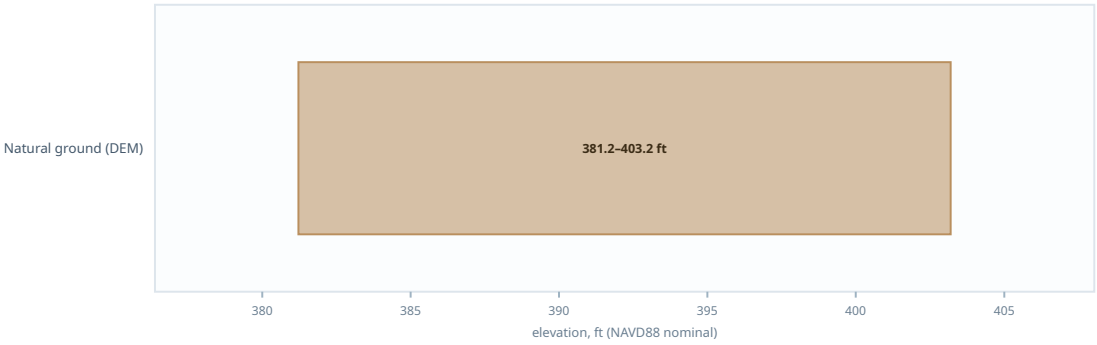
Zone: no polygon on parcel | Not in a mapped SFHA

Zones intersecting the parcel	no polygon on parcel
Special Flood Hazard Area	Not per the national layer
FIRM panel	48449C0350D · effective 2010-09-29
Mapping status	likely Zone X (unshaded) - area is mapped

No FEMA flood-hazard polygon intersects this parcel, but mapped flood zones exist within 1 mile(s), so the area carries effective NFHL coverage. This is consistent with unshaded Zone X - outside the 0.2%-annual-chance floodplain. Confirm against the effective FIRM panel.

C11-13 Ground elevation, BFE and minimum finished floor OFFICIAL MODEL-DERIVED

Lowest natural ground (DEM)	381.22 ft
Highest natural ground (DEM)	403.20 ft
Mean ground	392.90 ft
Indicative BFE	not published for this location



Freeboard scenarios. The brown band is the measured spread of natural ground across the lot. Red ticks are the minimum finished-floor elevation at BFE plus 0, 1, 2 and 3 ft of freeboard. **The freeboard the county actually enforces is not established in this report** — the scenarios show the consequence of each, they do not assert a requirement.

C · Flood items still to confirm NEEDS CONFIRMATION

Official flood determination and the effective FIRM panel read at the parcel.
Obtain from: County floodplain administrator or a flood certification service

Enforced freeboard above BFE, fill rules and compensatory-storage requirements.
Obtain from: County floodplain administrator

First Street / third-party forward-looking flood risk and estimated flood depth.
Obtain from: First Street Foundation (riskfactor.com) for the confirmed address

Flood insurance quote under current NFIP pricing.
Obtain from: Insurance agent, using an Elevation Certificate

Site drainage: swales, culverts, outfall and upstream runoff from neighbouring lots.
Obtain from: Civil engineer site visit; county drainage criteria

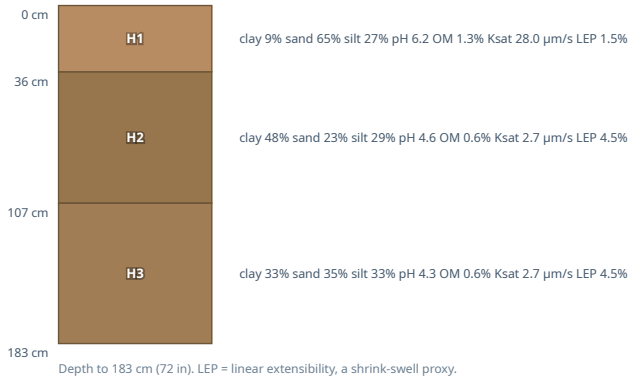
D14 Soil characteristics MODEL-DERIVED

NRCS map unit intersecting the lot	Share of lot (sampled)
Kirvin very fine sandy loam, 3 to 8 percent slopes	85.7%
Sacul fine sandy loam, 5 to 15 percent slopes	14.3%

Component	%	Drainage	Hyd.grp	LEP %	Shrink-swell	pH	Clay %	Flooding
Kirvin	85%	Well drained	C	4.5	Moderate	6.2	9	None
Sacul	90%	Moderately well drained	D	7.5	High	5.3	13	None

LEP = linear extensibility (shrink-swell proxy). Hydrologic group D indicates the slowest infiltration. Values are NRCS representative values for the map unit, not measurements on this lot.

Kirvin — representative profile



Representative horizon sequence for the dominant component (Kirvin).

D15 Foundation suitability MODEL-DERIVED

Dominant soil shrink-swell potential is **Moderate** (peak LEP 4.5%). Stiffened/post-tensioned slab or moisture-conditioned subgrade is typical.

Slab-on-grade	Stiffened / post-tensioned
Crawlspace	Suitable
Pier and beam	Suitable
Basement	Not advised
Stepped footing	Only if sloped
NRCS rating, dwellings without basements	—
Geotechnical investigation advised	Yes

Foundation selection is an engineering decision. NRCS map-unit shrink-swell indicates risk; it does not substitute for a geotechnical investigation.

D16 Septic / on-site sewage suitability MODEL-DERIVED

Septic Tank Absorption Fields	Very limited
Septic Tank Leaching Chamber (TX)	Very limited
Septic Tank, Subsurface Drip Irrigation (TX)	Somewhat limited

Constraints identified from the soil and terrain data

- Hydrologic group C indicates slow infiltration, which constrains conventional drainfields.
- Bedrock within 107.0 cm limits trench depth.

Permitted bedroom count: NOT ESTABLISHED
Permitted bedroom count follows from a site-specific soil evaluation (TCEQ Ch. 285 class, loading rate, usable drainfield area and setbacks). It cannot be derived from map-unit data. A licensed site evaluator must run the percolation/soil profile test.

NRCS ratings describe the map unit, not the lot. A TCEQ-compliant site evaluation and percolation test is required for any permit.

D17 Gardening suitability MODEL-DERIVED

SURFACE PH	CLAY	SAND	ORGANIC MATTER
6.2	9 %	65 %	1.3 %

Drainage class	Well drained
pH range through profile	4.3 – 6.2
NRCS lawn / landscape rating	Somewhat limited
Raised beds advised	Not necessary

Surface horizon is workable; in-ground beds are reasonable with normal organic amendment.

Likely amendments

- Compost / organic matter, surface organic matter is low
- Mulch, to buffer soil temperature and moisture

Based on NRCS map-unit averages. A lab soil test is required before committing to any amendment programme.

D · Soil and planting items to confirm NEEDS CONFIRMATION

Laboratory soil test: pH, macro and micronutrients, organic matter, texture.
Obtain from: Texas A&M AgriLife Soil Testing Lab, from samples taken on the lot

Plant hardiness zone confirmation and a species list matched to the tested soil.
Obtain from: USDA hardiness map plus the county AgriLife extension office

E19-23 Utilities and infrastructure NEEDS CONFIRMATION

Utility availability is not published as open geospatial data at parcel level. Nothing is inferred here: each item below must come from the serving utility in writing, because a missing water tap or a long electric extension can change the build budget materially.

Water supply: municipal, water-supply-corporation, or private well; is a tap at the road?
Obtain from: Serving water supply corporation / county; well logs via TX Water Development Board

Sewer: is public sewer available, or is an on-site facility required? Distance to any main.
Obtain from: County / city utility department or MUD

Electricity: provider, whether service exists at the lot line, overhead vs underground, connection cost.
Obtain from: Retail electric provider / local co-op (service availability letter)

Natural gas or propane availability and existing infrastructure.
Obtain from: Gas utility or local propane suppliers

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Internet: fibre, cable, fixed wireless, DSL or satellite, with advertised speeds.
Obtain from: FCC National Broadband Map plus a direct serviceability check by address

F24-25 Waterfront position MODEL-DERIVED

Near water - frontage unlikely; access likely via easement or common area | Monticello Reservoir

Nearest mapped waterbody	Monticello Reservoir
Distance from parcel boundary	652 ft
Waterbody area	42.928 km²

NHD polygons are hydrographic mapping, not a boundary survey. Whether the lot carries riparian frontage, deeded access or only common-area access is a title/plat question and is NOT established here.

F26-29 · Dock, depth, boating and fishing NEEDS CONFIRMATION

Whether a dock is permittable at this shoreline, and the governing authority.
Obtain from: Controlling lake / river authority, plus the HOA

Dock size limits, shoreline setbacks, shared-dock rules and permit cost.
Obtain from: Lake authority rules and permit schedule

Whether riparian or shoreline rights pass with the lot, or the shoreline is a common reserve.
Obtain from: Title commitment and recorded plat

Lake water-level regime and any shoreline-management plan.
Obtain from: Lake/river authority or reservoir operator

Bathymetry: average and maximum depth, and depth near this shoreline.
Obtain from: Lake authority survey, TPWD reservoir report, or a sonar survey

Seasonal water-level variation and drawdown history.
Obtain from: Reservoir operator gauge records / USGS gauges

Boating rules: permitted craft, motor limits, wake rules, ramps and marinas.
Obtain from: Lake authority and TPWD

Fishery: species present, stocking, licence requirements and catch limits.
Obtain from: Texas Parks & Wildlife reservoir survey reports

G30-33 HOA, covenants, zoning and a second dwelling NEEDS CONFIRMATION

The parcel sits in **Paradise Shores** according to the recorded legal description, so a declaration of covenants very likely exists. The documents themselves are not public geospatial data and were not retrieved, so no dues figure, restriction or amenity is asserted here.

Whether an HOA or POA exists and is mandatory, with its name and management contact.
Obtain from: Title commitment, recorded Declaration, seller disclosure

Zoning designation, or confirmation that the county is unzoned.
Obtain from: County or city development services

Minimum lot size, building coverage, setbacks, height limits, accessory-structure rules.
Obtain from: County/city development standards or subdivision regulations

Whether a second independent dwelling or in-law suite (ADU) is permitted, and on what terms.
Obtain from: County/city development services plus the HOA

Permitting path, driveway/culvert permit and any development fee.
Obtain from: County permitting office

... continued

Annual dues, monthly equivalent, transfer fee, and any special assessment.
Obtain from: HOA or management company estoppel / resale certificate

Declaration, CC&Rs, architectural guidelines, rules and all amendments.
Obtain from: HOA or title company

Restrictions on docks, fences, outbuildings, RVs, boats, livestock and poultry.
Obtain from: Recorded CC&Rs and architectural guidelines

Minimum dwelling size and any construction time limit after purchase.
Obtain from: Recorded CC&Rs

Amenities actually included: clubhouse, pool, courts, parks, lake and boat facilities.
Obtain from: HOA documents and budget

H34 / I35 Hobby farming and short-term rental NEEDS CONFIRMATION

Whether chickens, ducks, rabbits, goats or bees are allowed, and in what numbers.
Obtain from: County rules plus the recorded CC&Rs (either can prohibit)

Any agricultural or wildlife valuation currently on the parcel, and rollback-tax exposure if use changes.
Obtain from: County appraisal district

Whether short-term rental is permitted at this location.
Obtain from: County/city ordinance and the HOA

Any STR permit, occupancy cap, parking or noise condition.
Obtain from: Local ordinance

Hotel occupancy tax registration and remittance obligations.
Obtain from: Texas Comptroller plus local taxing authority

J36-42 Environmental and geological due diligence MODEL-DERIVED

Ground-penetrating radar penetration (NRCS rating)	Low penetration Indicates how useful GPR would be here; clay-rich soils attenuate the signal.
Sinkhole / karst exposure	Not indicated by the soil map units returned for this parcel. No restrictive bedrock or karst feature was reported. CONFIRM WITH TX BUREAU OF ECONOMIC GEOLOGY
Radon	EPA publishes radon zones by county, not by parcel. LOOK UP THE COUNTY ZONE AND TEST ANY BUILT STRUCTURE

Existing FEMA Elevation Certificate for the lot.
Obtain from: Seller, county floodplain office or a surveyor

Recent boundary, ALTA or topographic survey.
Obtain from: Seller or a licensed Texas surveyor (RPLS)

Geotechnical / soil boring report.
Obtain from: Seller or a geotechnical engineer

On-site sewage facility site evaluation and percolation test.
Obtain from: Licensed site evaluator / county OSSF office

Any ground-penetrating-radar or other subsurface investigation.
Obtain from: Seller; commission if subsurface features are suspected

K43-45 Residential construction feasibility MODEL-DERIVED

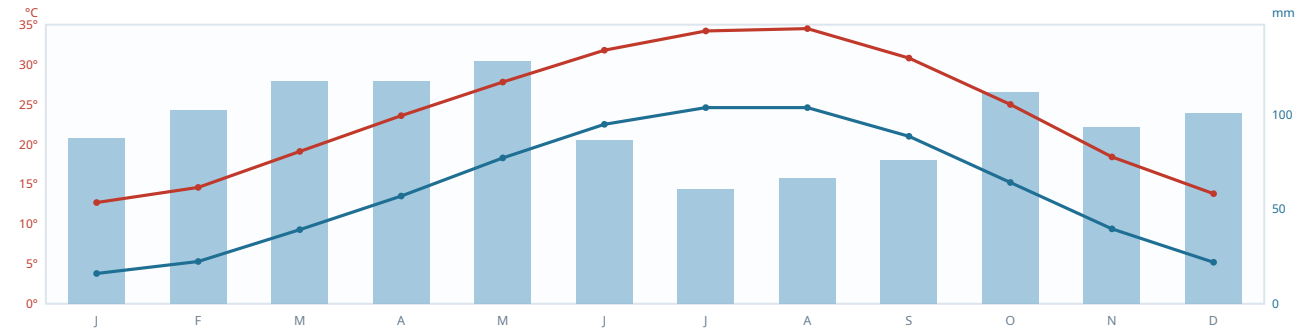
Lot area available	5.771 ac (251,373 sq ft)
Usable building envelope	NOT COMPUTABLE Setbacks, easements and any lake or drainage reserve come from the recorded plat and the HOA architectural guidelines, none of which were available.
Terrain constraint	Gentle slope — Minimal grading; Routine cut-and-fill
Foundation implication	Moderate shrink-swell. Stiffened/post-tensioned slab or moisture-conditioned subgrade is typical.
Flood implication	No mapped SFHA on the parcel per the national layer; confirm on the effective FIRM.

Construction cost
No build cost is estimated in this report. A defensible figure for a four-bedroom house depends on the HOA's minimum size and material rules, the finished-floor elevation forced by the flood zone, the foundation type the soil requires, and whether water and sewer must be created on site. Three of those four inputs are still unconfirmed for this lot, so any number quoted now would be arbitrary. Price it from a builder bid once the plat, CC&Rs, floodplain requirement and soil report are in hand.

L46 Climate MODEL-DERIVED

MEAN ANNUAL TEMP	MEAN SUMMER HIGH	MEAN WINTER LOW	ANNUAL RAINFALL
19.2 °C	33.5 °C	4.7 °C	1,147 mm

Record high	43.1 °C	**Record low**	-17.3 °C	**Rainfall**	45.2 in
Growing season	~285 days	**Days at or below 0 °C**	19.4 /yr	**Period**	1994-2023 (30 years)



Monthly normals, 1994-2023 (30 years). Bars are mean monthly rainfall (mm, right axis); lines are mean daily maximum and minimum temperature (°C, left axis). Source: ERA5 daily reanalysis via Open-Meteo archive API.

L47 Natural hazards

Flood	No mapped SFHA on the parcel
Other perils	Hurricane and tropical wind, tornado, hail, severe convective storm, lightning, extreme heat, drought and wildfire all vary by location within Texas. QUANTIFY FROM FEMA NATIONAL RISK INDEX FOR THE CONFIRMED COUNTY AND FROM AN INSURANCE QUOTE

L48-55 Community, schools, healthcare, shopping and access NEEDS CONFIRMATION

Demographics not retrieved
Census API requires an API key (keyless request rejected) Free key from https://api.census.gov/data/key_signup.html, then set CENSUS_API_KEY and re-run 05_context.py. Or read the figures directly from data.census.gov.

Driving distance and time to the nearest hospital, emergency department and urgent care.
Obtain from: Direct routing check from the confirmed parcel location

Distance to grocery, major retail, pharmacy, bank, fuel and hardware.
Obtain from: Direct routing check

Assigned school district and campuses, with distances and current ratings.
Obtain from: TEA school-district locator and the district attendance-zone map

Nearest Indian / South Asian grocery, restaurants and places of worship.
Obtain from: Direct search from the confirmed parcel location

Nearest international airport, driving time and route.
Obtain from: Direct routing check

Walk / Bike / Transit scores.
Obtain from: Walk Score for the confirmed address

M56 Documented strengths

- Lot of 5.77 acres, which is generous for a single dwelling and leaves room for outbuildings and a garden.
- Close to Monticello Reservoir at about 652 ft, so lake amenity is nearby without shoreline exposure.
- No Special Flood Hazard Area maps onto the parcel, which avoids mandatory flood insurance and floodplain construction rules.

M57 Limitations and risks

- Parcel identification is probable but not certain: Exactly one parcel in the county roll carries situs 'PR 52415': Lot 05, Paradise Shores, 5.77 acres. The listing gives no lot number, so confirm the acreage against the listing.
- Septic absorption is rated "very limited" for the dominant soil, so a conventional drainfield may not be permittable and an engineered or aerobic system may be required at higher cost.
- HOA covenants, zoning, utility availability, dock rights and short-term-rental rules are all unconfirmed. Any of them can independently prevent the intended use.

M59 Construction feasibility conclusion

Main house	Likely The lot is large enough and the terrain is workable. Finished-floor elevation and foundation type are the governing variables, not land area.
Second in-law suite / ADU	Unresolved Depends entirely on county rules and the HOA declaration; neither was available. Do not assume a second dwelling is permitted.
Garage	Likely No physical constraint identified, subject to setbacks and architectural guidelines.
Septic system	Unresolved Soil ratings are unfavourable; permittability and bedroom count require a site evaluation.
Garden	Likely Native surface soil is workable with normal organic amendment.
Hobby farming / poultry	Unresolved Governed by county rules and CC&Rs; unconfirmed.
Dock	Unresolved The lot does not adjoin mapped water, so a private dock is unlikely; shared access may exist.
Short-term rental	Unresolved Governed by local ordinance and CC&Rs; unconfirmed.
Pool / outdoor recreation	Conditional Space is available; flood-zone and soil-movement considerations affect pool design where applicable.

M60 Key measurements summary

Parameter	Finding
Lot size	5.771 ac (251,373 sq ft)
Lot dimensions (long axis × width)	959 ft × 526 ft
Lowest elevation	381.22 ft
Highest elevation	403.20 ft
Elevation difference	21.98 ft
Average slope	4.96% (2.84°)
Maximum slope	17.90% (10.15°)
Steepest 50 ft run	4.8% (2.7°)
FEMA zone	likely Zone X (unshaded) - area is mapped
BFE	not published
Minimum finished-floor elevation	NEEDS CONFIRMATION
Fill or elevation required?	Not indicated
Sewer	NEEDS CONFIRMATION
Water	NEEDS CONFIRMATION
Electricity	NEEDS CONFIRMATION
Gas	NEEDS CONFIRMATION
Internet	NEEDS CONFIRMATION

... continued

Parameter	Finding
Septic required?	NEEDS CONFIRMATION — soil absorption rated Very limited
Soil	Kirvin — Well drained, hydrologic group C, Moderate shrink-swell
HOA	Platted subdivision (Paradise Shores), so covenants are likely. NEEDS CONFIRMATION
HOA fee	NEEDS CONFIRMATION
Zoning	NEEDS CONFIRMATION
Dock possible?	NEEDS CONFIRMATION
Lake depth	NEEDS CONFIRMATION
Boating	NEEDS CONFIRMATION
Fishing	NEEDS CONFIRMATION
Hobby farming	NEEDS CONFIRMATION
Second in-law suite	NEEDS CONFIRMATION
Short-term rental	NEEDS CONFIRMATION
Nearest hospital	NEEDS CONFIRMATION
Nearest international airport	NEEDS CONFIRMATION
Distance to nearest mapped water	652 ft to Monticello Reservoir
Mean annual temperature	19.2 °C
Mean annual rainfall	1,147 mm (45.2 in)
Appraised market value	$80,444

119 Dolly Rd, Sulphur Springs, TX 75482

Lot 7 of 12 | Parcel not identified

This lot could not be located in the statewide parcel record
No parcel in the Texas StratMap statewide layer could be matched to this address. For this county the layer carries no situs address values, so address matching is impossible. The parcel must be identified from the county appraisal district record or the recorded plat before any site-specific analysis is meaningful.

Because the parcel boundary is unknown, no flood zone, elevation, slope or soil result can be attributed to this lot. Everything downstream of parcel identification is therefore withheld rather than estimated. Resolve identification first:

- Ask the listing agent for the county parcel ID / account number.
- Search the county appraisal district by owner or address.
- Obtain the recorded plat and the legal description from the title company.

Required before any analysis NEEDS CONFIRMATION

Parcel ID / account number and legal description for this address.
Obtain from: Listing agent or the county appraisal district

Recorded plat showing the lot within its subdivision.
Obtain from: County Clerk plat records

15 Harrell Cemetery Rd, Coldspring, TX 77331

Lot 8 of 12 | ID confidence: high | Creek Wood Bend | San Jacinto County

A1 Property identification & record data OFFICIAL

Situs address of record	15 Harrell Cemetery Rd, Coldspring, TX 77331
Legal description	Creek Wood Bend, Lot 15, Acres 1.399
Subdivision	Creek Wood Bend
County	San Jacinto County (FIPS 48407)
Parcel ID / account	96750
Geographic ID (APN)	1680-000-0150
Owner of record	Rutgers Adam Timothy & Jenkins Rose
Acreage — appraisal roll (legal)	1.40 ac
Acreage — GIS polygon	1.352 ac
Acreage — recomputed from boundary	1.357 ac
Appraised land value	$82,110
Appraised improvement value	$0
Appraised market value	$82,110
Centroid (WGS84)	30.702949, -95.176845
Longest axis / width	475 ft × 250 ft (axis bearing 335°)

Source: Texas StratMap statewide land parcels (county appraisal roll). Matched by situs address match. Appraisal-roll acreage and a GIS polygon rarely agree exactly; neither replaces a survey.

A2 · Listing and plat material not retrievable here NEEDS CONFIRMATION

Current asking price, days on market and listing history.
Obtain from: Listing agent / MLS

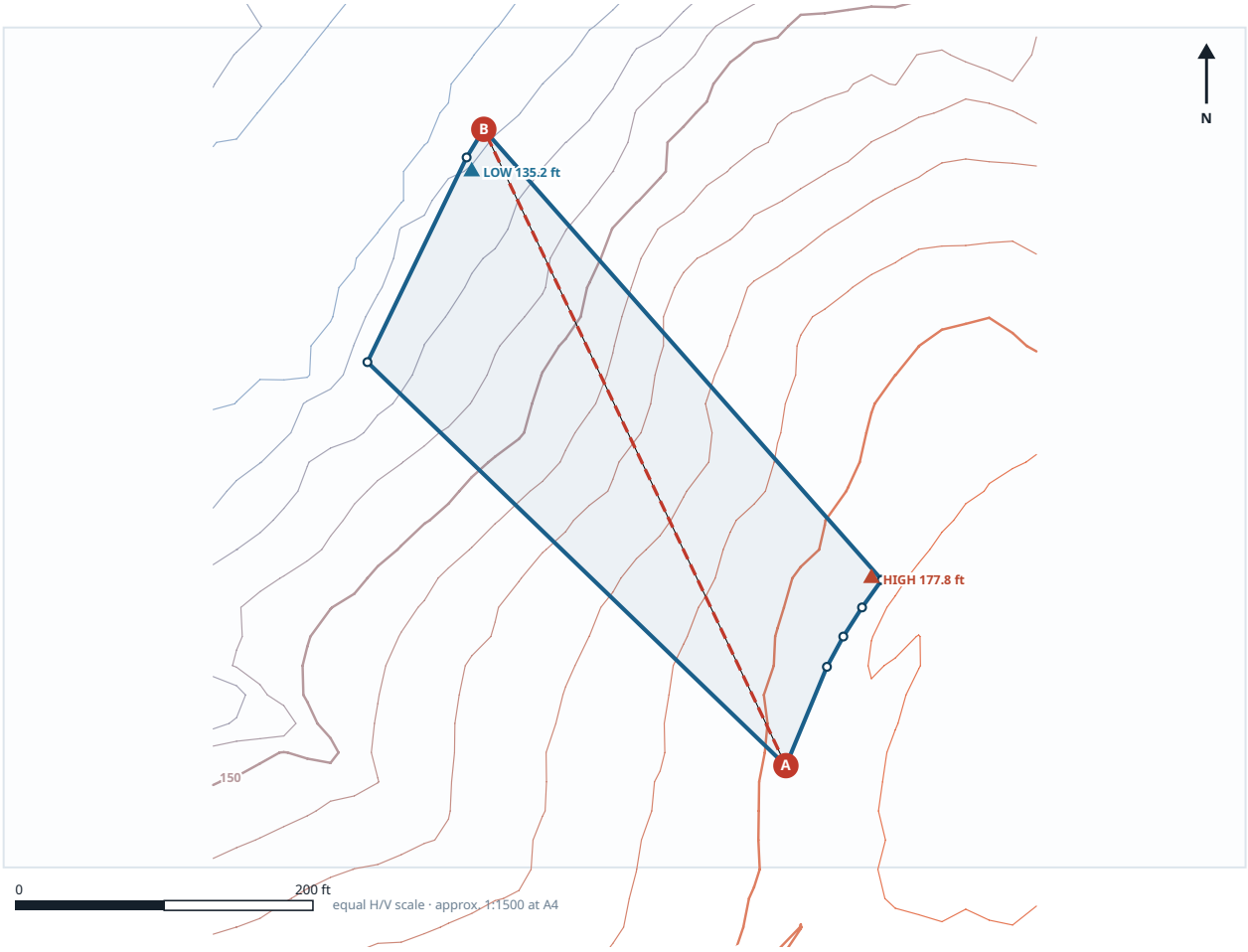
Listing photographs: lot frontage, shoreline, road access and surroundings.
Obtain from: Listing agent (Zillow blocks automated retrieval, so no listing imagery is reproduced here)

Seller disclosure notice and any known defects.
Obtain from: Listing agent

Recorded plat and Esri/county GIS parcel exhibit for comparison with this report.
Obtain from: County Clerk plat records and the county GIS portal

B5 Terrain, contours and true-scale site plan MODEL-DERIVED

LOWEST POINT	HIGHEST POINT	RELIEF	MEAN ELEVATION
135.2 ft	177.8 ft	42.6 ft	158.2 ft



True-scale site plan. Parcel boundary from the county GIS polygon; contours at 5.0 ft intervals derived from the USGS 3DEP 1 m bare-earth DEM. A–B is the cross-section line. Horizontal and vertical map scales are equal, so shapes and distances are undistorted.

MEAN GRADE	MEDIAN GRADE	90TH PCT GRADE	MAX GRADE
9.94 %	9.66 %	16.88 %	35.02 %

Mean slope angle	5.68°	**90th pct angle**	9.58°	**Max angle**	19.30°	**Grid cells**	1,156

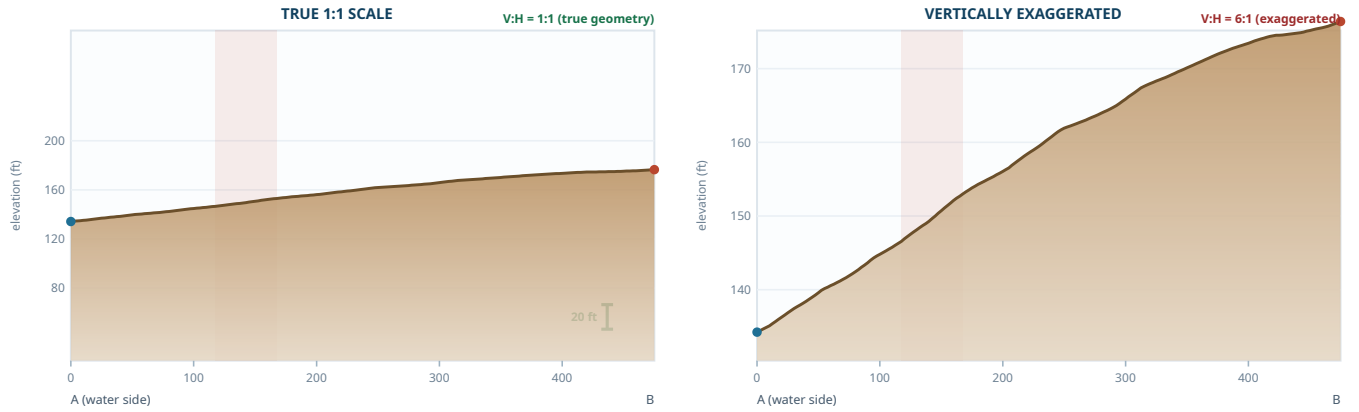
B7 · Terrain classification

Moderate slope. Expect cut-and-fill balancing and possibly a stepped footing.

Basis: mean DEM grade 9.94% (90th pct 16.88%), relief 42.61 ft

Grading expectation: Significant grading; Cut-and-fill

B6 Longitudinal cross-section MODEL-DERIVED

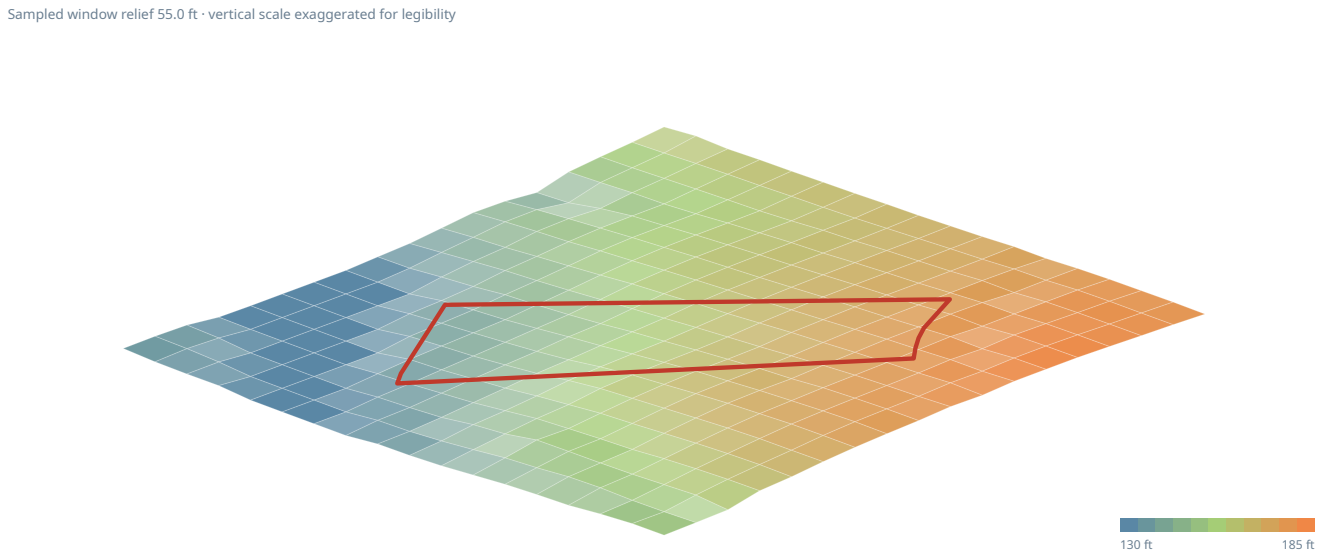


Run A–B	475 ft	**Lowest**	134.3 ft	**Highest**	176.4 ft
Elevation difference	42.1 ft	**Average grade**	8.9%	**Direct rising angle**	5.1°
Steepest 50 ft	**12.8%** (7.3°) over a 50 ft run				

Reading: The ground rises 42.1 ft over 475 ft, a direct angle of 5.1° (8.9% average grade). That is a gentle, buildable fall.

Left panel is the real geometry at 1:1 vertical-to-horizontal scale. The right panel exaggerates the vertical axis, which is how slope is conventionally drawn on maps but overstates steepness. The shaded band marks the steepest 50 ft run. Profile sampled from the 3DEP DEM along the lot's longest axis.

B8 Three-dimensional terrain MODEL-DERIVED



Isometric view of the sampled DEM window; the red outline is the parcel boundary draped on the surface. Vertical scale is exaggerated to reveal form — use the 1:1 cross-section above for true steepness.

C9 FEMA flood-zone determination OFFICIAL

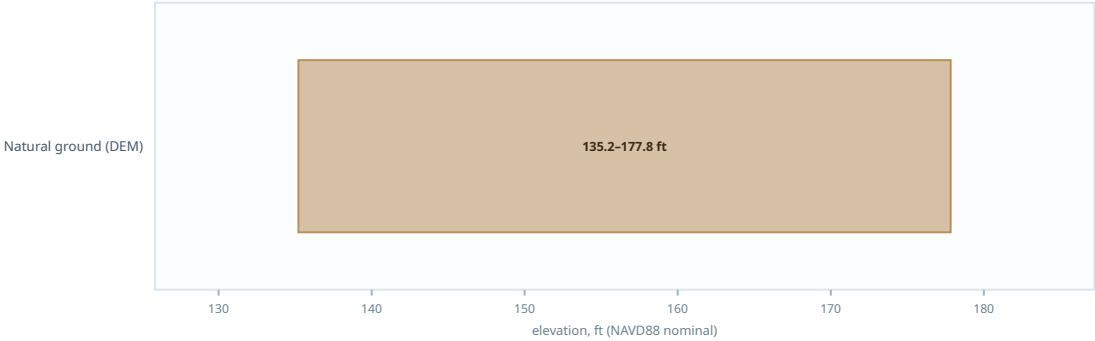
Zone: A + X | Inside Special Flood Hazard Area | Split zone

Zones intersecting the parcel	A + X
Special Flood Hazard Area	Yes — mandatory flood insurance for a federally backed mortgage
FIRM panel	48407C0150C · effective 2010-11-04
Mapping status	—

Parcel intersects mapped FEMA flood-hazard polygon(s); see zones above.

C11-13 Ground elevation, BFE and minimum finished floor OFFICIAL MODEL-DERIVED

Lowest natural ground (DEM)	135.22 ft
Highest natural ground (DEM)	177.84 ft
Mean ground	158.16 ft
Indicative BFE	not published for this location



Freeboard scenarios. The brown band is the measured spread of natural ground across the lot. Red ticks are the minimum finished-floor elevation at BFE plus 0, 1, 2 and 3 ft of freeboard. **The freeboard the county actually enforces is not established in this report** — the scenarios show the consequence of each, they do not assert a requirement.

C · Flood items still to confirm NEEDS CONFIRMATION

Official flood determination and the effective FIRM panel read at the parcel.
Obtain from: County floodplain administrator or a flood certification service

Enforced freeboard above BFE, fill rules and compensatory-storage requirements.
Obtain from: County floodplain administrator

First Street / third-party forward-looking flood risk and estimated flood depth.
Obtain from: First Street Foundation (riskfactor.com) for the confirmed address

Flood insurance quote under current NFIP pricing.
Obtain from: Insurance agent, using an Elevation Certificate

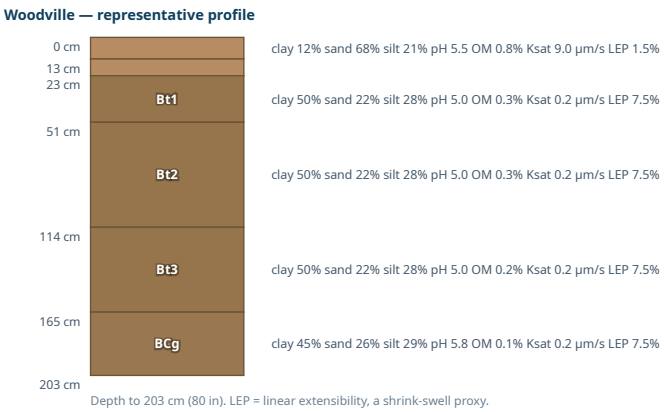
Site drainage: swales, culverts, outfall and upstream runoff from neighbouring lots.
Obtain from: Civil engineer site visit; county drainage criteria

D14 Soil characteristics MODEL-DERIVED

NRCS map unit intersecting the lot	Share of lot (sampled)
Woodville fine sandy loam, 5 to 12 percent slopes	100.0%

Component	%	Drainage	Hyd.grp	LEP %	Shrink-swell	pH	Clay %	Flooding
Woodville	90%	Somewhat poorly drained	D	7.5	High	5.5	12	None

LEP = linear extensibility (shrink-swell proxy). Hydrologic group D indicates the slowest infiltration. Values are NRCS representative values for the map unit, not measurements on this lot.



Representative horizon sequence for the dominant component (Woodville).

D15 Foundation suitability MODEL-DERIVED

Dominant soil shrink-swell potential is **High** (peak LEP 7.5%). Engineered foundation expected: post-tensioned slab or pier-and-beam.

Slab-on-grade	Engineered post-tensioned only
Crawlspace	Suitable with piers
Pier and beam	Preferred
Basement	Not advised
Stepped footing	Only if sloped
NRCS rating, dwellings without basements	—
Geotechnical investigation advised	Yes

Foundation selection is an engineering decision. NRCS map-unit shrink-swell indicates risk; it does not substitute for a geotechnical investigation.

D16 Septic / on-site sewage suitability MODEL-DERIVED

Septic Tank Absorption Fields	Very limited
Septic Tank Leaching Chamber (TX)	Very limited
Septic Tank, Subsurface Drip Irrigation (TX)	Very limited

Constraints identified from the soil and terrain data

- Hydrologic group D indicates slow infiltration, which constrains conventional drainfields.
- Seasonal water table within 51.0 cm of the surface.
- 90th-percentile grade of 16.88% complicates drainfield layout.
- Mapped water about 24.5 ft away; TCEQ/county setbacks from water will restrict drainfield placement.

Permitted bedroom count: NOT ESTABLISHED
Permitted bedroom count follows from a site-specific soil evaluation (TCEQ Ch. 285 class, loading rate, usable drainfield area and setbacks). It cannot be derived from map-unit data. A licensed site evaluator must run the percolation/soil profile test.

NRCS ratings describe the map unit, not the lot. A TCEQ-compliant site evaluation and percolation test is required for any permit.

D17 Gardening suitability MODEL-DERIVED

SURFACE PH	CLAY	SAND	ORGANIC MATTER
5.5	12 %	68 %	0.8 %

Drainage class	Somewhat poorly drained
pH range through profile	5.0 – 5.8
NRCS lawn / landscape rating	Somewhat limited
Raised beds advised	Yes

Heavy clay / restricted drainage / high shrink-swell in the surface horizon makes raised beds with imported loam the lower-risk option.

Likely amendments

- Lime, to raise pH toward 6.2-6.8 for most vegetables
- Compost / organic matter, surface organic matter is low
- Mulch, to buffer soil temperature and moisture

Based on NRCS map-unit averages. A lab soil test is required before committing to any amendment programme.

D · Soil and planting items to confirm NEEDS CONFIRMATION

Laboratory soil test: pH, macro and micronutrients, organic matter, texture.
Obtain from: Texas A&M AgriLife Soil Testing Lab, from samples taken on the lot

Plant hardiness zone confirmation and a species list matched to the tested soil.
Obtain from: USDA hardiness map plus the county AgriLife extension office

E19-23 Utilities and infrastructure NEEDS CONFIRMATION

Utility availability is not published as open geospatial data at parcel level. Nothing is inferred here: each item below must come from the serving utility in writing, because a missing water tap or a long electric extension can change the build budget materially.

Water supply: municipal, water-supply-corporation, or private well; is a tap at the road?
Obtain from: Serving water supply corporation / county; well logs via TX Water Development Board

Sewer: is public sewer available, or is an on-site facility required? Distance to any main.
Obtain from: County / city utility department or MUD

Electricity: provider, whether service exists at the lot line, overhead vs underground, connection cost.
Obtain from: Retail electric provider / local co-op (service availability letter)

Natural gas or propane availability and existing infrastructure.
Obtain from: Gas utility or local propane suppliers

Internet: fibre, cable, fixed wireless, DSL or satellite, with advertised speeds.
Obtain from: FCC National Broadband Map plus a direct serviceability check by address

F24-25 Waterfront position MODEL-DERIVED

Abuts mapped water - consistent with direct water frontage | Lake Livingston

Nearest mapped waterbody	Lake Livingston
Distance from parcel boundary	25 ft
Waterbody area	336.746 km²

NHD polygons are hydrographic mapping, not a boundary survey. Whether the lot carries riparian frontage, deeded access or only common-area access is a title/plat question and is NOT established here.

F26-29 · Dock, depth, boating and fishing NEEDS CONFIRMATION

Whether a dock is permittable at this shoreline, and the governing authority.
Obtain from: Controlling lake / river authority, plus the HOA

Dock size limits, shoreline setbacks, shared-dock rules and permit cost.
Obtain from: Lake authority rules and permit schedule

Whether riparian or shoreline rights pass with the lot, or the shoreline is a common reserve.
Obtain from: Title commitment and recorded plat

Lake water-level regime and any shoreline-management plan.
Obtain from: Lake/river authority or reservoir operator

Bathymetry: average and maximum depth, and depth near this shoreline.
Obtain from: Lake authority survey, TPWD reservoir report, or a sonar survey

Seasonal water-level variation and drawdown history.
Obtain from: Reservoir operator gauge records / USGS gauges

Boating rules: permitted craft, motor limits, wake rules, ramps and marinas.
Obtain from: Lake authority and TPWD

Fishery: species present, stocking, licence requirements and catch limits.
Obtain from: Texas Parks & Wildlife reservoir survey reports

G30-33 HOA, covenants, zoning and a second dwelling NEEDS CONFIRMATION

The parcel sits in **Creek Wood Bend** according to the recorded legal description, so a declaration of covenants very likely exists. The documents themselves are not public geospatial data and were not retrieved, so no dues figure, restriction or amenity is asserted here.

Whether an HOA or POA exists and is mandatory, with its name and management contact.
Obtain from: Title commitment, recorded Declaration, seller disclosure

Annual dues, monthly equivalent, transfer fee, and any special assessment.
Obtain from: HOA or management company estoppel / resale certificate

Declaration, CC&Rs, architectural guidelines, rules and all amendments.
Obtain from: HOA or title company

Zoning designation, or confirmation that the county is unzoned.
Obtain from: County or city development services

Minimum lot size, building coverage, setbacks, height limits, accessory-structure rules.
Obtain from: County/city development standards or subdivision regulations

Whether a second independent dwelling or in-law suite (ADU) is permitted, and on what terms.
Obtain from: County/city development services plus the HOA

Permitting path, driveway/culvert permit and any development fee.
Obtain from: County permitting office

... continued

Restrictions on docks, fences, outbuildings, RVs, boats, livestock and poultry.
Obtain from: Recorded CC&Rs and architectural guidelines

Minimum dwelling size and any construction time limit after purchase.
Obtain from: Recorded CC&Rs

Amenities actually included: clubhouse, pool, courts, parks, lake and boat facilities.
Obtain from: HOA documents and budget

H34 / I35 Hobby farming and short-term rental NEEDS CONFIRMATION

Whether chickens, ducks, rabbits, goats or bees are allowed, and in what numbers.
Obtain from: County rules plus the recorded CC&Rs (either can prohibit)

Any agricultural or wildlife valuation currently on the parcel, and rollback-tax exposure if use changes.
Obtain from: County appraisal district

Whether short-term rental is permitted at this location.
Obtain from: County/city ordinance and the HOA

Any STR permit, occupancy cap, parking or noise condition.
Obtain from: Local ordinance

Hotel occupancy tax registration and remittance obligations.
Obtain from: Texas Comptroller plus local taxing authority

J36-42 Environmental and geological due diligence MODEL-DERIVED

Ground-penetrating radar penetration (NRCS rating)	Low penetration Indicates how useful GPR would be here; clay-rich soils attenuate the signal.
Sinkhole / karst exposure	Not indicated by the soil map units returned for this parcel. No restrictive bedrock or karst feature was reported. CONFIRM WITH TX BUREAU OF ECONOMIC GEOLOGY
Radon	EPA publishes radon zones by county, not by parcel. LOOK UP THE COUNTY ZONE AND TEST ANY BUILT STRUCTURE

Existing FEMA Elevation Certificate for the lot.
Obtain from: Seller, county floodplain office or a surveyor

Recent boundary, ALTA or topographic survey.
Obtain from: Seller or a licensed Texas surveyor (RPLS)

Geotechnical / soil boring report.
Obtain from: Seller or a geotechnical engineer

On-site sewage facility site evaluation and percolation test.
Obtain from: Licensed site evaluator / county OSSF office

Any ground-penetrating-radar or other subsurface investigation.
Obtain from: Seller; commission if subsurface features are suspected

K43-45 Residential construction feasibility MODEL-DERIVED

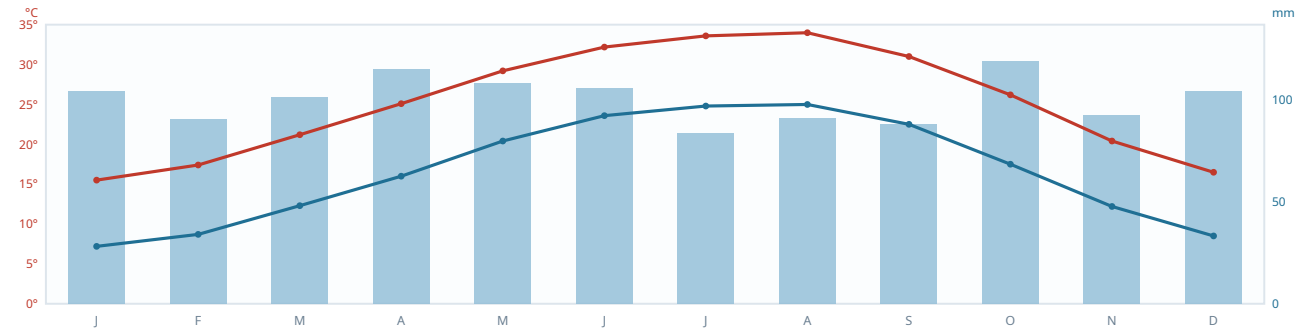
Lot area available	1.352 ac (58,898 sq ft)
Usable building envelope	NOT COMPUTABLE Setbacks, easements and any lake or drainage reserve come from the recorded plat and the HOA architectural guidelines, none of which were available.
Terrain constraint	Moderate slope — Significant grading; Cut-and-fill
Foundation implication	High shrink-swell. Engineered foundation expected: post-tensioned slab or pier-and-beam.
Flood implication	Inside a mapped SFHA, so finished-floor elevation, flood venting and insurance all bind the design.

Construction cost
No build cost is estimated in this report. A defensible figure for a four-bedroom house depends on the HOA's minimum size and material rules, the finished-floor elevation forced by the flood zone, the foundation type the soil requires, and whether water and sewer must be created on site. Three of those four inputs are still unconfirmed for this lot, so any number quoted now would be arbitrary. Price it from a builder bid once the plat, CC&Rs, floodplain requirement and soil report are in hand.

L46 Climate MODEL-DERIVED

MEAN ANNUAL TEMP	MEAN SUMMER HIGH	MEAN WINTER LOW	ANNUAL RAINFALL
20.9 °C	33.3 °C	8.1 °C	1,204 mm

Record high	42.5 °C	**Record low**	-11.1 °C	**Rainfall**	47.4 in
Growing season	~335 days	**Days at or below 0 °C**	4.1 /yr	**Period**	1994-2023 (30 years)



Monthly normals, 1994-2023 (30 years). Bars are mean monthly rainfall (mm, right axis); lines are mean daily maximum and minimum temperature (°C, left axis). Source: ERA5 daily reanalysis via Open-Meteo archive API.

L47 Natural hazards

Flood	Mapped Special Flood Hazard Area
Other perils	Hurricane and tropical wind, tornado, hail, severe convective storm, lightning, extreme heat, drought and wildfire all vary by location within Texas. QUANTIFY FROM FEMA NATIONAL RISK INDEX FOR THE CONFIRMED COUNTY AND FROM AN INSURANCE QUOTE

L48-55 Community, schools, healthcare, shopping and access NEEDS CONFIRMATION

Demographics not retrieved
Census API requires an API key (keyless request rejected) Free key from https://api.census.gov/data/key_signup.html, then set CENSUS_API_KEY and re-run 05_context.py. Or read the figures directly from data.census.gov.

Driving distance and time to the nearest hospital, emergency department and urgent care.
Obtain from: Direct routing check from the confirmed parcel location

Distance to grocery, major retail, pharmacy, bank, fuel and hardware.
Obtain from: Direct routing check

Assigned school district and campuses, with distances and current ratings.
Obtain from: TEA school-district locator and the district attendance-zone map

Nearest Indian / South Asian grocery, restaurants and places of worship.
Obtain from: Direct search from the confirmed parcel location

Nearest international airport, driving time and route.
Obtain from: Direct routing check

Walk / Bike / Transit scores.
Obtain from: Walk Score for the confirmed address

M56 Documented strengths

- Lot of 1.35 acres, which is generous for a single dwelling and leaves room for outbuildings and a garden.
- Abuts mapped water at roughly 25 ft (Lake Livingston), consistent with genuine water frontage subject to title confirmation.
- Long growing season of roughly 335 frost-free days supports year-round production.

M57 Limitations and risks

- **Inside a FEMA Special Flood Hazard Area (A+X).** This forces an elevated finished floor, triggers floodplain permitting, and makes flood insurance mandatory with a federally backed mortgage.
- The lot straddles more than one flood zone, so the building position materially changes the flood obligation.
- **High soil shrink-swell.** Expansive clay is the leading cause of foundation distress in this region; an engineered foundation and a geotechnical report should be assumed.
- Septic absorption is rated "very limited" for the dominant soil, so a conventional drainfield may not be permittable and an engineered or aerobic system may be required at higher cost.
- Seasonal water table within 51.0 cm of the surface.
- HOA covenants, zoning, utility availability, dock rights and short-term-rental rules are all unconfirmed. Any of them can independently prevent the intended use.

M59 Construction feasibility conclusion

Main house	Likely The lot is large enough and the terrain is workable. Finished-floor elevation and foundation type are the governing variables, not land area.
Second in-law suite / ADU	Unresolved Depends entirely on county rules and the HOA declaration; neither was available. Do not assume a second dwelling is permitted.
Garage	Likely No physical constraint identified, subject to setbacks and architectural guidelines.
Septic system	Unresolved Soil ratings are unfavourable; permittability and bedroom count require a site evaluation.
Garden	Conditional Workable with raised beds and imported loam; native surface soil is restrictive.
Hobby farming / poultry	Unresolved Governed by county rules and CC&Rs; unconfirmed.
Dock	Unresolved The lot adjoins water, but dock rights are a title and permitting question that was not resolved.
Short-term rental	Unresolved Governed by local ordinance and CC&Rs; unconfirmed.
Pool / outdoor recreation	Conditional Space is available; flood-zone and soil-movement considerations affect pool design where applicable.

M60 Key measurements summary

Parameter	Finding
Lot size	1.352 ac (58,898 sq ft)
Lot dimensions (long axis × width)	475 ft × 250 ft
Lowest elevation	135.22 ft
Highest elevation	177.84 ft
Elevation difference	42.61 ft
Average slope	9.94% (5.68°)
Maximum slope	35.02% (19.30°)
Steepest 50 ft run	12.8% (7.3°)
FEMA zone	A + X
BFE	not published
Minimum finished-floor elevation	NEEDS CONFIRMATION
Fill or elevation required?	Probably, confirm against the governing BFE
Sewer	NEEDS CONFIRMATION
Water	NEEDS CONFIRMATION

... continued

Parameter	Finding
Electricity	NEEDS CONFIRMATION
Gas	NEEDS CONFIRMATION
Internet	NEEDS CONFIRMATION
Septic required?	NEEDS CONFIRMATION — soil absorption rated Very limited
Soil	Woodville — Somewhat poorly drained, hydrologic group D, High shrink-swell
HOA	Platted subdivision (Creek Wood Bend), so covenants are likely. NEEDS CONFIRMATION
HOA fee	NEEDS CONFIRMATION
Zoning	NEEDS CONFIRMATION
Dock possible?	NEEDS CONFIRMATION
Lake depth	NEEDS CONFIRMATION
Boating	NEEDS CONFIRMATION
Fishing	NEEDS CONFIRMATION
Hobby farming	NEEDS CONFIRMATION
Second in-law suite	NEEDS CONFIRMATION
Short-term rental	NEEDS CONFIRMATION
Nearest hospital	NEEDS CONFIRMATION
Nearest international airport	NEEDS CONFIRMATION
Distance to nearest mapped water	25 ft to Lake Livingston
Mean annual temperature	20.9 °C
Mean annual rainfall	1,204 mm (47.4 in)
Appraised market value	$82,110

0 Saint Lawrence River Rd, Montgomery, TX 77316

Lot 9 of 12 | ID confidence: LOW - DO NOT RELY | Lake Conroe Forest | Montgomery County

Read this before anything else on this lot
The address '0 Saint Lawrence River Rd' carries no house number, and 37 parcels front that road inside Lake Conroe Forest Section 2. The parcel analysed here (Block 41, Lot 10) could not be tied to the listing and is almost certainly NOT it: the appraisal record carries an improvement value of about $265,000, meaning a house already stands on it, whereas the listing is offered as a vacant lot. Its GIS acreage is also recorded as 0. These pages should be read as a demonstration of the method over a neighbouring parcel in the same subdivision, not as findings for the lot you are buying. Obtain the parcel ID from the listing agent and re-run before relying on a single figure here.

A1 Property identification & record data OFFICIAL

Situs address of record	1202 SAINT LAWRENCE RIVER RD, MONTGOMERY, TX 77316
Legal description	LAKE CONROE FOREST 02, BLOCK 41, LOT 10
Subdivision	Lake Conroe Forest
County	Montgomery County (FIPS 48339)
Parcel ID / account	110944.00000000
Geographic ID (APN)	6580-02-73200
Owner of record	ELV PROPERTY HOLDINGS LLC SERIES I
Acreage — appraisal roll (legal)	0.00 ac
Acreage — GIS polygon	0.000 ac
Acreage — recomputed from boundary	0.229 ac
Appraised land value	$29,850
Appraised improvement value	$265,120
Appraised market value	$294,970
Centroid (WGS84)	30.349985, -95.607197
Longest axis / width	170 ft × 116 ft (axis bearing 305°)

Source: Texas StratMap statewide land parcels (county appraisal roll). Matched by situs address match. Appraisal-roll acreage and a GIS polygon rarely agree exactly; neither replaces a survey.

A2 · Listing and plat material not retrievable here NEEDS CONFIRMATION

Current asking price, days on market and listing history.
Obtain from: Listing agent / MLS

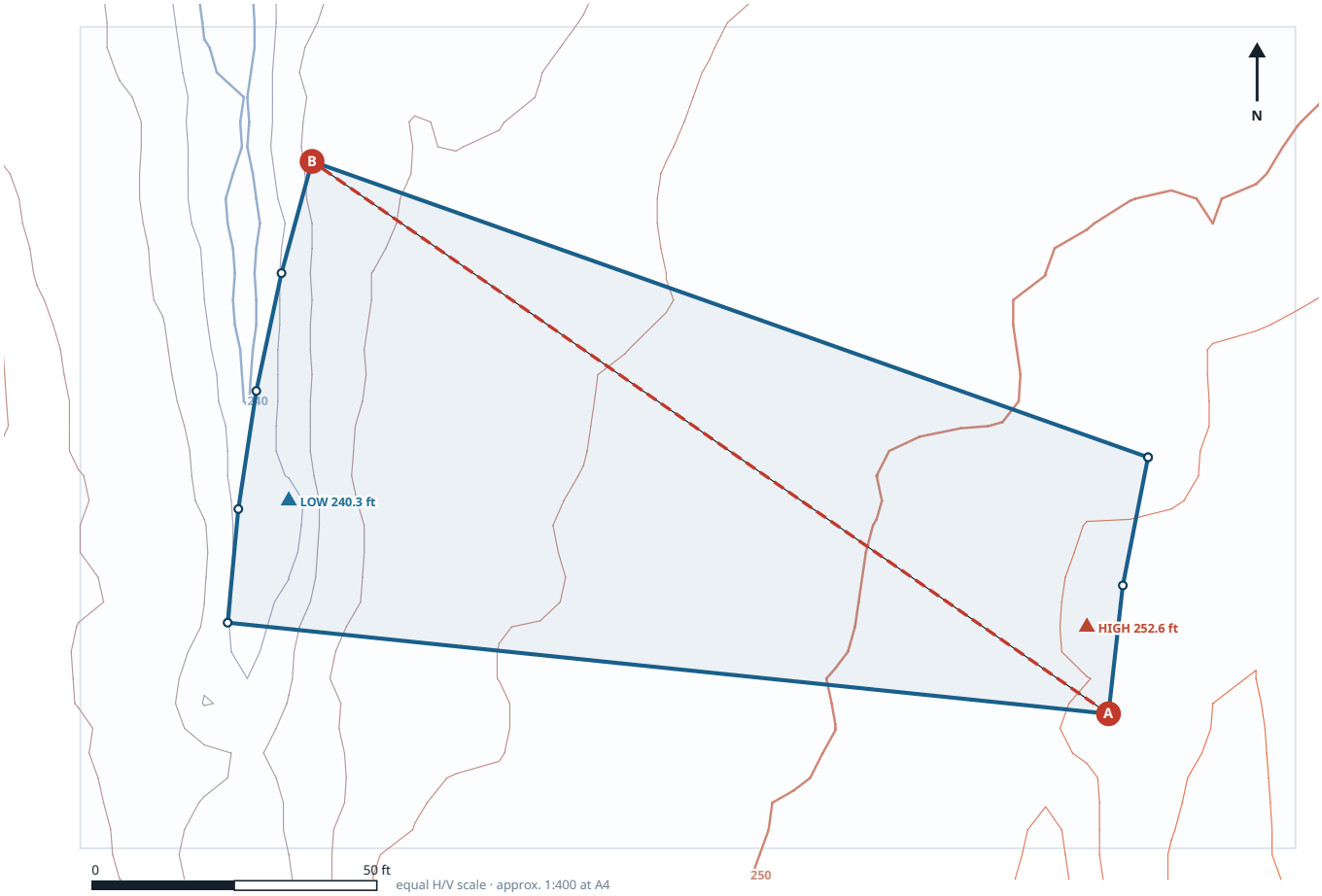
Listing photographs: lot frontage, shoreline, road access and surroundings.
Obtain from: Listing agent (Zillow blocks automated retrieval, so no listing imagery is reproduced here)

Seller disclosure notice and any known defects.
Obtain from: Listing agent

Recorded plat and Esri/county GIS parcel exhibit for comparison with this report.
Obtain from: County Clerk plat records and the county GIS portal

B5 Terrain, contours and true-scale site plan MODEL-DERIVED

LOWEST POINT	HIGHEST POINT	RELIEF	MEAN ELEVATION
240.3 ft	252.8 ft	12.4 ft	248.1 ft



True-scale site plan. Parcel boundary from the county GIS polygon; contours at 2.0 ft intervals derived from the USGS 3DEP 1 m bare-earth DEM. A–B is the cross-section line. Horizontal and vertical map scales are equal, so shapes and distances are undistorted.

MEAN GRADE	MEDIAN GRADE	90TH PCT GRADE	MAX GRADE
9.09 %	6.29 %	19.69 %	43.32 %

Mean slope angle	5.20°	**90th pct angle**	11.14°	**Max angle**	23.42°	**Grid cells**	1,156

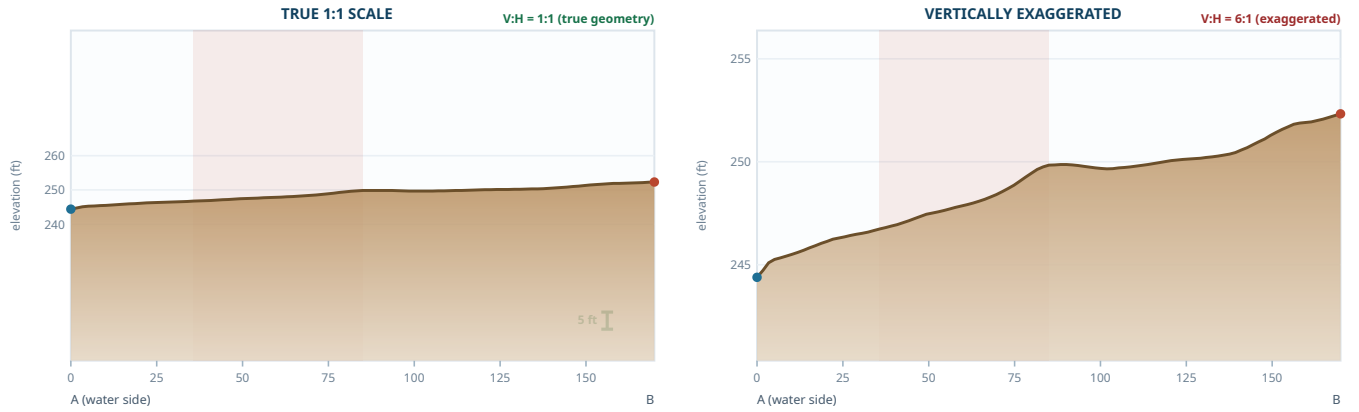
B7 · Terrain classification

Moderate slope. Expect cut-and-fill balancing and possibly a stepped footing.

Basis: mean DEM grade 9.09% (90th pct 19.69%), relief 12.42 ft

Grading expectation: Significant grading; Cut-and-fill

B6 Longitudinal cross-section MODEL-DERIVED

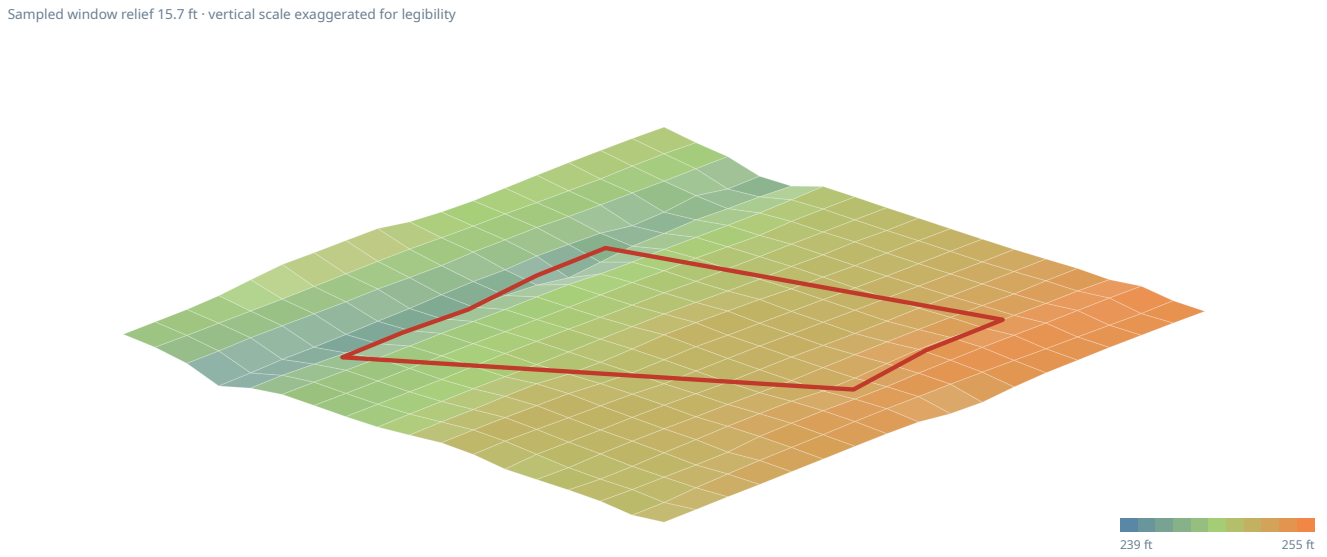


Run A–B	170 ft	**Lowest**	244.4 ft	**Highest**	252.3 ft
Elevation difference	7.9 ft	**Average grade**	4.7%	**Direct rising angle**	2.7°
Steepest 50 ft	**6.3%** (3.6°) over a 49 ft run				

Reading: The ground rises 7.9 ft over 170 ft, a direct angle of 2.7° (4.7% average grade). That is a gentle, buildable fall.

Left panel is the real geometry at 1:1 vertical-to-horizontal scale. The right panel exaggerates the vertical axis, which is how slope is conventionally drawn on maps but overstates steepness. The shaded band marks the steepest 50 ft run. Profile sampled from the 3DEP DEM along the lot's longest axis.

B8 Three-dimensional terrain MODEL-DERIVED



Isometric view of the sampled DEM window; the red outline is the parcel boundary draped on the surface. Vertical scale is exaggerated to reveal form — use the 1:1 cross-section above for true steepness.

C9 FEMA flood-zone determination OFFICIAL

Zone: AE + X | Inside Special Flood Hazard Area | Split zone

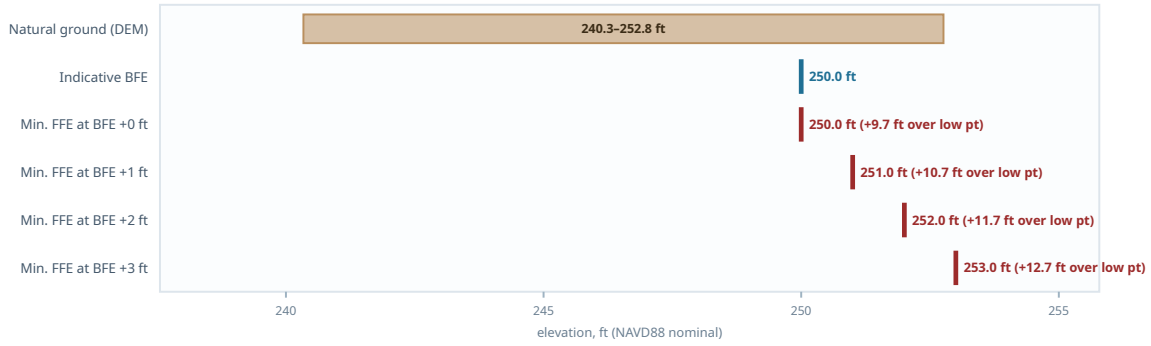
Zones intersecting the parcel	AE + X
Special Flood Hazard Area	Yes — mandatory flood insurance for a federally backed mortgage
FIRM panel	48339C0375G · effective 2014-08-18
Mapping status	—

Parcel intersects mapped FEMA flood-hazard polygon(s); see zones above.

C11-13 Ground elevation, BFE and minimum finished floor OFFICIAL MODEL-DERIVED

Lowest natural ground (DEM)	240.34 ft
Highest natural ground (DEM)	252.76 ft
Mean ground	248.14 ft
Indicative BFE	250.0 ft
Lowest ground relative to BFE	**-9.66 ft**
Highest ground relative to BFE	+2.76 ft

BFE basis: highest of 7 FEMA BFE line value(s) within 2,000 ft ([234.0, 240.0, 241.0, 243.0, 244.0, 249.0, 250.0]) - indicative only, the governing BFE must be read at the site from the FIRM/FIS profile



Freeboard scenarios. The brown band is the measured spread of natural ground across the lot. Red ticks are the minimum finished-floor elevation at BFE plus 0, 1, 2 and 3 ft of freeboard. **The freeboard the county actually enforces is not established in this report** — the scenarios show the consequence of each, they do not assert a requirement.

Scenario	Min. finished floor (ft)	Rise above lowest ground (ft)	Rise above highest ground (ft)
BFE + 0 ft	250.00	9.66	-2.76
BFE + 1 ft	251.00	10.66	-1.76
BFE + 2 ft	252.00	11.66	-0.76
BFE + 3 ft	253.00	12.66	0.24

DEM ground across the lot runs 240.34-252.76 ft against an indicative BFE of 250.0 ft. The lot therefore sits 9.7 ft below BFE at its lowest point. Any habitable floor must be set above BFE plus the freeboard the county actually enforces - that freeboard figure is NOT established here.

Do not assume fill is permitted
Raising a pad with fill inside a mapped floodplain is frequently restricted or requires compensatory storage and a floodplain development permit. Elevating on piers or a stem wall is often the permitted route instead. This must be confirmed with the county floodplain administrator before any design assumption is made.

C · Flood items still to confirm NEEDS CONFIRMATION

Official flood determination and the effective FIRM panel read at the parcel.
Obtain from: County floodplain administrator or a flood certification service

... continued

Enforced freeboard above BFE, fill rules and compensatory-storage requirements.
Obtain from: County floodplain administrator

First Street / third-party forward-looking flood risk and estimated flood depth.
Obtain from: First Street Foundation (riskfactor.com) for the confirmed address

Flood insurance quote under current NFIP pricing.
Obtain from: Insurance agent, using an Elevation Certificate

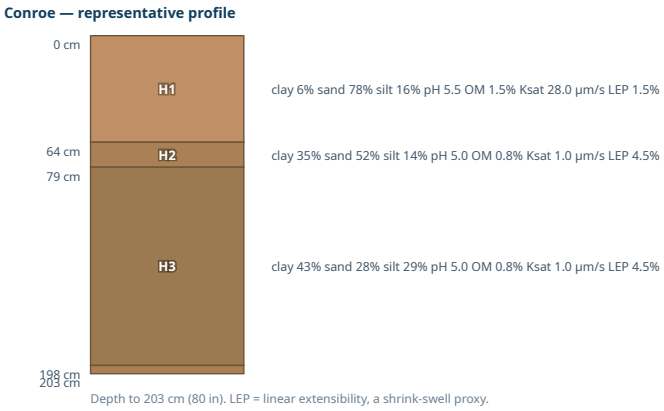
Site drainage: swales, culverts, outfall and upstream runoff from neighbouring lots.
Obtain from: Civil engineer site visit; county drainage criteria

D14 Soil characteristics MODEL-DERIVED

NRCS map unit intersecting the lot	Share of lot (sampled)
Conroe loamy fine sand, 5 to 12 percent slopes	—
Woodville fine sandy loam, 1 to 5 percent slopes	100.0%

Component	%	Drainage	Hyd.grp	LEP %	Shrink-swell	pH	Clay %	Flooding
Conroe	100%	Moderately well drained	B	4.5	Moderate	5.5	6	None
Woodville	90%	Somewhat poorly drained	D	7.5	High	5.5	12	None

LEP = linear extensibility (shrink-swell proxy). Hydrologic group D indicates the slowest infiltration. Values are NRCS representative values for the map unit, not measurements on this lot.



Representative horizon sequence for the dominant component (Woodville).

D15 Foundation suitability MODEL-DERIVED

Dominant soil shrink-swell potential is **Moderate** (peak LEP 4.5%). Stiffened/post-tensioned slab or moisture-conditioned subgrade is typical.

Slab-on-grade	Stiffened / post-tensioned
Crawlspace	Suitable
Pier and beam	Suitable
Basement	Not advised
Stepped footing	Only if sloped
NRCS rating, dwellings without basements	—
Geotechnical investigation advised	Yes

Foundation selection is an engineering decision. NRCS map-unit shrink-swell indicates risk; it does not substitute for a geotechnical investigation.

D16 Septic / on-site sewage suitability MODEL-DERIVED

Septic Tank Absorption Fields	Very limited
Septic Tank Leaching Chamber (TX)	Somewhat limited
Septic Tank, Subsurface Drip Irrigation (TX)	Very limited

Constraints identified from the soil and terrain data

- Seasonal water table within 84.0 cm of the surface.
- 90th-percentile grade of 19.69% complicates drainfield layout.

Permitted bedroom count: NOT ESTABLISHED
Permitted bedroom count follows from a site-specific soil evaluation (TCEQ Ch. 285 class, loading rate, usable drainfield area and setbacks). It cannot be derived from map-unit data. A licensed site evaluator must run the percolation/soil profile test.

NRCS ratings describe the map unit, not the lot. A TCEQ-compliant site evaluation and percolation test is required for any permit.

D17 Gardening suitability MODEL-DERIVED

SURFACE PH	CLAY	SAND	ORGANIC MATTER
5.5	6 %	78 %	1.5 %

Drainage class	Moderately well drained
pH range through profile	5.0 – 5.5
NRCS lawn / landscape rating	Very limited
Raised beds advised	Not necessary

Surface horizon is workable; in-ground beds are reasonable with normal organic amendment.

Likely amendments

- Lime, to raise pH toward 6.2-6.8 for most vegetables
- Compost / organic matter, surface organic matter is low
- Mulch, to buffer soil temperature and moisture

Based on NRCS map-unit averages. A lab soil test is required before committing to any amendment programme.

D · Soil and planting items to confirm NEEDS CONFIRMATION

Laboratory soil test: pH, macro and micronutrients, organic matter, texture.
Obtain from: Texas A&M AgriLife Soil Testing Lab, from samples taken on the lot

Plant hardiness zone confirmation and a species list matched to the tested soil.
Obtain from: USDA hardiness map plus the county AgriLife extension office

E19-23 Utilities and infrastructure NEEDS CONFIRMATION

Utility availability is not published as open geospatial data at parcel level. Nothing is inferred here: each item below must come from the serving utility in writing, because a missing water tap or a long electric extension can change the build budget materially.

Water supply: municipal, water-supply-corporation, or private well; is a tap at the road?
Obtain from: Serving water supply corporation / county; well logs via TX Water Development Board

Sewer: is public sewer available, or is an on-site facility required? Distance to any main.
Obtain from: County / city utility department or MUD

Electricity: provider, whether service exists at the lot line, overhead vs underground, connection cost.
Obtain from: Retail electric provider / local co-op (service availability letter)

Natural gas or propane availability and existing infrastructure.
Obtain from: Gas utility or local propane suppliers

... continued

Internet: fibre, cable, fixed wireless, DSL or satellite, with advertised speeds.
Obtain from: FCC National Broadband Map plus a direct serviceability check by address

F24-25 Waterfront position MODEL-DERIVED

Near water - frontage unlikely; access likely via easement or common area | Rushing Springs Lake

Nearest mapped waterbody	Rushing Springs Lake
Distance from parcel boundary	336 ft
Waterbody area	0.074 km²

NHD polygons are hydrographic mapping, not a boundary survey. Whether the lot carries riparian frontage, deeded access or only common-area access is a title/plat question and is NOT established here.

F26-29 · Dock, depth, boating and fishing NEEDS CONFIRMATION

Whether a dock is permittable at this shoreline, and the governing authority.
Obtain from: Controlling lake / river authority, plus the HOA

Dock size limits, shoreline setbacks, shared-dock rules and permit cost.
Obtain from: Lake authority rules and permit schedule

Whether riparian or shoreline rights pass with the lot, or the shoreline is a common reserve.
Obtain from: Title commitment and recorded plat

Lake water-level regime and any shoreline-management plan.
Obtain from: Lake/river authority or reservoir operator

Bathymetry: average and maximum depth, and depth near this shoreline.
Obtain from: Lake authority survey, TPWD reservoir report, or a sonar survey

Seasonal water-level variation and drawdown history.
Obtain from: Reservoir operator gauge records / USGS gauges

Boating rules: permitted craft, motor limits, wake rules, ramps and marinas.
Obtain from: Lake authority and TPWD

Fishery: species present, stocking, licence requirements and catch limits.
Obtain from: Texas Parks & Wildlife reservoir survey reports

G30-33 HOA, covenants, zoning and a second dwelling NEEDS CONFIRMATION

The parcel sits in **Lake Conroe Forest** according to the recorded legal description, so a declaration of covenants very likely exists. The documents themselves are not public geospatial data and were not retrieved, so no dues figure, restriction or amenity is asserted here.

Whether an HOA or POA exists and is mandatory, with its name and management contact.
Obtain from: Title commitment, recorded Declaration, seller disclosure

Zoning designation, or confirmation that the county is unzoned.
Obtain from: County or city development services

Minimum lot size, building coverage, setbacks, height limits, accessory-structure rules.
Obtain from: County/city development standards or subdivision regulations

Whether a second independent dwelling or in-law suite (ADU) is permitted, and on what terms.
Obtain from: County/city development services plus the HOA

Permitting path, driveway/culvert permit and any development fee.
Obtain from: County permitting office

... continued

Annual dues, monthly equivalent, transfer fee, and any special assessment.
Obtain from: HOA or management company estoppel / resale certificate

Declaration, CC&Rs, architectural guidelines, rules and all amendments.
Obtain from: HOA or title company

Restrictions on docks, fences, outbuildings, RVs, boats, livestock and poultry.
Obtain from: Recorded CC&Rs and architectural guidelines

Minimum dwelling size and any construction time limit after purchase.
Obtain from: Recorded CC&Rs

Amenities actually included: clubhouse, pool, courts, parks, lake and boat facilities.
Obtain from: HOA documents and budget

H34 / I35 Hobby farming and short-term rental NEEDS CONFIRMATION

Whether chickens, ducks, rabbits, goats or bees are allowed, and in what numbers.
Obtain from: County rules plus the recorded CC&Rs (either can prohibit)

Any agricultural or wildlife valuation currently on the parcel, and rollback-tax exposure if use changes.
Obtain from: County appraisal district

Whether short-term rental is permitted at this location.
Obtain from: County/city ordinance and the HOA

Any STR permit, occupancy cap, parking or noise condition.
Obtain from: Local ordinance

Hotel occupancy tax registration and remittance obligations.
Obtain from: Texas Comptroller plus local taxing authority

J36-42 Environmental and geological due diligence MODEL-DERIVED

Ground-penetrating radar penetration (NRCS rating)	Low penetration Indicates how useful GPR would be here; clay-rich soils attenuate the signal.
Sinkhole / karst exposure	Not indicated by the soil map units returned for this parcel. No restrictive bedrock or karst feature was reported. CONFIRM WITH TX BUREAU OF ECONOMIC GEOLOGY
Radon	EPA publishes radon zones by county, not by parcel. LOOK UP THE COUNTY ZONE AND TEST ANY BUILT STRUCTURE

Existing FEMA Elevation Certificate for the lot.
Obtain from: Seller, county floodplain office or a surveyor

Recent boundary, ALTA or topographic survey.
Obtain from: Seller or a licensed Texas surveyor (RPLS)

Geotechnical / soil boring report.
Obtain from: Seller or a geotechnical engineer

On-site sewage facility site evaluation and percolation test.
Obtain from: Licensed site evaluator / county OSSF office

Any ground-penetrating-radar or other subsurface investigation.
Obtain from: Seller; commission if subsurface features are suspected

K43-45 Residential construction feasibility MODEL-DERIVED

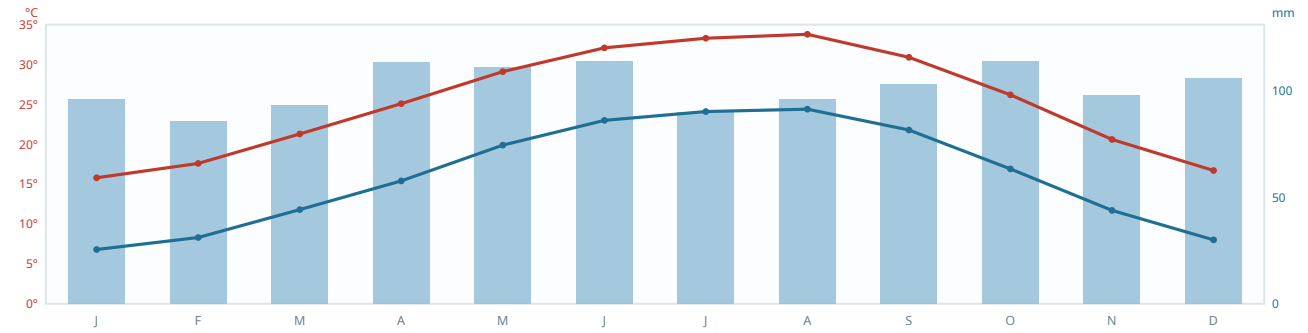
Lot area available	0.000 ac (0 sq ft)
Usable building envelope	NOT COMPUTABLE Setbacks, easements and any lake or drainage reserve come from the recorded plat and the HOA architectural guidelines, none of which were available.
Terrain constraint	Moderate slope — Significant grading; Cut-and-fill
Foundation implication	Moderate shrink-swell. Stiffened/post-tensioned slab or moisture-conditioned subgrade is typical.
Flood implication	Inside a mapped SFHA, so finished-floor elevation, flood venting and insurance all bind the design.

Construction cost
No build cost is estimated in this report. A defensible figure for a four-bedroom house depends on the HOA's minimum size and material rules, the finished-floor elevation forced by the flood zone, the foundation type the soil requires, and whether water and sewer must be created on site. Three of those four inputs are still unconfirmed for this lot, so any number quoted now would be arbitrary. Price it from a builder bid once the plat, CC&Rs, floodplain requirement and soil report are in hand.

L46 Climate MODEL-DERIVED

MEAN ANNUAL TEMP	MEAN SUMMER HIGH	MEAN WINTER LOW	ANNUAL RAINFALL
20.7 °C	33.1 °C	7.7 °C	1,222 mm

Record high	41.9 °C	**Record low**	-12.3 °C	**Rainfall**	48.1 in
Growing season	~324 days	**Days at or below 0 °C**	5.5 /yr	**Period**	1994-2023 (30 years)



Monthly normals, 1994-2023 (30 years). Bars are mean monthly rainfall (mm, right axis); lines are mean daily maximum and minimum temperature (°C, left axis). Source: ERA5 daily reanalysis via Open-Meteo archive API.

L47 Natural hazards

Flood	Mapped Special Flood Hazard Area
Other perils	Hurricane and tropical wind, tornado, hail, severe convective storm, lightning, extreme heat, drought and wildfire all vary by location within Texas. QUANTIFY FROM FEMA NATIONAL RISK INDEX FOR THE CONFIRMED COUNTY AND FROM AN INSURANCE QUOTE

L48-55 Community, schools, healthcare, shopping and access NEEDS CONFIRMATION

Demographics not retrieved
Census API requires an API key (keyless request rejected) Free key from https://api.census.gov/data/key_signup.html, then set CENSUS_API_KEY and re-run 05_context.py. Or read the figures directly from data.census.gov.

Driving distance and time to the nearest hospital, emergency department and urgent care.
Obtain from: Direct routing check from the confirmed parcel location

Distance to grocery, major retail, pharmacy, bank, fuel and hardware.
Obtain from: Direct routing check

Assigned school district and campuses, with distances and current ratings.
Obtain from: TEA school-district locator and the district attendance-zone map

Nearest Indian / South Asian grocery, restaurants and places of worship.
Obtain from: Direct search from the confirmed parcel location

Nearest international airport, driving time and route.
Obtain from: Direct routing check

Walk / Bike / Transit scores.
Obtain from: Walk Score for the confirmed address

M56 Documented strengths

- Close to Rushing Springs Lake at about 336 ft, so lake amenity is nearby without shoreline exposure.
- Long growing season of roughly 324 frost-free days supports year-round production.

M57 Limitations and risks

- **Parcel identity is not established.** The address '0 Saint Lawrence River Rd' carries no house number, and 37 parcels front that road inside Lake Conroe Forest Section 2. The parcel analysed here (Block 41, Lot 10) could not be tied to the listing and is almost certainly NOT it: the appraisal record carries an improvement value of about $265,000, meaning a house already stands on it, whereas the listing is offered as a vacant lot. Its GIS acreage is also recorded as 0. These pages should be read as a demonstration of the method over a neighbouring parcel in the same subdivision, not as findings for the lot you are buying. Obtain the parcel ID from the listing agent and re-run before relying on a single figure here.
- **Inside a FEMA Special Flood Hazard Area (AE+X).** This forces an elevated finished floor, triggers floodplain permitting, and makes flood insurance mandatory with a federally backed mortgage.
- Lowest natural ground sits about 9.7 ft below the indicative BFE, so the pad must be raised or the house elevated.
- The lot straddles more than one flood zone, so the building position materially changes the flood obligation.
- Septic absorption is rated "very limited" for the dominant soil, so a conventional drainfield may not be permittable and an engineered or aerobic system may be required at higher cost.
- Seasonal water table within 84.0 cm of the surface.
- The appraisal record carries a zero value or zero area, which signals a questionable or superseded parcel record.
- HOA covenants, zoning, utility availability, dock rights and short-term-rental rules are all unconfirmed. Any of them can independently prevent the intended use.

M59 Construction feasibility conclusion

Main house	Likely The lot is large enough and the terrain is workable. Finished-floor elevation and foundation type are the governing variables, not land area.
Second in-law suite / ADU	Unresolved Depends entirely on county rules and the HOA declaration; neither was available. Do not assume a second dwelling is permitted.
Garage	Likely No physical constraint identified, subject to setbacks and architectural guidelines.
Septic system	Unresolved Soil ratings are unfavourable; permittability and bedroom count require a site evaluation.
Garden	Likely Native surface soil is workable with normal organic amendment.
Hobby farming / poultry	Unresolved Governed by county rules and CC&Rs; unconfirmed.
Dock	Unresolved The lot does not adjoin mapped water, so a private dock is unlikely; shared access may exist.
Short-term rental	Unresolved Governed by local ordinance and CC&Rs; unconfirmed.
Pool / outdoor recreation	Conditional Space is available; flood-zone and soil-movement considerations affect pool design where applicable.

M60 Key measurements summary

Parameter	Finding
Lot size	0.000 ac (0 sq ft)
Lot dimensions (long axis × width)	170 ft × 116 ft
Lowest elevation	240.34 ft
Highest elevation	252.76 ft
Elevation difference	12.42 ft
Average slope	9.09% (5.20°)
Maximum slope	43.32% (23.42°)
Steepest 50 ft run	6.3% (3.6°)
FEMA zone	AE + X
BFE	250.0 ft (indicative)

... continued

Parameter	Finding
Minimum finished-floor elevation	251.0 ft at +1 ft / 252.0 ft at +2 ft freeboard COUNTY FIGURE UNCONFIRMED
Fill or elevation required?	Yes — lowest ground is 9.7 ft below BFE
Sewer	NEEDS CONFIRMATION
Water	NEEDS CONFIRMATION
Electricity	NEEDS CONFIRMATION
Gas	NEEDS CONFIRMATION
Internet	NEEDS CONFIRMATION
Septic required?	NEEDS CONFIRMATION — soil absorption rated Very limited
Soil	Woodville — Moderately well drained, hydrologic group B, Moderate shrink-swell
HOA	Platted subdivision (Lake Conroe Forest), so covenants are likely. NEEDS CONFIRMATION
HOA fee	NEEDS CONFIRMATION
Zoning	NEEDS CONFIRMATION
Dock possible?	NEEDS CONFIRMATION
Lake depth	NEEDS CONFIRMATION
Boating	NEEDS CONFIRMATION
Fishing	NEEDS CONFIRMATION
Hobby farming	NEEDS CONFIRMATION
Second in-law suite	NEEDS CONFIRMATION
Short-term rental	NEEDS CONFIRMATION
Nearest hospital	NEEDS CONFIRMATION
Nearest international airport	NEEDS CONFIRMATION
Distance to nearest mapped water	336 ft to Rushing Springs Lake
Mean annual temperature	20.7 °C
Mean annual rainfall	1,222 mm (48.1 in)
Appraised market value	$294,970

141 Ward Rd, Justiceburg, TX 79330

Lot 10 of 12 | ID confidence: medium | Oak Canyon | Garza County

Identification caveat
No situs address is recorded. Matched to Oak Canyon Subdivision Lot 141 by applying the same lot-number-as-address convention seen on neighbouring parcels. Two records exist for Lot 141, which suggests a split interest. Confirm against the listing's parcel ID.

A1 Property identification & record data OFFICIAL

Situs address of record	, , TX
Legal description	LOT 141 OAK CANYON SUBDIVISION LOCATED IN AB 935 SEC 68 (PHASE 2) 2 ACRES OAK CANYON 50% INT .500000 INT IN 2 ACRES
Subdivision	Oak Canyon
County	Garza County (FIPS 48169)
Parcel ID / account	24780
Geographic ID (APN)	380-0000-0141-000-00-00
Owner of record	DEVIL DOG FISHING DYNASTY LLC OR
Acreage — appraisal roll (legal)	2.00 ac
Acreage — GIS polygon	1.938 ac
Acreage — recomputed from boundary	1.944 ac
Appraised land value	$0
Appraised improvement value	$0
Appraised market value	$0
Centroid (WGS84)	33.034547, -101.042511
Longest axis / width	521 ft × 325 ft (axis bearing 116°)

Source: Texas StratMap statewide land parcels (county appraisal roll). Matched by manual pin / legal-desc match. Appraisal-roll acreage and a GIS polygon rarely agree exactly; neither replaces a survey.

A2 · Listing and plat material not retrievable here NEEDS CONFIRMATION

Current asking price, days on market and listing history.
Obtain from: Listing agent / MLS

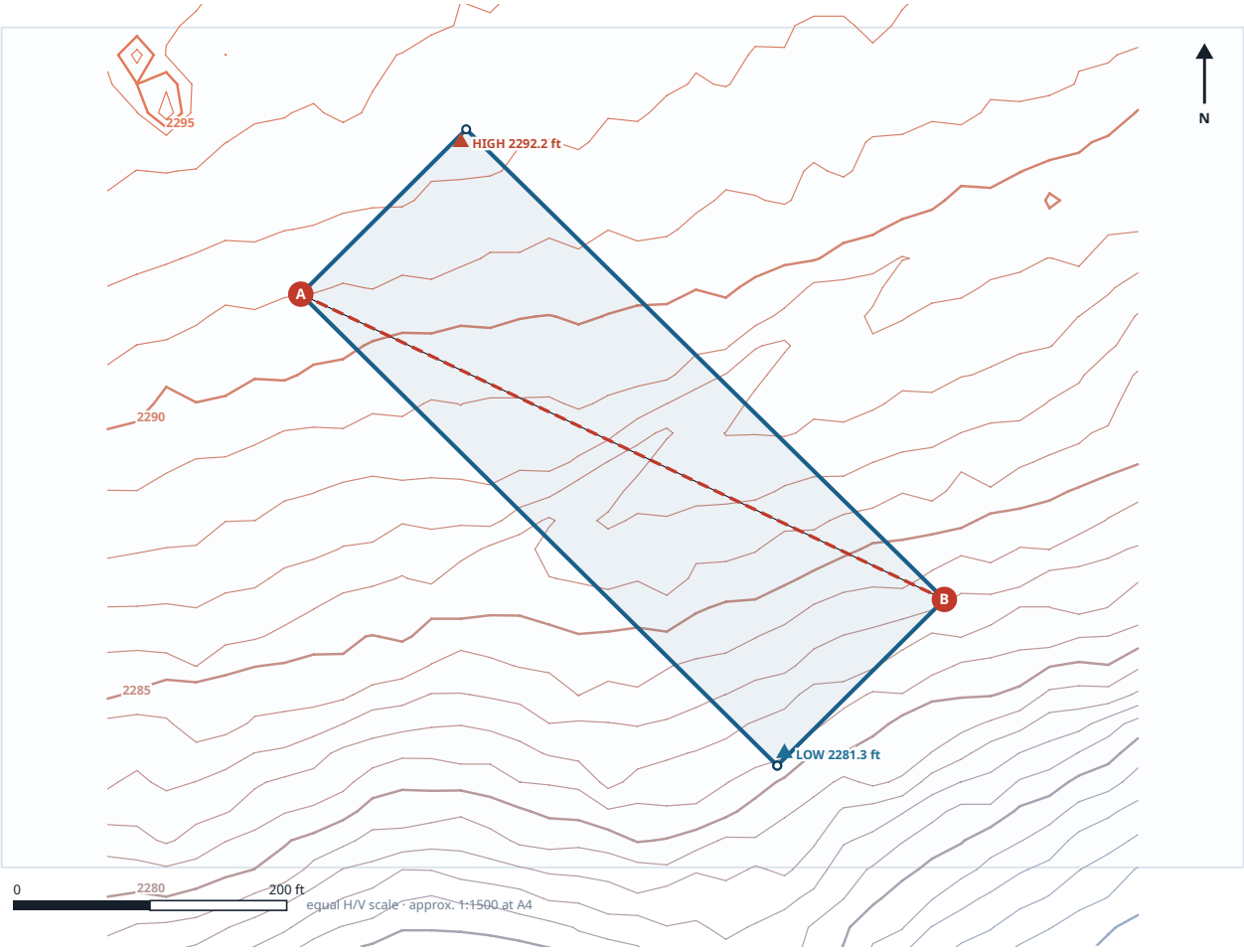
Listing photographs: lot frontage, shoreline, road access and surroundings.
Obtain from: Listing agent (Zillow blocks automated retrieval, so no listing imagery is reproduced here)

Seller disclosure notice and any known defects.
Obtain from: Listing agent

Recorded plat and Esri/county GIS parcel exhibit for comparison with this report.
Obtain from: County Clerk plat records and the county GIS portal

B5 Terrain, contours and true-scale site plan MODEL-DERIVED

LOWEST POINT	HIGHEST POINT	RELIEF	MEAN ELEVATION
2,281.3 ft	2,292.2 ft	10.9 ft	2,287.6 ft



True-scale site plan. Parcel boundary from the county GIS polygon; contours at 1.0 ft intervals derived from the USGS 3DEP 1 m bare-earth DEM. A–B is the cross-section line. Horizontal and vertical map scales are equal, so shapes and distances are undistorted.

MEAN GRADE	MEDIAN GRADE	90TH PCT GRADE	MAX GRADE
2.82 %	2.49 %	5.18 %	8.90 %

Mean slope angle	1.62°	**90th pct angle**	2.97°	**Max angle**	5.09°	**Grid cells**	1,156

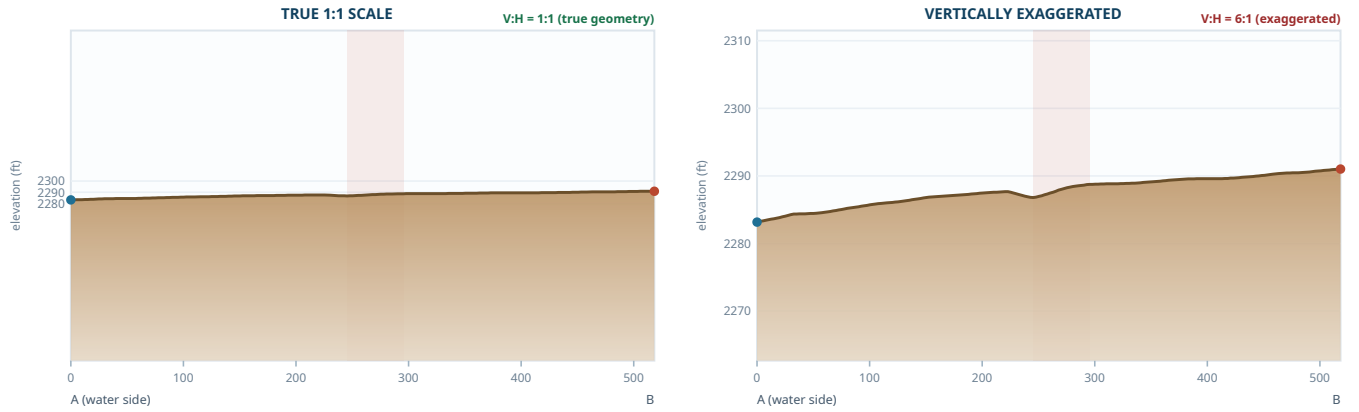
B7 · Terrain classification

Gentle slope. Standard residential grading is normally sufficient.

Basis: mean DEM grade 2.82% (90th pct 5.18%), relief 10.92 ft

Grading expectation: Minimal grading; Routine cut-and-fill

B6 Longitudinal cross-section MODEL-DERIVED

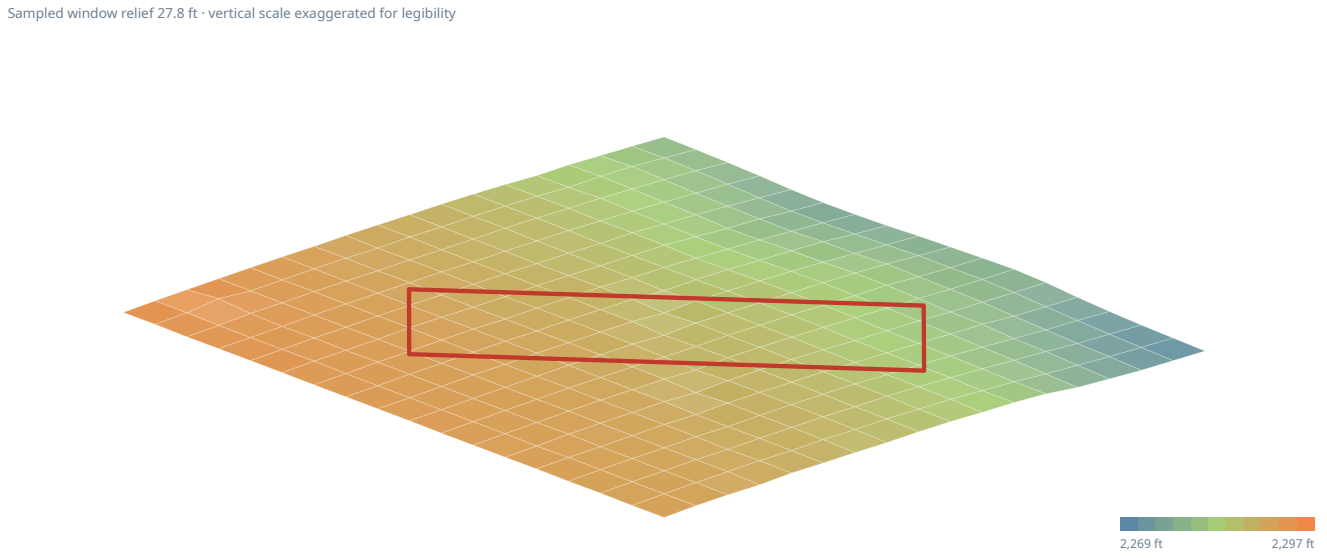


Run A–B	518 ft	**Lowest**	2,283.2 ft	**Highest**	2,291.0 ft
Elevation difference	7.8 ft	**Average grade**	1.5%	**Direct rising angle**	0.9°
Steepest 50 ft					**3.9%** (2.2°) over a 50 ft run

Reading: Over 518 ft the ground rises only 7.8 ft, a direct angle of 0.9°. At true 1:1 scale the profile is almost a straight line, which is the correct reading: this lot is effectively flat. The exaggerated panel is the same data and should not be read as slope.

Left panel is the real geometry at 1:1 vertical-to-horizontal scale. The right panel exaggerates the vertical axis, which is how slope is conventionally drawn on maps but overstates steepness. The shaded band marks the steepest 50 ft run. Profile sampled from the 3DEP DEM along the lot's longest axis.

B8 Three-dimensional terrain MODEL-DERIVED



Isometric view of the sampled DEM window; the red outline is the parcel boundary draped on the surface. Vertical scale is exaggerated to reveal form — use the 1:1 cross-section above for true steepness.

C9 FEMA flood-zone determination OFFICIAL

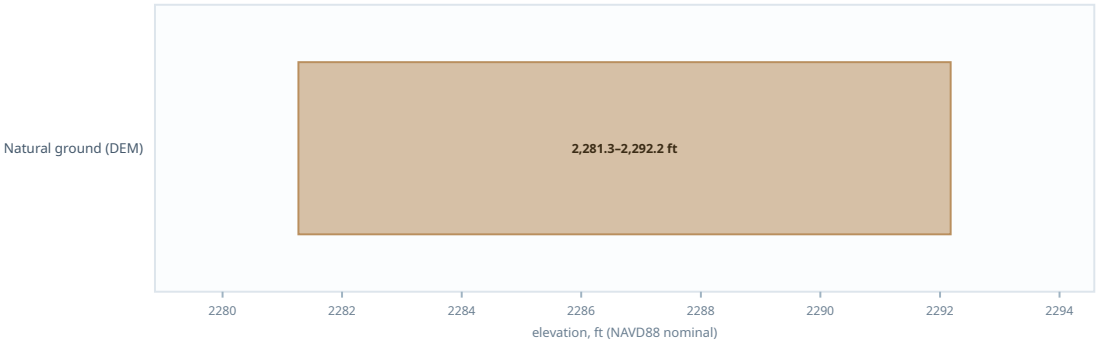
Zone: no polygon on parcel | Not in a mapped SFHA

Zones intersecting the parcel	no polygon on parcel
Special Flood Hazard Area	Not per the national layer
FIRM panel	—
Mapping status	unmapped / Zone D - undetermined

No effective NFHL flood-hazard polygons were returned within 15 miles. This location appears to lie outside FEMA's digital mapping. The flood zone is UNDETERMINED (functionally Zone D); obtain a county floodplain determination before relying on any flood conclusion.

C11-13 Ground elevation, BFE and minimum finished floor OFFICIAL MODEL-DERIVED

Lowest natural ground (DEM)	2,281.27 ft
Highest natural ground (DEM)	2,292.18 ft
Mean ground	2,287.62 ft
Indicative BFE	not published for this location



Freeboard scenarios. The brown band is the measured spread of natural ground across the lot. Red ticks are the minimum finished-floor elevation at BFE plus 0, 1, 2 and 3 ft of freeboard. **The freeboard the county actually enforces is not established in this report** — the scenarios show the consequence of each, they do not assert a requirement.

C · Flood items still to confirm NEEDS CONFIRMATION

Official flood determination and the effective FIRM panel read at the parcel.
Obtain from: County floodplain administrator or a flood certification service

Enforced freeboard above BFE, fill rules and compensatory-storage requirements.
Obtain from: County floodplain administrator

First Street / third-party forward-looking flood risk and estimated flood depth.
Obtain from: First Street Foundation (riskfactor.com) for the confirmed address

Flood insurance quote under current NFIP pricing.
Obtain from: Insurance agent, using an Elevation Certificate

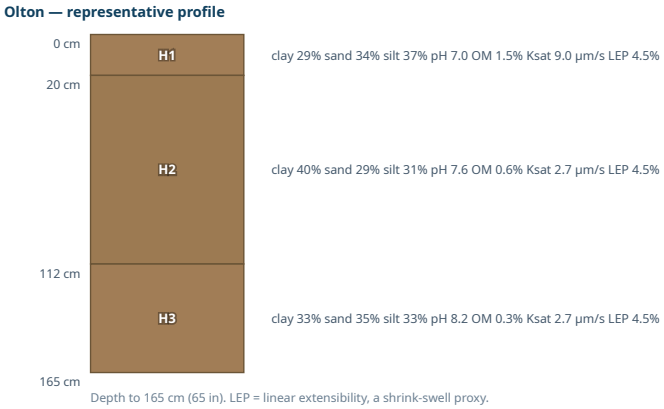
Site drainage: swales, culverts, outfall and upstream runoff from neighbouring lots.
Obtain from: Civil engineer site visit; county drainage criteria

D14 Soil characteristics MODEL-DERIVED

NRCS map unit intersecting the lot	Share of lot (sampled)
Olton clay loam, 1 to 3 percent slopes	100.0%

Component	%	Drainage	Hyd.grp	LEP %	Shrink-swell	pH	Clay %	Flooding
Olton	100%	Well drained	C	4.5	Moderate	7.0	29	None

LEP = linear extensibility (shrink-swell proxy). Hydrologic group D indicates the slowest infiltration. Values are NRCS representative values for the map unit, not measurements on this lot.



Representative horizon sequence for the dominant component (Olton).

D15 Foundation suitability MODEL-DERIVED

Dominant soil shrink-swell potential is **Moderate** (peak LEP 4.5%). Stiffened/post-tensioned slab or moisture-conditioned subgrade is typical.

Slab-on-grade	Stiffened / post-tensioned
Crawlspace	Suitable
Pier and beam	Suitable
Basement	Not advised
Stepped footing	Only if sloped
NRCS rating, dwellings without basements	—
Geotechnical investigation advised	Yes

Foundation selection is an engineering decision. NRCS map-unit shrink-swell indicates risk; it does not substitute for a geotechnical investigation.

D16 Septic / on-site sewage suitability MODEL-DERIVED

Septic Tank Absorption Fields	Very limited
Septic Tank Leaching Chamber (TX)	Very limited
Septic Tank, Subsurface Drip Irrigation (TX)	Not limited

Constraints identified from the soil and terrain data

- Hydrologic group C indicates slow infiltration, which constrains conventional drainfields.

Permitted bedroom count: NOT ESTABLISHED
Permitted bedroom count follows from a site-specific soil evaluation (TCEQ Ch. 285 class, loading rate, usable drainfield area and setbacks). It cannot be derived from map-unit data. A licensed site evaluator must run the percolation/soil profile test.

NRCS ratings describe the map unit, not the lot. A TCEQ-compliant site evaluation and percolation test is required for any permit.

D17 Gardening suitability MODEL-DERIVED

SURFACE PH	CLAY	SAND	ORGANIC MATTER
7.0	29 %	34 %	1.5 %

Drainage class	Well drained
pH range through profile	7.0 – 8.2
NRCS lawn / landscape rating	Somewhat limited
Raised beds advised	Not necessary

Surface horizon is workable; in-ground beds are reasonable with normal organic amendment.

Likely amendments

- Compost / organic matter, surface organic matter is low
- Mulch, to buffer soil temperature and moisture

> Based on NRCS map-unit averages. A lab soil test is required before committing to any amendment programme.

D · Soil and planting items to confirm NEEDS CONFIRMATION

Laboratory soil test: pH, macro and micronutrients, organic matter, texture.
Obtain from: Texas A&M AgriLife Soil Testing Lab, from samples taken on the lot

Plant hardiness zone confirmation and a species list matched to the tested soil.
Obtain from: USDA hardiness map plus the county AgriLife extension office

E19-23 Utilities and infrastructure NEEDS CONFIRMATION

Utility availability is not published as open geospatial data at parcel level. Nothing is inferred here: each item below must come from the serving utility in writing, because a missing water tap or a long electric extension can change the build budget materially.

Water supply: municipal, water-supply-corporation, or private well; is a tap at the road?
Obtain from: Serving water supply corporation / county; well logs via TX Water Development Board

Sewer: is public sewer available, or is an on-site facility required? Distance to any main.
Obtain from: County / city utility department or MUD

Electricity: provider, whether service exists at the lot line, overhead vs underground, connection cost.
Obtain from: Retail electric provider / local co-op (service availability letter)

Natural gas or propane availability and existing infrastructure.
Obtain from: Gas utility or local propane suppliers

Internet: fibre, cable, fixed wireless, DSL or satellite, with advertised speeds.
Obtain from: FCC National Broadband Map plus a direct serviceability check by address

F24-25 Waterfront position MODEL-DERIVED

Near water - frontage unlikely; access likely via easement or common area | Lake Alan Henry

Nearest mapped waterbody	Lake Alan Henry
Distance from parcel boundary	587 ft
Waterbody area	10.775 km²

> NHD polygons are hydrographic mapping, not a boundary survey. Whether the lot carries riparian frontage, deeded access or only common-area access is a title/plat question and is NOT established here.

F26-29 · Dock, depth, boating and fishing NEEDS CONFIRMATION

Whether a dock is permittable at this shoreline, and the governing authority.
Obtain from: Controlling lake / river authority, plus the HOA

Dock size limits, shoreline setbacks, shared-dock rules and permit cost.
Obtain from: Lake authority rules and permit schedule

Whether riparian or shoreline rights pass with the lot, or the shoreline is a common reserve.
Obtain from: Title commitment and recorded plat

Lake water-level regime and any shoreline-management plan.
Obtain from: Lake/river authority or reservoir operator

Bathymetry: average and maximum depth, and depth near this shoreline.
Obtain from: Lake authority survey, TPWD reservoir report, or a sonar survey

Seasonal water-level variation and drawdown history.
Obtain from: Reservoir operator gauge records / USGS gauges

Boating rules: permitted craft, motor limits, wake rules, ramps and marinas.
Obtain from: Lake authority and TPWD

Fishery: species present, stocking, licence requirements and catch limits.
Obtain from: Texas Parks & Wildlife reservoir survey reports

G30-33 HOA, covenants, zoning and a second dwelling NEEDS CONFIRMATION

The parcel sits in **Oak Canyon** according to the recorded legal description, so a declaration of covenants very likely exists. The documents themselves are not public geospatial data and were not retrieved, so no dues figure, restriction or amenity is asserted here.

Whether an HOA or POA exists and is mandatory, with its name and management contact.
Obtain from: Title commitment, recorded Declaration, seller disclosure

Annual dues, monthly equivalent, transfer fee, and any special assessment.
Obtain from: HOA or management company estoppel / resale certificate

Declaration, CC&Rs, architectural guidelines, rules and all amendments.
Obtain from: HOA or title company

Zoning designation, or confirmation that the county is unzoned.
Obtain from: County or city development services

Minimum lot size, building coverage, setbacks, height limits, accessory-structure rules.
Obtain from: County/city development standards or subdivision regulations

Whether a second independent dwelling or in-law suite (ADU) is permitted, and on what terms.
Obtain from: County/city development services plus the HOA

Permitting path, driveway/culvert permit and any development fee.
Obtain from: County permitting office

... continued

Restrictions on docks, fences, outbuildings, RVs, boats, livestock and poultry.
Obtain from: Recorded CC&Rs and architectural guidelines

Minimum dwelling size and any construction time limit after purchase.
Obtain from: Recorded CC&Rs

Amenities actually included: clubhouse, pool, courts, parks, lake and boat facilities.
Obtain from: HOA documents and budget

H34 / I35 Hobby farming and short-term rental

NEEDS CONFIRMATION

Whether chickens, ducks, rabbits, goats or bees are allowed, and in what numbers.
Obtain from: County rules plus the recorded CC&Rs (either can prohibit)

Any agricultural or wildlife valuation currently on the parcel, and rollback-tax exposure if use changes.
Obtain from: County appraisal district

Whether short-term rental is permitted at this location.
Obtain from: County/city ordinance and the HOA

Any STR permit, occupancy cap, parking or noise condition.
Obtain from: Local ordinance

Hotel occupancy tax registration and remittance obligations.
Obtain from: Texas Comptroller plus local taxing authority

J36-42 Environmental and geological due diligence MODEL-DERIVED

Ground-penetrating radar penetration (NRCS rating)	Low penetration Indicates how useful GPR would be here; clay-rich soils attenuate the signal.
Sinkhole / karst exposure	Not indicated by the soil map units returned for this parcel. No restrictive bedrock or karst feature was reported. CONFIRM WITH TX BUREAU OF ECONOMIC GEOLOGY
Radon	EPA publishes radon zones by county, not by parcel. LOOK UP THE COUNTY ZONE AND TEST ANY BUILT STRUCTURE

Existing FEMA Elevation Certificate for the lot.
Obtain from: Seller, county floodplain office or a surveyor

Recent boundary, ALTA or topographic survey.
Obtain from: Seller or a licensed Texas surveyor (RPLS)

Geotechnical / soil boring report.
Obtain from: Seller or a geotechnical engineer

On-site sewage facility site evaluation and percolation test.
Obtain from: Licensed site evaluator / county OSSF office

Any ground-penetrating-radar or other subsurface investigation.
Obtain from: Seller; commission if subsurface features are suspected

K43-45 Residential construction feasibility MODEL-DERIVED

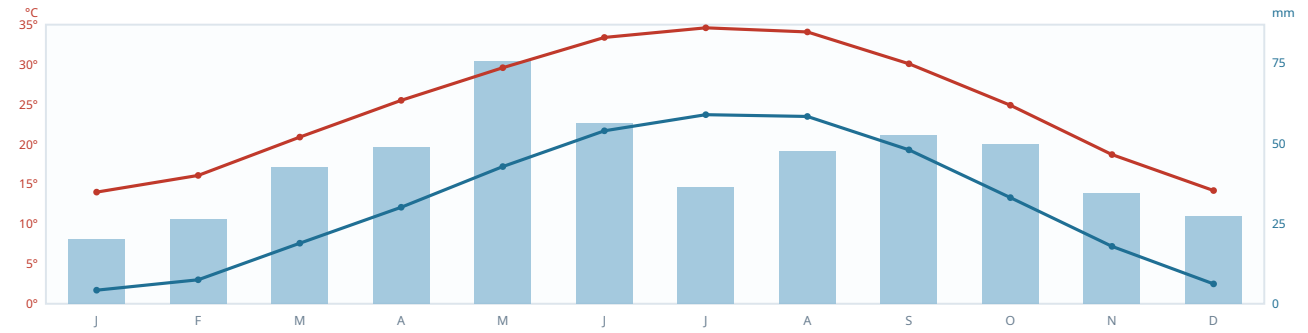
Lot area available	1.938 ac (84,437 sq ft)
Usable building envelope	NOT COMPUTABLE Setbacks, easements and any lake or drainage reserve come from the recorded plat and the HOA architectural guidelines, none of which were available.
Terrain constraint	Gentle slope — Minimal grading; Routine cut-and-fill
Foundation implication	Moderate shrink-swell. Stiffened/post-tensioned slab or moisture-conditioned subgrade is typical.
Flood implication	No mapped SFHA on the parcel per the national layer; confirm on the effective FIRM.

Construction cost
No build cost is estimated in this report. A defensible figure for a four-bedroom house depends on the HOA's minimum size and material rules, the finished-floor elevation forced by the flood zone, the foundation type the soil requires, and whether water and sewer must be created on site. Three of those four inputs are still unconfirmed for this lot, so any number quoted now would be arbitrary. Price it from a builder bid once the plat, CC&Rs, floodplain requirement and soil report are in hand.

L46 Climate MODEL-DERIVED

MEAN ANNUAL TEMP	MEAN SUMMER HIGH	MEAN WINTER LOW	ANNUAL RAINFALL
18.7 °C	**34.0** °C	**2.4** °C	**518** mm

Record high	45.4 °C	**Record low**	-16.7 °C	**Rainfall**	20.4 in
Growing season	~254 days	**Days at or below 0 °C**	34.3 /yr	**Period**	1994-2023 (30 years)



Monthly normals, 1994-2023 (30 years). Bars are mean monthly rainfall (mm, right axis); lines are mean daily maximum and minimum temperature (°C, left axis). Source: ERA5 daily reanalysis via Open-Meteo archive API.

L47 Natural hazards

Flood	No mapped SFHA on the parcel
Other perils	Hurricane and tropical wind, tornado, hail, severe convective storm, lightning, extreme heat, drought and wildfire all vary by location within Texas. QUANTIFY FROM FEMA NATIONAL RISK INDEX FOR THE CONFIRMED COUNTY AND FROM AN INSURANCE QUOTE

L48-55 Community, schools, healthcare, shopping and access NEEDS CONFIRMATION

Demographics not retrieved
Census API requires an API key (keyless request rejected) Free key from https://api.census.gov/data/key_signup.html, then set CENSUS_API_KEY and re-run 05_context.py. Or read the figures directly from data.census.gov.

Driving distance and time to the nearest hospital, emergency department and urgent care.
Obtain from: Direct routing check from the confirmed parcel location

Distance to grocery, major retail, pharmacy, bank, fuel and hardware.
Obtain from: Direct routing check

Assigned school district and campuses, with distances and current ratings.
Obtain from: TEA school-district locator and the district attendance-zone map

Nearest Indian / South Asian grocery, restaurants and places of worship.
Obtain from: Direct search from the confirmed parcel location

Nearest international airport, driving time and route.
Obtain from: Direct routing check

Walk / Bike / Transit scores.
Obtain from: Walk Score for the confirmed address

M56 Documented strengths

- Lot of 1.94 acres, which is generous for a single dwelling and leaves room for outbuildings and a garden.
- Close to Lake Alan Henry at about 587 ft, so lake amenity is nearby without shoreline exposure.
- Gentle mean grade of 2.82%, so routine grading should suffice.

M57 Limitations and risks

- Parcel identification is probable but not certain: No situs address is recorded. Matched to Oak Canyon Subdivision Lot 141 by applying the same lot-number-as-address convention seen on neighbouring parcels. Two records exist for Lot 141, which suggests a split interest. Confirm against the listing's parcel ID.
- FEMA has no effective digital flood mapping here, so flood risk is formally undetermined rather than low. Lenders and insurers may treat it conservatively.
- Septic absorption is rated "very limited" for the dominant soil, so a conventional drainfield may not be permittable and an engineered or aerobic system may be required at higher cost.
- The appraisal record carries a zero value or zero area, which signals a questionable or superseded parcel record.
- HOA covenants, zoning, utility availability, dock rights and short-term-rental rules are all unconfirmed. Any of them can independently prevent the intended use.

M59 Construction feasibility conclusion

Main house	Likely The lot is large enough and the terrain is workable. Finished-floor elevation and foundation type are the governing variables, not land area.
Second in-law suite / ADU	Unresolved Depends entirely on county rules and the HOA declaration; neither was available. Do not assume a second dwelling is permitted.
Garage	Likely No physical constraint identified, subject to setbacks and architectural guidelines.
Septic system	Unresolved Soil ratings are unfavourable; permittability and bedroom count require a site evaluation.
Garden	Likely Native surface soil is workable with normal organic amendment.
Hobby farming / poultry	Unresolved Governed by county rules and CC&Rs; unconfirmed.
Dock	Unresolved The lot does not adjoin mapped water, so a private dock is unlikely; shared access may exist.
Short-term rental	Unresolved Governed by local ordinance and CC&Rs; unconfirmed.
Pool / outdoor recreation	Conditional Space is available; flood-zone and soil-movement considerations affect pool design where applicable.

M60 Key measurements summary

Parameter	Finding
Lot size	1.938 ac (84,437 sq ft)
Lot dimensions (long axis × width)	521 ft × 325 ft
Lowest elevation	2,281.27 ft
Highest elevation	2,292.18 ft
Elevation difference	10.92 ft
Average slope	2.82% (1.62°)
Maximum slope	8.90% (5.09°)
Steepest 50 ft run	3.9% (2.2°)
FEMA zone	unmapped / Zone D - undetermined
BFE	not published
Minimum finished-floor elevation	NEEDS CONFIRMATION
Fill or elevation required?	Not indicated
Sewer	NEEDS CONFIRMATION
Water	NEEDS CONFIRMATION
Electricity	NEEDS CONFIRMATION

... continued

Parameter	Finding
Gas	NEEDS CONFIRMATION
Internet	NEEDS CONFIRMATION
Septic required?	NEEDS CONFIRMATION — soil absorption rated Very limited
Soil	Olton — Well drained, hydrologic group C, Moderate shrink-swell
HOA	Platted subdivision (Oak Canyon), so covenants are likely. NEEDS CONFIRMATION
HOA fee	NEEDS CONFIRMATION
Zoning	NEEDS CONFIRMATION
Dock possible?	NEEDS CONFIRMATION
Lake depth	NEEDS CONFIRMATION
Boating	NEEDS CONFIRMATION
Fishing	NEEDS CONFIRMATION
Hobby farming	NEEDS CONFIRMATION
Second in-law suite	NEEDS CONFIRMATION
Short-term rental	NEEDS CONFIRMATION
Nearest hospital	NEEDS CONFIRMATION
Nearest international airport	NEEDS CONFIRMATION
Distance to nearest mapped water	587 ft to Lake Alan Henry
Mean annual temperature	18.7 °C
Mean annual rainfall	518 mm (20.4 in)
Appraised market value	$0

80 Ward Rd, Snyder, TX 79549

Lot 11 of 12 | ID confidence: medium | Oak Canyon | Kent County

Identification caveat
Situs reads '80 WARD RD' and the parcel is Oak Canyon Estates Lot 80. In this subdivision the number in the address is the LOT number, and the appraisal roll records the city inconsistently as Snyder or Justiceburg. The parcel is in Kent County. Confirm against the listing's parcel ID.

A1 Property identification & record data OFFICIAL

Situs address of record	80 WARD RD , ,
Legal description	OAK CANYON ESTATES LOT 80
Subdivision	Oak Canyon
County	Kent County (FIPS 48263)
Parcel ID / account	51098
Geographic ID (APN)	05059-00200-00122-000030
Owner of record	THOMPSON TYE ALAN & TARA
Acreage — appraisal roll (legal)	2.00 ac
Acreage — GIS polygon	1.913 ac
Acreage — recomputed from boundary	1.919 ac
Appraised land value	$15,600
Appraised improvement value	$0
Appraised market value	$15,600
Centroid (WGS84)	33.019703, -101.036427
Longest axis / width	694 ft × 158 ft (axis bearing 65°)

Source: Texas StratMap statewide land parcels (county appraisal roll). Matched by situs address match. Appraisal-roll acreage and a GIS polygon rarely agree exactly; neither replaces a survey.

A2 · Listing and plat material not retrievable here NEEDS CONFIRMATION

Current asking price, days on market and listing history.
Obtain from: Listing agent / MLS

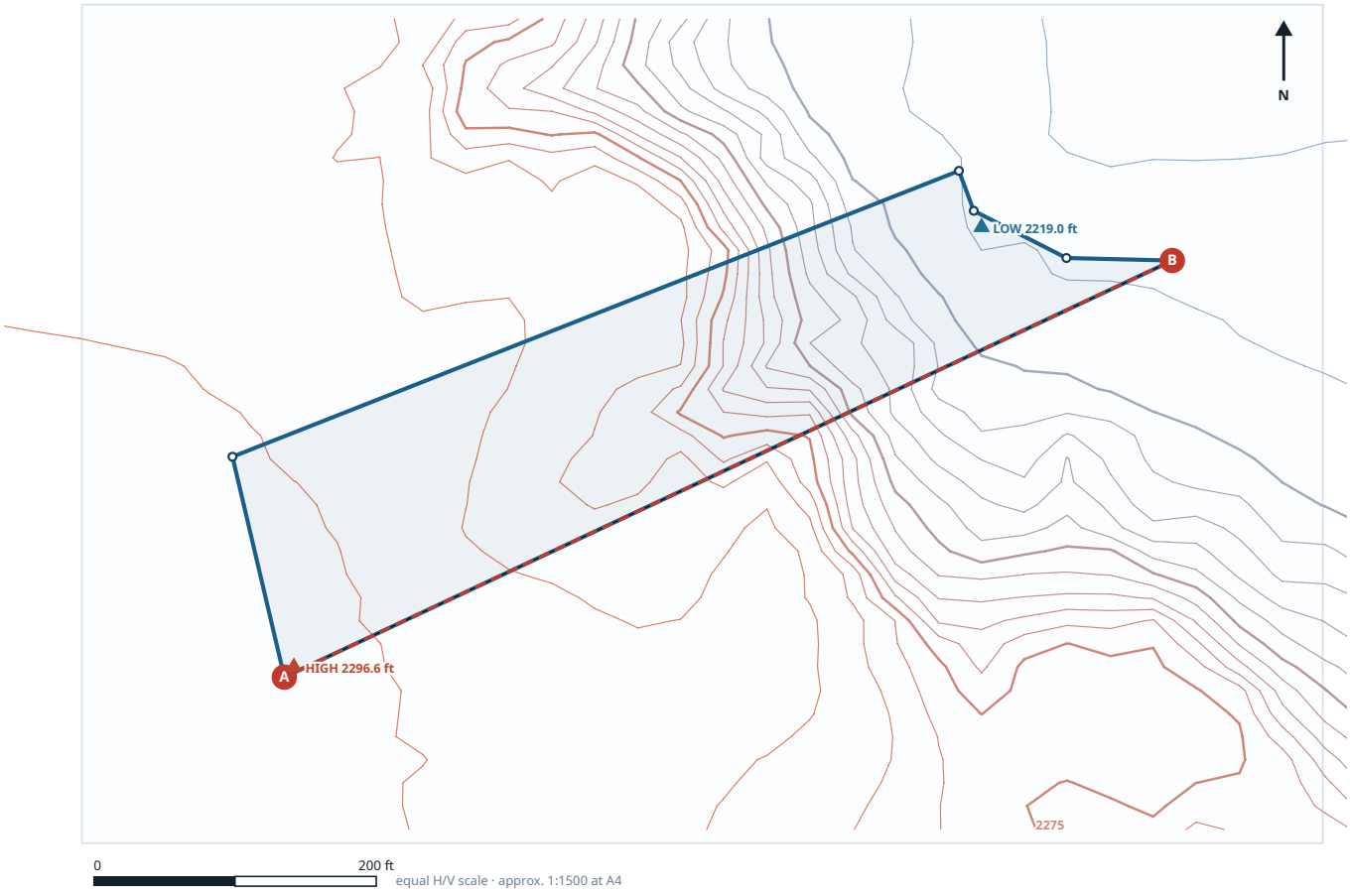
Listing photographs: lot frontage, shoreline, road access and surroundings.
Obtain from: Listing agent (Zillow blocks automated retrieval, so no listing imagery is reproduced here)

Seller disclosure notice and any known defects.
Obtain from: Listing agent

Recorded plat and Esri/county GIS parcel exhibit for comparison with this report.
Obtain from: County Clerk plat records and the county GIS portal

B5 Terrain, contours and true-scale site plan MODEL-DERIVED

LOWEST POINT	HIGHEST POINT	RELIEF	MEAN ELEVATION
2,219.0 ft	2,296.6 ft	77.6 ft	2,271.2 ft



True-scale site plan. Parcel boundary from the county GIS polygon; contours at 5.0 ft intervals derived from the USGS 3DEP 1 m bare-earth DEM. A–B is the cross-section line. Horizontal and vertical map scales are equal, so shapes and distances are undistorted.

MEAN GRADE	MEDIAN GRADE	90TH PCT GRADE	MAX GRADE
11.99 %	5.09 %	36.28 %	78.35 %

Mean slope angle	6.84°	**90th pct angle**	19.94°	**Max angle**	38.08°	**Grid cells**	1,156

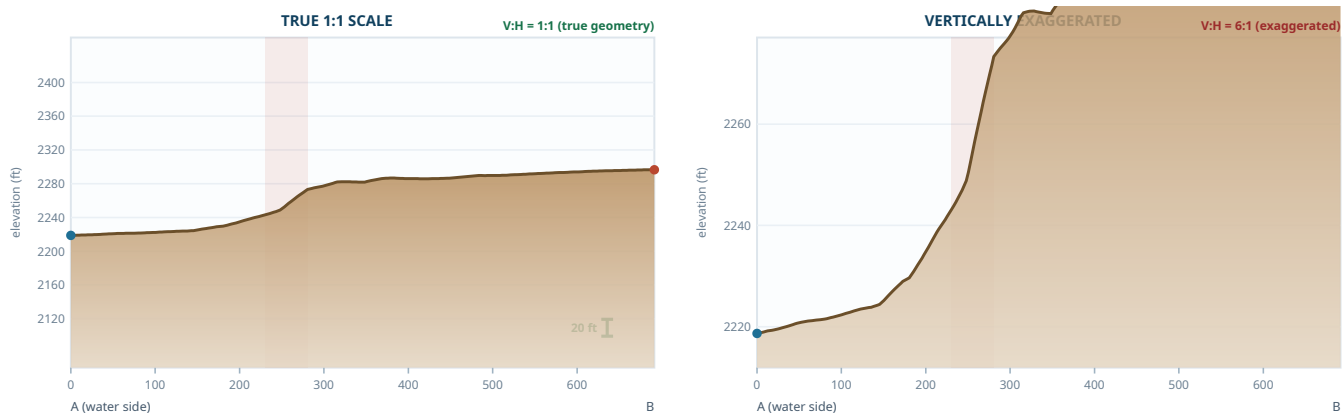
B7 · Terrain classification

Steep slope. Significant site engineering: retaining walls, terracing, erosion control.

Basis: mean DEM grade 11.99% (90th pct 36.28%), relief 77.59 ft

Grading expectation: Cut-and-fill; Retaining walls; Extensive site engineering

B6 Longitudinal cross-section MODEL-DERIVED



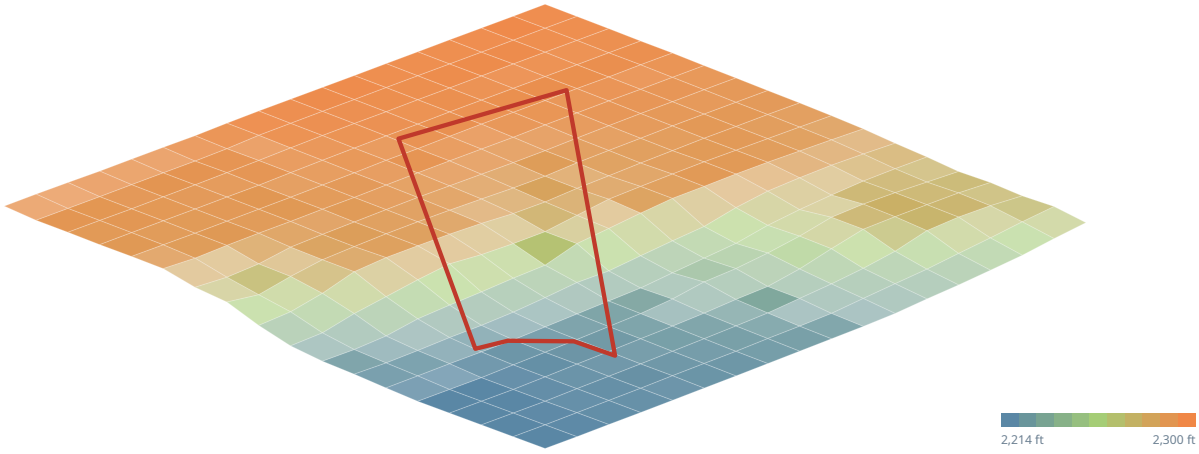
Run A–B	692 ft	**Lowest**	2,218.7 ft	**Highest**	2,296.6 ft
Elevation difference	78.0 ft	**Average grade**	11.3%	**Direct rising angle**	6.4°
Steepest 50 ft					**60.3%** (31.1°) over a 50 ft run

Reading: The ground rises 78.0 ft over 692 ft, a direct angle of 6.4° (11.3% average), and the steepest 50 ft run reaches 60% (31°). This is real, visible slope that will drive access and retaining cost.

Left panel is the real geometry at 1:1 vertical-to-horizontal scale. The right panel exaggerates the vertical axis, which is how slope is conventionally drawn on maps but overstates steepness. The shaded band marks the steepest 50 ft run. Profile sampled from the 3DEP DEM along the lot's longest axis.

B8 Three-dimensional terrain MODEL-DERIVED

Sampled window relief 85.9 ft · vertical scale exaggerated for legibility



Isometric view of the sampled DEM window; the red outline is the parcel boundary draped on the surface. Vertical scale is exaggerated to reveal form — use the 1:1 cross-section above for true steepness.

C9 FEMA flood-zone determination OFFICIAL

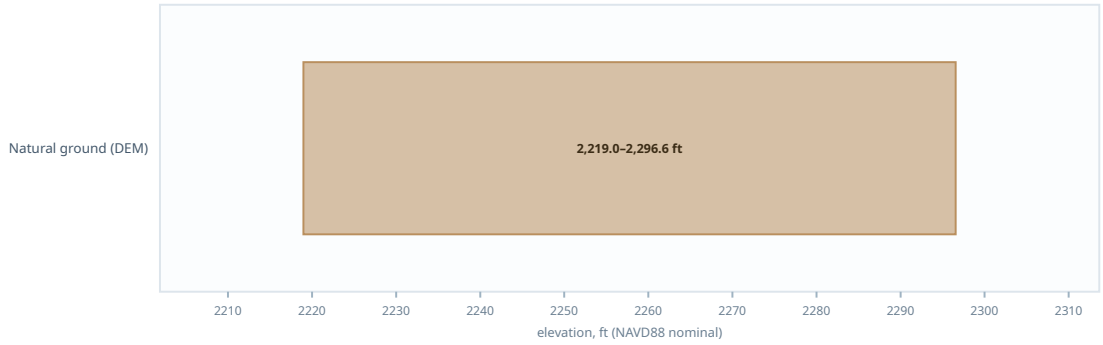
Zone: no polygon on parcel | Not in a mapped SFHA

Zones intersecting the parcel	no polygon on parcel
Special Flood Hazard Area	Not per the national layer
FIRM panel	—
Mapping status	unmapped / Zone D - undetermined

No effective NFHL flood-hazard polygons were returned within 15 miles. This location appears to lie outside FEMA's digital mapping. The flood zone is UNDETERMINED (functionally Zone D); obtain a county floodplain determination before relying on any flood conclusion.

C11-13 Ground elevation, BFE and minimum finished floor OFFICIAL MODEL-DERIVED

Lowest natural ground (DEM)	2,218.98 ft
Highest natural ground (DEM)	2,296.57 ft
Mean ground	2,271.20 ft
Indicative BFE	not published for this location



Freeboard scenarios. The brown band is the measured spread of natural ground across the lot. Red ticks are the minimum finished-floor elevation at BFE plus 0, 1, 2 and 3 ft of freeboard. **The freeboard the county actually enforces is not established in this report** — the scenarios show the consequence of each, they do not assert a requirement.

C · Flood items still to confirm NEEDS CONFIRMATION

Official flood determination and the effective FIRM panel read at the parcel.
Obtain from: County floodplain administrator or a flood certification service

Enforced freeboard above BFE, fill rules and compensatory-storage requirements.
Obtain from: County floodplain administrator

First Street / third-party forward-looking flood risk and estimated flood depth.
Obtain from: First Street Foundation (riskfactor.com) for the confirmed address

Flood insurance quote under current NFIP pricing.
Obtain from: Insurance agent, using an Elevation Certificate

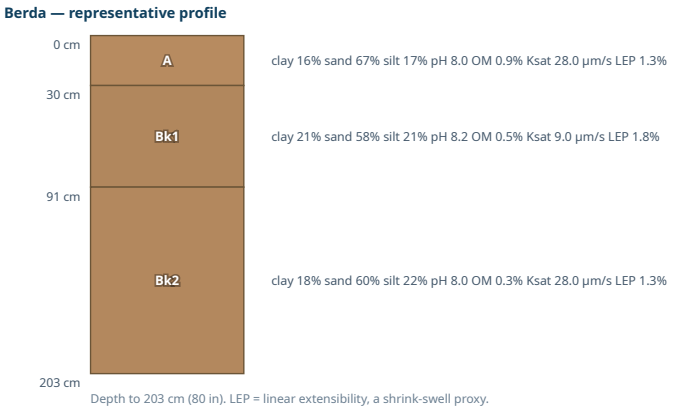
Site drainage: swales, culverts, outfall and upstream runoff from neighbouring lots.
Obtain from: Civil engineer site visit; county drainage criteria

D14 Soil characteristics MODEL-DERIVED

NRCS map unit intersecting the lot	Share of lot (sampled)
Berda fine sandy loam, moist, 3 to 5 percent slopes	16.7%
Miles-Cobb complex, 1 to 3 percent slopes	41.7%
Rough broken land	41.7%

Component	%	Drainage	Hyd.grp	LEP %	Shrink-swell	pH	Clay %	Flooding
Berda	90%	Well drained	B	1.8	Low	8.0	16	None
Miles	57%	Well drained	B	2.8	Low	7.0	13	None
Cobb	35%	Well drained	C	2.6	Low	7.0	13	None
Rough broken land	100%	Well drained	C	1.5	Low	7.9	19	None

LEP = linear extensibility (shrink-swell proxy). Hydrologic group D indicates the slowest infiltration. Values are NRCS representative values for the map unit, not measurements on this lot.



Representative horizon sequence for the dominant component (Miles).

D15 Foundation suitability MODEL-DERIVED

Dominant soil shrink-swell potential is **Low** (peak LEP 1.8%). Conventional slab-on-grade is normally viable.

Slab-on-grade	Suitable
Crawlspace	Suitable
Pier and beam	Suitable
Basement	Site-specific
Stepped footing	Only if sloped
NRCS rating, dwellings without basements	—
Geotechnical investigation advised	Not indicated

> Foundation selection is an engineering decision. NRCS map-unit shrink-swell indicates risk; it does not substitute for a geotechnical investigation.

D16 Septic / on-site sewage suitability MODEL-DERIVED

Septic Tank Absorption Fields	Somewhat limited
Septic Tank Leaching Chamber (TX)	Not limited
Septic Tank, Subsurface Drip Irrigation (TX)	Not limited

Constraints identified from the soil and terrain data

- 90th-percentile grade of 36.28% complicates drainfield layout.
- Mapped water about 56.2 ft away; TCEQ/county setbacks from water will restrict drainfield placement.

Permitted bedroom count: NOT ESTABLISHED
Permitted bedroom count follows from a site-specific soil evaluation (TCEQ Ch. 285 class, loading rate, usable drainfield area and setbacks). It cannot be derived from map-unit data. A licensed site evaluator must run the percolation/soil profile test.

NRCS ratings describe the map unit, not the lot. A TCEQ-compliant site evaluation and percolation test is required for any permit.

D17 Gardening suitability MODEL-DERIVED

SURFACE PH	CLAY	SAND	ORGANIC MATTER
8.0	16 %	67 %	0.9 %

Drainage class	Well drained
pH range through profile	8.0 – 8.2
NRCS lawn / landscape rating	Somewhat limited
Raised beds advised	Not necessary

Surface horizon is workable; in-ground beds are reasonable with normal organic amendment.

Likely amendments

- Elemental sulphur or acidifying organic matter to lower pH
- Compost / organic matter, surface organic matter is low
- Mulch, to buffer soil temperature and moisture

Based on NRCS map-unit averages. A lab soil test is required before committing to any amendment programme.

D · Soil and planting items to confirm NEEDS CONFIRMATION

Laboratory soil test: pH, macro and micronutrients, organic matter, texture.
Obtain from: Texas A&M AgriLife Soil Testing Lab, from samples taken on the lot

Plant hardiness zone confirmation and a species list matched to the tested soil.
Obtain from: USDA hardiness map plus the county AgriLife extension office

E19-23 Utilities and infrastructure NEEDS CONFIRMATION

Utility availability is not published as open geospatial data at parcel level. Nothing is inferred here: each item below must come from the serving utility in writing, because a missing water tap or a long electric extension can change the build budget materially.

Water supply: municipal, water-supply-corporation, or private well; is a tap at the road?
Obtain from: Serving water supply corporation / county; well logs via TX Water Development Board

Sewer: is public sewer available, or is an on-site facility required? Distance to any main.
Obtain from: County / city utility department or MUD

Electricity: provider, whether service exists at the lot line, overhead vs underground, connection cost.
Obtain from: Retail electric provider / local co-op (service availability letter)

Natural gas or propane availability and existing infrastructure.
Obtain from: Gas utility or local propane suppliers

... continued

Internet: fibre, cable, fixed wireless, DSL or satellite, with advertised speeds.
Obtain from: FCC National Broadband Map plus a direct serviceability check by address

F24-25 Waterfront position MODEL-DERIVED

Very close to mapped water (not necessarily frontage) | Lake Alan Henry

Nearest mapped waterbody	Lake Alan Henry
Distance from parcel boundary	56 ft
Waterbody area	10.775 km²

> NHD polygons are hydrographic mapping, not a boundary survey. Whether the lot carries riparian frontage, deeded access or only common-area access is a title/plat question and is NOT established here.

F26-29 · Dock, depth, boating and fishing NEEDS CONFIRMATION

Whether a dock is permittable at this shoreline, and the governing authority.
Obtain from: Controlling lake / river authority, plus the HOA

Dock size limits, shoreline setbacks, shared-dock rules and permit cost.
Obtain from: Lake authority rules and permit schedule

Whether riparian or shoreline rights pass with the lot, or the shoreline is a common reserve.
Obtain from: Title commitment and recorded plat

Lake water-level regime and any shoreline-management plan.
Obtain from: Lake/river authority or reservoir operator

Bathymetry: average and maximum depth, and depth near this shoreline.
Obtain from: Lake authority survey, TPWD reservoir report, or a sonar survey

Seasonal water-level variation and drawdown history.
Obtain from: Reservoir operator gauge records / USGS gauges

Boating rules: permitted craft, motor limits, wake rules, ramps and marinas.
Obtain from: Lake authority and TPWD

Fishery: species present, stocking, licence requirements and catch limits.
Obtain from: Texas Parks & Wildlife reservoir survey reports

G30-33 HOA, covenants, zoning and a second dwelling NEEDS CONFIRMATION

The parcel sits in **Oak Canyon** according to the recorded legal description, so a declaration of covenants very likely exists. The documents themselves are not public geospatial data and were not retrieved, so no dues figure, restriction or amenity is asserted here.

Whether an HOA or POA exists and is mandatory, with its name and management contact.
Obtain from: Title commitment, recorded Declaration, seller disclosure

Zoning designation, or confirmation that the county is unzoned.
Obtain from: County or city development services

Minimum lot size, building coverage, setbacks, height limits, accessory-structure rules.
Obtain from: County/city development standards or subdivision regulations

Whether a second independent dwelling or in-law suite (ADU) is permitted, and on what terms.
Obtain from: County/city development services plus the HOA

Permitting path, driveway/culvert permit and any development fee.
Obtain from: County permitting office

... continued

Annual dues, monthly equivalent, transfer fee, and any special assessment.
Obtain from: HOA or management company estoppel / resale certificate

Declaration, CC&Rs, architectural guidelines, rules and all amendments.
Obtain from: HOA or title company

Restrictions on docks, fences, outbuildings, RVs, boats, livestock and poultry.
Obtain from: Recorded CC&Rs and architectural guidelines

Minimum dwelling size and any construction time limit after purchase.
Obtain from: Recorded CC&Rs

Amenities actually included: clubhouse, pool, courts, parks, lake and boat facilities.
Obtain from: HOA documents and budget

H34 / I35 Hobby farming and short-term rental NEEDS CONFIRMATION

Whether chickens, ducks, rabbits, goats or bees are allowed, and in what numbers.
Obtain from: County rules plus the recorded CC&Rs (either can prohibit)

Any agricultural or wildlife valuation currently on the parcel, and rollback-tax exposure if use changes.
Obtain from: County appraisal district

Whether short-term rental is permitted at this location.
Obtain from: County/city ordinance and the HOA

Any STR permit, occupancy cap, parking or noise condition.
Obtain from: Local ordinance

Hotel occupancy tax registration and remittance obligations.
Obtain from: Texas Comptroller plus local taxing authority

J36-42 Environmental and geological due diligence MODEL-DERIVED

Ground-penetrating radar penetration (NRCS rating)	Moderate penetration Indicates how useful GPR would be here; clay-rich soils attenuate the signal.
Sinkhole / karst exposure	Not indicated by the soil map units returned for this parcel. No restrictive bedrock or karst feature was reported. CONFIRM WITH TX BUREAU OF ECONOMIC GEOLOGY
Radon	EPA publishes radon zones by county, not by parcel. LOOK UP THE COUNTY ZONE AND TEST ANY BUILT STRUCTURE

Existing FEMA Elevation Certificate for the lot.
Obtain from: Seller, county floodplain office or a surveyor

Recent boundary, ALTA or topographic survey.
Obtain from: Seller or a licensed Texas surveyor (RPLS)

Geotechnical / soil boring report.
Obtain from: Seller or a geotechnical engineer

On-site sewage facility site evaluation and percolation test.
Obtain from: Licensed site evaluator / county OSSF office

Any ground-penetrating-radar or other subsurface investigation.
Obtain from: Seller; commission if subsurface features are suspected

K43-45 Residential construction feasibility MODEL-DERIVED

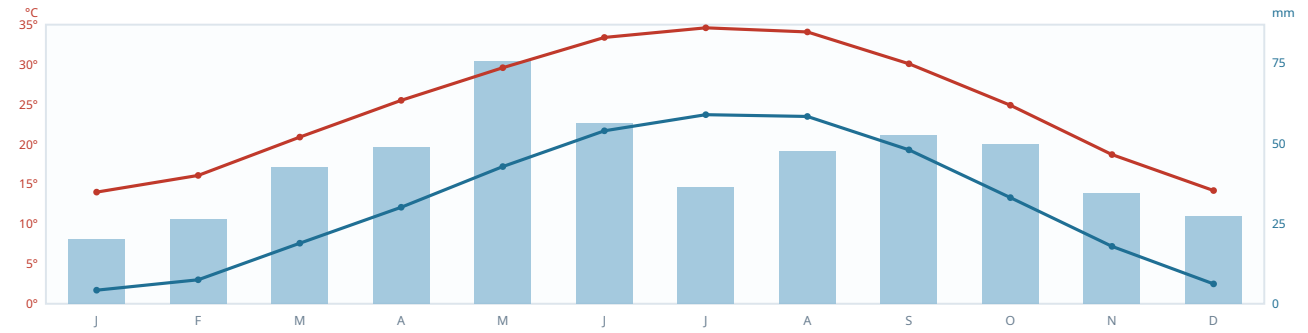
Lot area available	1.913 ac (83,317 sq ft)
Usable building envelope	NOT COMPUTABLE Setbacks, easements and any lake or drainage reserve come from the recorded plat and the HOA architectural guidelines, none of which were available.
Terrain constraint	Steep slope — Cut-and-fill; Retaining walls; Extensive site engineering
Foundation implication	Low shrink-swell. Conventional slab-on-grade is normally viable.
Flood implication	No mapped SFHA on the parcel per the national layer; confirm on the effective FIRM.

Construction cost
No build cost is estimated in this report. A defensible figure for a four-bedroom house depends on the HOA's minimum size and material rules, the finished-floor elevation forced by the flood zone, the foundation type the soil requires, and whether water and sewer must be created on site. Three of those four inputs are still unconfirmed for this lot, so any number quoted now would be arbitrary. Price it from a builder bid once the plat, CC&Rs, floodplain requirement and soil report are in hand.

L46 Climate MODEL-DERIVED

MEAN ANNUAL TEMP	MEAN SUMMER HIGH	MEAN WINTER LOW	ANNUAL RAINFALL
18.7 °C	**34.0** °C	**2.4** °C	**518** mm

Record high	45.4 °C	**Record low**	-16.7 °C	**Rainfall**	20.4 in
Growing season	~254 days	**Days at or below 0 °C**	34.3 /yr	**Period**	1994-2023 (30 years)



Monthly normals, 1994-2023 (30 years). Bars are mean monthly rainfall (mm, right axis); lines are mean daily maximum and minimum temperature (°C, left axis). Source: ERA5 daily reanalysis via Open-Meteo archive API.

L47 Natural hazards

Flood	No mapped SFHA on the parcel
Other perils	Hurricane and tropical wind, tornado, hail, severe convective storm, lightning, extreme heat, drought and wildfire all vary by location within Texas. QUANTIFY FROM FEMA NATIONAL RISK INDEX FOR THE CONFIRMED COUNTY AND FROM AN INSURANCE QUOTE

L48-55 Community, schools, healthcare, shopping and access NEEDS CONFIRMATION

Demographics not retrieved
Census API requires an API key (keyless request rejected) Free key from https://api.census.gov/data/key_signup.html, then set CENSUS_API_KEY and re-run 05_context.py. Or read the figures directly from data.census.gov.

Driving distance and time to the nearest hospital, emergency department and urgent care.
Obtain from: Direct routing check from the confirmed parcel location

Distance to grocery, major retail, pharmacy, bank, fuel and hardware.
Obtain from: Direct routing check

Assigned school district and campuses, with distances and current ratings.
Obtain from: TEA school-district locator and the district attendance-zone map

Nearest Indian / South Asian grocery, restaurants and places of worship.
Obtain from: Direct search from the confirmed parcel location

Nearest international airport, driving time and route.
Obtain from: Direct routing check

Walk / Bike / Transit scores.
Obtain from: Walk Score for the confirmed address

M56 Documented strengths

- Lot of 1.91 acres, which is generous for a single dwelling and leaves room for outbuildings and a garden.
- Abuts mapped water at roughly 56 ft (Lake Alan Henry), consistent with genuine water frontage subject to title confirmation.
- Low soil shrink-swell, the most forgiving case for a conventional slab.
- Septic absorption is rated only "somewhat limited" rather than "very limited", which is the better of the two common outcomes here.

M57 Limitations and risks

- Parcel identification is probable but not certain: Situs reads '80 WARD RD' and the parcel is Oak Canyon Estates Lot 80. In this subdivision the number in the address is the LOT number, and the appraisal roll records the city inconsistently as Snyder or Justiceburg. The parcel is in Kent County. Confirm against the listing's parcel ID.
- FEMA has no effective digital flood mapping here, so flood risk is formally undetermined rather than low. Lenders and insurers may treat it conservatively.
- Steep sub-areas: the 90th-percentile grade reaches 36.3%, so siting, driveway grade and retaining structures need care.
- A 50 ft run reaches 60% (31°), which is steep enough to drive retaining-wall and access cost.
- HOA covenants, zoning, utility availability, dock rights and short-term-rental rules are all unconfirmed. Any of them can independently prevent the intended use.

M59 Construction feasibility conclusion

Main house	Likely The lot is large enough and the terrain is workable. Finished-floor elevation and foundation type are the governing variables, not land area.
Second in-law suite / ADU	Unresolved Depends entirely on county rules and the HOA declaration; neither was available. Do not assume a second dwelling is permitted.
Garage	Likely No physical constraint identified, subject to setbacks and architectural guidelines.
Septic system	Conditional Plausible but unproven. A percolation test and site evaluation decide it.
Garden	Likely Native surface soil is workable with normal organic amendment.
Hobby farming / poultry	Unresolved Governed by county rules and CC&Rs; unconfirmed.

... continued

Dock	Unresolved The lot adjoins water, but dock rights are a title and permitting question that was not resolved.
Short-term rental	Unresolved Governed by local ordinance and CC&Rs; unconfirmed.
Pool / outdoor recreation	Conditional Space is available; flood-zone and soil-movement considerations affect pool design where applicable.

M60 Key measurements summary

Parameter	Finding
Lot size	1.913 ac (83,317 sq ft)
Lot dimensions (long axis × width)	694 ft × 158 ft
Lowest elevation	2,218.98 ft
Highest elevation	2,296.57 ft
Elevation difference	77.59 ft
Average slope	11.99% (6.84°)
Maximum slope	78.35% (38.08°)
Steepest 50 ft run	60.3% (31.1°)
FEMA zone	unmapped / Zone D - undetermined
BFE	not published
Minimum finished-floor elevation	NEEDS CONFIRMATION
Fill or elevation required?	Not indicated
Sewer	NEEDS CONFIRMATION
Water	NEEDS CONFIRMATION
Electricity	NEEDS CONFIRMATION
Gas	NEEDS CONFIRMATION
Internet	NEEDS CONFIRMATION
Septic required?	NEEDS CONFIRMATION — soil absorption rated Somewhat limited
Soil	Miles — Well drained, hydrologic group B, Low shrink-swell
HOA	Platted subdivision (Oak Canyon), so covenants are likely. NEEDS CONFIRMATION
HOA fee	NEEDS CONFIRMATION
Zoning	NEEDS CONFIRMATION
Dock possible?	NEEDS CONFIRMATION
Lake depth	NEEDS CONFIRMATION
Boating	NEEDS CONFIRMATION
Fishing	NEEDS CONFIRMATION
Hobby farming	NEEDS CONFIRMATION
Second in-law suite	NEEDS CONFIRMATION
Short-term rental	NEEDS CONFIRMATION
Nearest hospital	NEEDS CONFIRMATION
Nearest international airport	NEEDS CONFIRMATION
Distance to nearest mapped water	56 ft to Lake Alan Henry
Mean annual temperature	18.7 °C
Mean annual rainfall	518 mm (20.4 in)
Appraised market value	$15,600

2883 W Arrowhead Dr, Wichita Falls, TX 76310

Lot 12 of 12 | ID confidence: high | West Blk | Clay County

A1 Property identification & record data OFFICIAL

Situs address of record	2883 W ARROWHEAD DR , WICHITA FALLS, TX 76310
Legal description	LT E-15 WEST BLK
Subdivision	West Blk
County	Clay County (FIPS 48077)
Parcel ID / account	8297
Geographic ID (APN)	04180-00001-51500-000000
Owner of record	LINDEMANN COREY & LAURA
Acreage — appraisal roll (legal)	0.23 ac
Acreage — GIS polygon	1.571 ac
Acreage — recomputed from boundary	1.575 ac
Appraised land value	$9,000
Appraised improvement value	$0
Appraised market value	$9,000
Centroid (WGS84)	33.744103, -98.382137
Longest axis / width	775 ft × 177 ft (axis bearing 87°)

Source: Texas StratMap statewide land parcels (county appraisal roll). Matched by situs address match. Appraisal-roll acreage and a GIS polygon rarely agree exactly; neither replaces a survey.

A2 · Listing and plat material not retrievable here NEEDS CONFIRMATION

Current asking price, days on market and listing history.
Obtain from: Listing agent / MLS

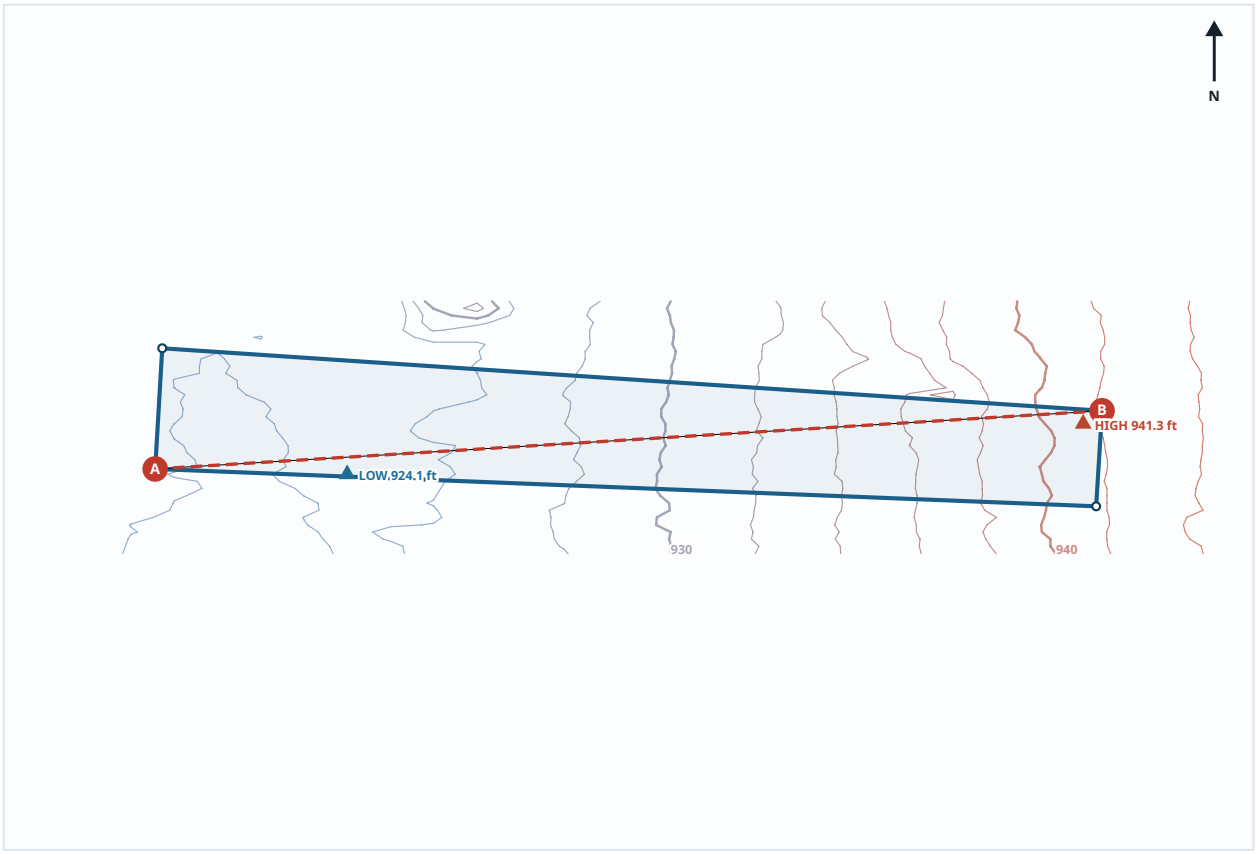
Listing photographs: lot frontage, shoreline, road access and surroundings.
Obtain from: Listing agent (Zillow blocks automated retrieval, so no listing imagery is reproduced here)

Seller disclosure notice and any known defects.
Obtain from: Listing agent

Recorded plat and Esri/county GIS parcel exhibit for comparison with this report.
Obtain from: County Clerk plat records and the county GIS portal

B5 Terrain, contours and true-scale site plan MODEL-DERIVED

LOWEST POINT	HIGHEST POINT	RELIEF	MEAN ELEVATION
924.1 ft	941.3 ft	17.2 ft	930.2 ft



True-scale site plan. Parcel boundary from the county GIS polygon; contours at 2.0 ft intervals derived from the USGS 3DEP 1 m bare-earth DEM. A–B is the cross-section line. Horizontal and vertical map scales are equal, so shapes and distances are undistorted.

MEAN GRADE	MEDIAN GRADE	90TH PCT GRADE	MAX GRADE
3.06 %	2.83 %	4.92 %	37.30 %

Mean slope angle	1.75°	**90th pct angle**	2.82°	**Max angle**	20.46°	**Grid cells**	1,156

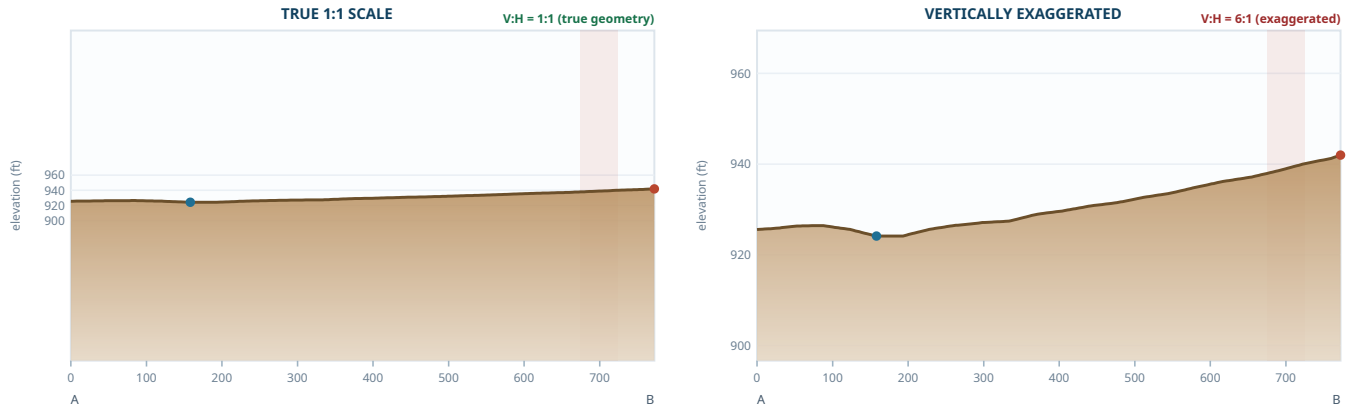
B7 · Terrain classification

Gentle slope. Standard residential grading is normally sufficient.

Basis: mean DEM grade 3.06% (90th pct 4.92%), relief 17.15 ft

Grading expectation: Minimal grading; Routine cut-and-fill

B6 Longitudinal cross-section MODEL-DERIVED



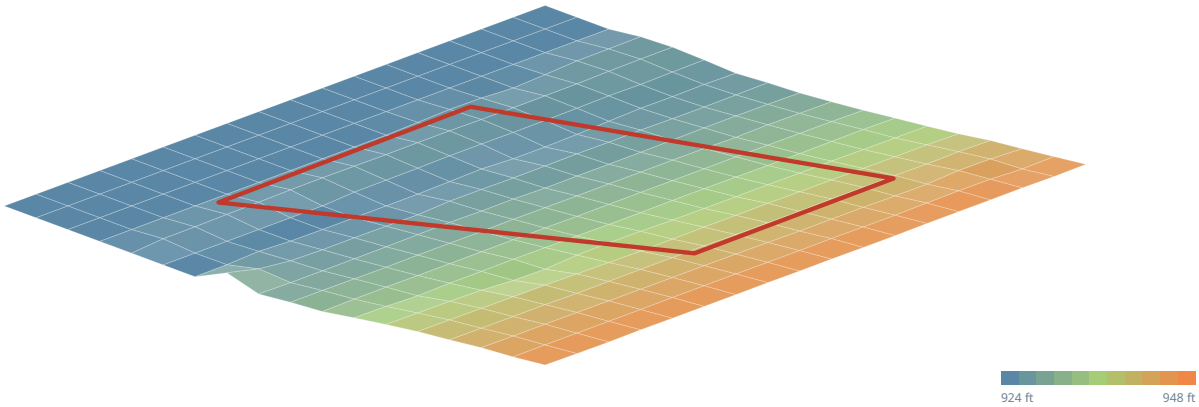
Run A–B	772 ft	**Lowest**	924.1 ft	**Highest**	942.0 ft
Elevation difference	17.9 ft	**Average grade**	2.3%	**Direct rising angle**	1.3°
Steepest 50 ft	**4.3%** (2.5°) over a 50 ft run				

Reading: Over 772 ft the ground rises only 17.9 ft, a direct angle of 1.3°. At true 1:1 scale the profile is almost a straight line, which is the correct reading: this lot is effectively flat. The exaggerated panel is the same data and should not be read as slope.

Left panel is the real geometry at 1:1 vertical-to-horizontal scale. The right panel exaggerates the vertical axis, which is how slope is conventionally drawn on maps but overstates steepness. The shaded band marks the steepest 50 ft run. Profile sampled from the 3DEP DEM along the lot's longest axis.

B8 Three-dimensional terrain MODEL-DERIVED

Sampled window relief 24.0 ft · vertical scale exaggerated for legibility



Isometric view of the sampled DEM window; the red outline is the parcel boundary draped on the surface. Vertical scale is exaggerated to reveal form — use the 1:1 cross-section above for true steepness.

C9 FEMA flood-zone determination OFFICIAL

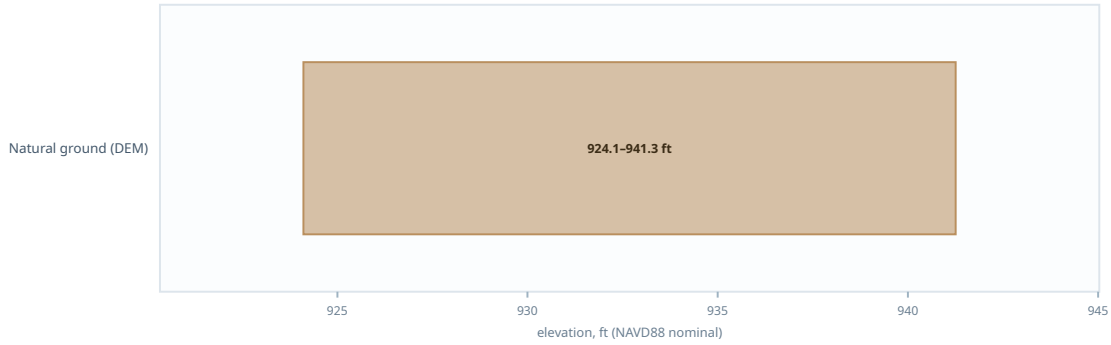
Zone: no polygon on parcel | Not in a mapped SFHA

Zones intersecting the parcel	no polygon on parcel
Special Flood Hazard Area	Not per the national layer
FIRM panel	48009C0250C · effective 2021-02-12
Mapping status	likely Zone X (unshaded) - area is mapped

No FEMA flood-hazard polygon intersects this parcel, but mapped flood zones exist within 5 mile(s), so the area carries effective NFHL coverage. This is consistent with unshaded Zone X - outside the 0.2%-annual-chance floodplain. Confirm against the effective FIRM panel.

C11-13 Ground elevation, BFE and minimum finished floor OFFICIAL MODEL-DERIVED

Lowest natural ground (DEM)	924.11 ft
Highest natural ground (DEM)	941.26 ft
Mean ground	930.22 ft
Indicative BFE	not published for this location



Freeboard scenarios. The brown band is the measured spread of natural ground across the lot. Red ticks are the minimum finished-floor elevation at BFE plus 0, 1, 2 and 3 ft of freeboard. **The freeboard the county actually enforces is not established in this report** — the scenarios show the consequence of each, they do not assert a requirement.

C · Flood items still to confirm NEEDS CONFIRMATION

Official flood determination and the effective FIRM panel read at the parcel.
Obtain from: County floodplain administrator or a flood certification service

Enforced freeboard above BFE, fill rules and compensatory-storage requirements.
Obtain from: County floodplain administrator

First Street / third-party forward-looking flood risk and estimated flood depth.
Obtain from: First Street Foundation (riskfactor.com) for the confirmed address

Flood insurance quote under current NFIP pricing.
Obtain from: Insurance agent, using an Elevation Certificate

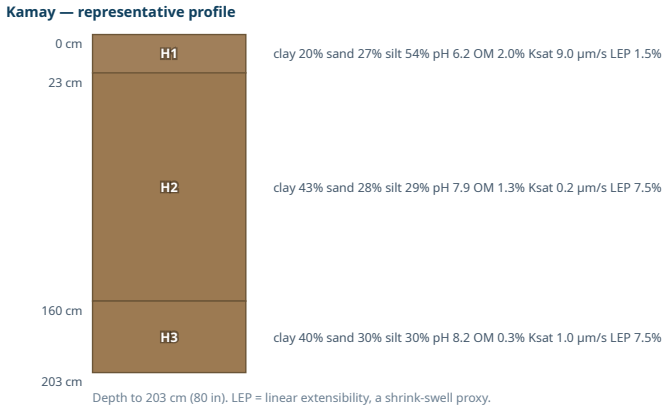
Site drainage: swales, culverts, outfall and upstream runoff from neighbouring lots.
Obtain from: Civil engineer site visit; county drainage criteria

D14 Soil characteristics MODEL-DERIVED

NRCS map unit intersecting the lot	Share of lot (sampled)
Kamay silt loam, 1 to 3 percent slopes	50.0%
Bluegrove-Stoneburg association, gently sloping	50.0%

Component	%	Drainage	Hyd.grp	LEP %	Shrink-swell	pH	Clay %	Flooding
Kamay	100%	Well drained	D	7.5	High	6.2	20	None
Bluegrove	70%	Well drained	C	4.5	Moderate	6.5	20	None
Stoneburg	20%	Well drained	C	4.5	Moderate	6.5	16	None

LEP = linear extensibility (shrink-swell proxy). Hydrologic group D indicates the slowest infiltration. Values are NRCS representative values for the map unit, not measurements on this lot.



Representative horizon sequence for the dominant component (Bluegrove).

D15 Foundation suitability MODEL-DERIVED

Dominant soil shrink-swell potential is **High** (peak LEP 7.5%). Engineered foundation expected: post-tensioned slab or pier-and-beam.

Slab-on-grade	Engineered post-tensioned only
Crawlspace	Suitable with piers
Pier and beam	Preferred
Basement	Not advised
Stepped footing	Only if sloped
NRCS rating, dwellings without basements	—
Geotechnical investigation advised	Yes

> Foundation selection is an engineering decision. NRCS map-unit shrink-swell indicates risk; it does not substitute for a geotechnical investigation.

D16 Septic / on-site sewage suitability MODEL-DERIVED

Septic Tank Absorption Fields	Very limited
Septic Tank Leaching Chamber (TX)	Very limited
Septic Tank, Subsurface Drip Irrigation (TX)	Not limited

Constraints identified from the soil and terrain data

- Hydrologic group D indicates slow infiltration, which constrains conventional drainfields.
- Mapped water about 26.4 ft away; TCEQ/county setbacks from water will restrict drainfield placement.

Permitted bedroom count: NOT ESTABLISHED
Permitted bedroom count follows from a site-specific soil evaluation (TCEQ Ch. 285 class, loading rate, usable drainfield area and setbacks). It cannot be derived from map-unit data. A licensed site evaluator must run the percolation/soil profile test.

NRCS ratings describe the map unit, not the lot. A TCEQ-compliant site evaluation and percolation test is required for any permit.

D17 Gardening suitability MODEL-DERIVED

SURFACE PH	CLAY	SAND	ORGANIC MATTER
6.2	20 %	27 %	2.0 %

Drainage class	Well drained
pH range through profile	6.2 – 8.2
NRCS lawn / landscape rating	Very limited
Raised beds advised	Yes

Heavy clay / restricted drainage / high shrink-swell in the surface horizon makes raised beds with imported loam the lower-risk option.

Likely amendments

- Mulch, to buffer soil temperature and moisture

Based on NRCS map-unit averages. A lab soil test is required before committing to any amendment programme.

D · Soil and planting items to confirm NEEDS CONFIRMATION

Laboratory soil test: pH, macro and micronutrients, organic matter, texture.
Obtain from: Texas A&M AgriLife Soil Testing Lab, from samples taken on the lot

Plant hardiness zone confirmation and a species list matched to the tested soil.
Obtain from: USDA hardiness map plus the county AgriLife extension office

E19-23 Utilities and infrastructure NEEDS CONFIRMATION

Utility availability is not published as open geospatial data at parcel level. Nothing is inferred here: each item below must come from the serving utility in writing, because a missing water tap or a long electric extension can change the build budget materially.

Water supply: municipal, water-supply-corporation, or private well; is a tap at the road?
Obtain from: Serving water supply corporation / county; well logs via TX Water Development Board

Sewer: is public sewer available, or is an on-site facility required? Distance to any main.
Obtain from: County / city utility department or MUD

Electricity: provider, whether service exists at the lot line, overhead vs underground, connection cost.
Obtain from: Retail electric provider / local co-op (service availability letter)

Natural gas or propane availability and existing infrastructure.
Obtain from: Gas utility or local propane suppliers

Internet: fibre, cable, fixed wireless, DSL or satellite, with advertised speeds.
Obtain from: FCC National Broadband Map plus a direct serviceability check by address

F24-25 Waterfront position MODEL-DERIVED

Abuts mapped water - consistent with direct water frontage | Lake Arrowhead

Nearest mapped waterbody	Lake Arrowhead
Distance from parcel boundary	26 ft
Waterbody area	55.774 km²

NHD polygons are hydrographic mapping, not a boundary survey. Whether the lot carries riparian frontage, deeded access or only common-area access is a title/plat question and is NOT established here.

F26-29 · Dock, depth, boating and fishing NEEDS CONFIRMATION

Whether a dock is permittable at this shoreline, and the governing authority.
Obtain from: Controlling lake / river authority, plus the HOA

Dock size limits, shoreline setbacks, shared-dock rules and permit cost.
Obtain from: Lake authority rules and permit schedule

Whether riparian or shoreline rights pass with the lot, or the shoreline is a common reserve.
Obtain from: Title commitment and recorded plat

Lake water-level regime and any shoreline-management plan.
Obtain from: Lake/river authority or reservoir operator

Bathymetry: average and maximum depth, and depth near this shoreline.
Obtain from: Lake authority survey, TPWD reservoir report, or a sonar survey

Seasonal water-level variation and drawdown history.
Obtain from: Reservoir operator gauge records / USGS gauges

Boating rules: permitted craft, motor limits, wake rules, ramps and marinas.
Obtain from: Lake authority and TPWD

Fishery: species present, stocking, licence requirements and catch limits.
Obtain from: Texas Parks & Wildlife reservoir survey reports

G30-33 HOA, covenants, zoning and a second dwelling NEEDS CONFIRMATION

The parcel sits in **West Blk** according to the recorded legal description, so a declaration of covenants very likely exists. The documents themselves are not public geospatial data and were not retrieved, so no dues figure, restriction or amenity is asserted here.

Whether an HOA or POA exists and is mandatory, with its name and management contact.
Obtain from: Title commitment, recorded Declaration, seller disclosure

Annual dues, monthly equivalent, transfer fee, and any special assessment.
Obtain from: HOA or management company estoppel / resale certificate

Declaration, CC&Rs, architectural guidelines, rules and all amendments.
Obtain from: HOA or title company

Restrictions on docks, fences, outbuildings, RVs, boats, livestock and poultry.
Obtain from: Recorded CC&Rs and architectural guidelines

Minimum dwelling size and any construction time limit after purchase.
Obtain from: Recorded CC&Rs

Amenities actually included: clubhouse, pool, courts, parks, lake and boat facilities.
Obtain from: HOA documents and budget

Zoning designation, or confirmation that the county is unzoned.
Obtain from: County or city development services

Minimum lot size, building coverage, setbacks, height limits, accessory-structure rules.
Obtain from: County/city development standards or subdivision regulations

Whether a second independent dwelling or in-law suite (ADU) is permitted, and on what terms.
Obtain from: County/city development services plus the HOA

Permitting path, driveway/culvert permit and any development fee.
Obtain from: County permitting office

H34 / I35 Hobby farming and short-term rental NEEDS CONFIRMATION

Whether chickens, ducks, rabbits, goats or bees are allowed, and in what numbers.
Obtain from: County rules plus the recorded CC&Rs (either can prohibit)

Any agricultural or wildlife valuation currently on the parcel, and rollback-tax exposure if use changes.
Obtain from: County appraisal district

Whether short-term rental is permitted at this location.
Obtain from: County/city ordinance and the HOA

Any STR permit, occupancy cap, parking or noise condition.
Obtain from: Local ordinance

Hotel occupancy tax registration and remittance obligations.
Obtain from: Texas Comptroller plus local taxing authority

J36-42 Environmental and geological due diligence MODEL-DERIVED

Ground-penetrating radar penetration (NRCS rating)	Unsuited Indicates how useful GPR would be here; clay-rich soils attenuate the signal.
Sinkhole / karst exposure	Not indicated by the soil map units returned for this parcel. No restrictive bedrock or karst feature was reported. CONFIRM WITH TX BUREAU OF ECONOMIC GEOLOGY
Radon	EPA publishes radon zones by county, not by parcel. LOOK UP THE COUNTY ZONE AND TEST ANY BUILT STRUCTURE

Existing FEMA Elevation Certificate for the lot.
Obtain from: Seller, county floodplain office or a surveyor

Recent boundary, ALTA or topographic survey.
Obtain from: Seller or a licensed Texas surveyor (RPLS)

Geotechnical / soil boring report.
Obtain from: Seller or a geotechnical engineer

On-site sewage facility site evaluation and percolation test.
Obtain from: Licensed site evaluator / county OSSF office

Any ground-penetrating-radar or other subsurface investigation.
Obtain from: Seller; commission if subsurface features are suspected

K43-45 Residential construction feasibility MODEL-DERIVED

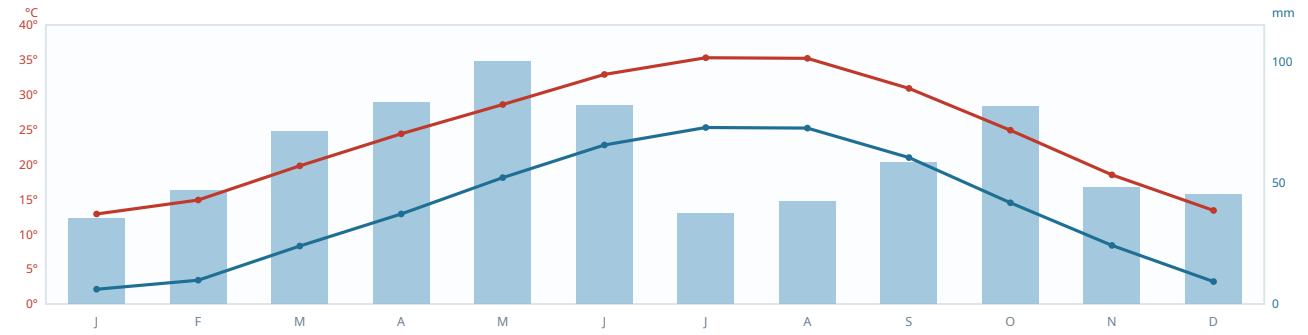
Lot area available	1.571 ac (68,437 sq ft)
Usable building envelope	NOT COMPUTABLE Setbacks, easements and any lake or drainage reserve come from the recorded plat and the HOA architectural guidelines, none of which were available.
Terrain constraint	Gentle slope — Minimal grading; Routine cut-and-fill
Foundation implication	High shrink-swell. Engineered foundation expected: post-tensioned slab or pier-and-beam.
Flood implication	No mapped SFHA on the parcel per the national layer; confirm on the effective FIRM.

Construction cost
No build cost is estimated in this report. A defensible figure for a four-bedroom house depends on the HOA's minimum size and material rules, the finished-floor elevation forced by the flood zone, the foundation type the soil requires, and whether water and sewer must be created on site. Three of those four inputs are still unconfirmed for this lot, so any number quoted now would be arbitrary. Price it from a builder bid once the plat, CC&Rs, floodplain requirement and soil report are in hand.

L46 Climate MODEL-DERIVED

MEAN ANNUAL TEMP	MEAN SUMMER HIGH	MEAN WINTER LOW	ANNUAL RAINFALL
19.1 °C	**34.5** °C	**2.9** °C	**731** mm

Record high	44.8 °C	**Record low**	-18.9 °C	**Rainfall**	28.8 in
Growing season	~262 days	**Days at or below 0 °C**	31.2 /yr	**Period**	1994-2023 (30 years)



Monthly normals, 1994-2023 (30 years). Bars are mean monthly rainfall (mm, right axis); lines are mean daily maximum and minimum temperature (°C, left axis). Source: ERA5 daily reanalysis via Open-Meteo archive API.

L47 Natural hazards

Flood	No mapped SFHA on the parcel
Other perils	Hurricane and tropical wind, tornado, hail, severe convective storm, lightning, extreme heat, drought and wildfire all vary by location within Texas. QUANTIFY FROM FEMA NATIONAL RISK INDEX FOR THE CONFIRMED COUNTY AND FROM AN INSURANCE QUOTE

L48-55 Community, schools, healthcare, shopping and access NEEDS CONFIRMATION

Demographics not retrieved
Census API requires an API key (keyless request rejected) Free key from https://api.census.gov/data/key_signup.html, then set CENSUS_API_KEY and re-run 05_context.py. Or read the figures directly from data.census.gov.

Driving distance and time to the nearest hospital, emergency department and urgent care.
Obtain from: Direct routing check from the confirmed parcel location

Distance to grocery, major retail, pharmacy, bank, fuel and hardware.
Obtain from: Direct routing check

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Assigned school district and campuses, with distances and current ratings.
Obtain from: TEA school-district locator and the district attendance-zone map

Nearest Indian / South Asian grocery, restaurants and places of worship.
Obtain from: Direct search from the confirmed parcel location

Nearest international airport, driving time and route.
Obtain from: Direct routing check

Walk / Bike / Transit scores.
Obtain from: Walk Score for the confirmed address

M56 Documented strengths

- Lot of 1.57 acres, which is generous for a single dwelling and leaves room for outbuildings and a garden.
- Abuts mapped water at roughly 26 ft (Lake Arrowhead), consistent with genuine water frontage subject to title confirmation.
- No Special Flood Hazard Area maps onto the parcel, which avoids mandatory flood insurance and floodplain construction rules.

M57 Limitations and risks

- **High soil shrink-swell.** Expansive clay is the leading cause of foundation distress in this region; an engineered foundation and a geotechnical report should be assumed.
- Septic absorption is rated "very limited" for the dominant soil, so a conventional drainfield may not be permittable and an engineered or aerobic system may be required at higher cost.
- HOA covenants, zoning, utility availability, dock rights and short-term-rental rules are all unconfirmed. Any of them can independently prevent the intended use.

M59 Construction feasibility conclusion

Main house	Likely The lot is large enough and the terrain is workable. Finished-floor elevation and foundation type are the governing variables, not land area.
Second in-law suite / ADU	Unresolved Depends entirely on county rules and the HOA declaration; neither was available. Do not assume a second dwelling is permitted.
Garage	Likely No physical constraint identified, subject to setbacks and architectural guidelines.
Septic system	Unresolved Soil ratings are unfavourable; permittability and bedroom count require a site evaluation.
Garden	Conditional Workable with raised beds and imported loam; native surface soil is restrictive.
Hobby farming / poultry	Unresolved Governed by county rules and CC&Rs; unconfirmed.
Dock	Unresolved The lot adjoins water, but dock rights are a title and permitting question that was not resolved.
Short-term rental	Unresolved Governed by local ordinance and CC&Rs; unconfirmed.
Pool / outdoor recreation	Conditional Space is available; flood-zone and soil-movement considerations affect pool design where applicable.

M60 Key measurements summary

Parameter	Finding
Lot size	1.571 ac (68,437 sq ft)
Lot dimensions (long axis × width)	775 ft × 177 ft
Lowest elevation	924.11 ft
Highest elevation	941.26 ft
Elevation difference	17.15 ft
Average slope	3.06% (1.75°)
Maximum slope	37.30% (20.46°)

... continued

Parameter	Finding
Steepest 50 ft run	4.3% (2.5°)
FEMA zone	likely Zone X (unshaded) - area is mapped
BFE	not published
Minimum finished-floor elevation	NEEDS CONFIRMATION
Fill or elevation required?	Not indicated
Sewer	NEEDS CONFIRMATION
Water	NEEDS CONFIRMATION
Electricity	NEEDS CONFIRMATION
Gas	NEEDS CONFIRMATION
Internet	NEEDS CONFIRMATION
Septic required?	NEEDS CONFIRMATION — soil absorption rated Very limited
Soil	Bluegrove — Well drained, hydrologic group D, High shrink-swell
HOA	Platted subdivision (West Blk), so covenants are likely. NEEDS CONFIRMATION
HOA fee	NEEDS CONFIRMATION
Zoning	NEEDS CONFIRMATION
Dock possible?	NEEDS CONFIRMATION
Lake depth	NEEDS CONFIRMATION
Boating	NEEDS CONFIRMATION
Fishing	NEEDS CONFIRMATION
Hobby farming	NEEDS CONFIRMATION
Second in-law suite	NEEDS CONFIRMATION
Short-term rental	NEEDS CONFIRMATION
Nearest hospital	NEEDS CONFIRMATION
Nearest international airport	NEEDS CONFIRMATION
Distance to nearest mapped water	26 ft to Lake Arrowhead
Mean annual temperature	19.1 °C
Mean annual rainfall	731 mm (28.8 in)
Appraised market value	$9,000

Documents to obtain before committing

This list applies to every lot. It is ordered roughly by how often it changes a purchase decision.

Document	Source	Why it matters
Recorded subdivision plat	County Clerk / Plat records	Confirms lot dimensions, easements, setbacks, and any lake or drainage reserve.
Boundary survey (and topographic survey)	Licensed Texas surveyor (RPLS)	Only a survey establishes boundaries, acreage and corner coordinates. DEM-derived contours in this report are not survey grade.
FEMA Elevation Certificate	Surveyor / prior owner / county floodplain office	Required to establish lowest adjacent grade and finished-floor elevation against BFE; governs flood insurance rating.
County floodplain development requirements	County floodplain administrator	Establishes the enforced freeboard above BFE, fill restrictions and permitting path.
Geotechnical / soil report	Geotechnical engineer	Needed for foundation design where shrink-swell is moderate or higher.
On-site sewage facility site evaluation and percolation test	Licensed site evaluator / county OSSF office	Determines whether septic is permittable and the bedroom count it supports.
Water service confirmation or well log	Water supply corporation / TX Water Development Board	Confirms whether a tap is available at the road, or the depth and yield of a well.
Utility availability letters (electric, gas, broadband)	Serving utilities	Confirms service at the lot line and connection cost.
HOA governing documents: Declaration/CC&Rs, architectural guidelines, rules, budget, dues and transfer fees	HOA / management company / title company	Controls buildability, minimum house size, outbuildings, livestock, rentals and dock rights.
Zoning / land-use confirmation and whether an ADU or second dwelling is permitted	County or city development services	Determines whether a second in-law suite is lawful.
Dock / shoreline permit requirements	Lake authority or river authority with jurisdiction, plus HOA	Dock feasibility is a permitting question, not a terrain question.
Short-term-rental rules	County/city ordinance plus HOA	Determines whether Airbnb/VRBO use is lawful.
Title commitment and recorded easements	Title company	Reveals access easements, utility easements, mineral reservations and restrictions.
Tax certificate / current assessment	County appraisal district / tax office	Confirms assessed value, exemptions and any special assessment.

Limitations of this report

- **Boundaries are not surveyed.** Parcel polygons come from county appraisal GIS, which is maintained for taxation and routinely differs from the recorded plat by several feet. Acreage shown from three sources (legal, GIS, recomputed) will not agree exactly.
- **Elevations are model values.** The 3DEP 1 m DEM is lidar-derived bare earth with vertical accuracy typically within about a foot, worse under dense canopy or where the ground was wet at acquisition. Contours here are interpolated from a sampled grid, not surveyed.
- **Base Flood Elevations are indicative.** Where the flood polygon carried no static BFE, the nearest published BFE line was used. The governing BFE must be read at the structure location from the FIRM and Flood Insurance Study profile.
- **Soil data describes map units, not lots.** NRCS polygons are generalised at roughly 1:24,000 and a single lot often spans more than one unit. Published values are representative, not measured.
- **No county, HOA or utility document was obtained.** Everything in that category is marked as needing confirmation rather than estimated.
- **No listing data.** Prices, photographs and listing history were not retrieved because the source blocks automated access.
- **Demographics were not retrieved** because the Census API now requires a key. The report states where to get the figures rather than substituting an estimate.
- **Nothing here is professional advice.** Flood, foundation, septic and land-use conclusions require a licensed surveyor, engineer, site evaluator or attorney as applicable.